TU WIEN Informatics

Vergleichende visuelle Analyse für multimodale Leberdaten

DIPLOMARBEIT

zur Erlangung des akademischen Grades

Diplom-Ingenieur

im Rahmen des Studiums

Biomedical Engineering

eingereicht von

Hugo Hernández Alemán, BSc
Matrikelnummer 12302816

an der Fakultät für Informatik

der Technischen Universität Wien

Betreuung: Assistant. Prof. Dr. Renata Georgia Raidou

Wien, 1. Mai 2024

Hugo Hernández Alemán	Renata Georgia Raidou

TU WIEN Informatics

Comparative Visual Analysis for Multi-Modal Liver Data

DIPLOMA THESIS

submitted in partial fulfillment of the requirements for the degree of

Diplom-Ingenieur

in

Biomedical Engineering

by

Hugo Hernández Alemán, BSc
Registration Number 12302816

to the Faculty of Informatics

at the TU Wien

Advisor: Assistant. Prof. Dr. Renata Georgia Raidou

Vienna, May 1, 2024

Hugo Hernández Alemán	Renata Georgia Raidou

Erklärung zur Verfassung der Arbeit

Hugo Hernández Alemán, BSc

Hiermit erkläre ich, dass ich diese Arbeit selbständig verfasst habe, dass ich die verwendeten Quellen und Hilfsmittel vollständig angegeben habe und dass ich die Stellen der Arbeit – einschließlich Tabellen, Karten und Abbildungen –, die anderen Werken oder dem Internet im Wortlaut oder dem Sinn nach entnommen sind, auf jeden Fall unter Angabe der Quelle als Entlehnung kenntlich gemacht habe.

Wien, 1. Mai 2024

Hugo Hernández Alemán

Danksagung

In dieser kurzen, aber intensiven Phase möchte ich allen Menschen danken, die mich direkt und indirekt auf dieser Reise begleitet haben:

Zuerst möchte ich meiner Betreuerin, Professorin Renata Raidou, meine immense Dankbarkeit für all die Unterstützung, Anleitung, Mentorschaft, Flexibilität, und für ihre Ratschläge und ihre Professionalität ausdrücken, die sie mir während der Entwicklung meiner Abschlussarbeit zur Verfügung gestellt hat. Vor allem möchte ich ihre Geduld mit einem Studenten hervorheben, dessen Hauptinteressensgebiet manchmal weit entfernt von dem war, was nötig ist, um die Arbeit zu bewältigen - immer mit einem Lächeln und mit dem in mich gesetzten Vertrauen über die geleistete Arbeit. Ich bin ihr zutiefst dankbar für die Zeit und Mühe, die sie in mich investiert hat, um mir zu helfen, die Komplexitäten der Arbeit zu lösen und alle detaillierten und akribischen Korrekturen vorzunehmen, die dazu beigetragen haben, dass es letztlich eine qualitativ hochwertige Arbeit geworden ist. Mein einziger Kritikpunkt ist, nicht länger unter ihrer Anleitung lernen zu können, viel mehr über ihr Arbeitsfeld, ihre Professionalität und ihr Wissen zu erfahren.

Zweitens danke ich meiner Mutter, meiner Patin, meinem Onkel, meiner Tante, Santi, Jochen, Jesús und Borja, kurz gesagt, meiner Familie. Ihre Unterstützung aus der Ferne war in dieser Erfahrung substantiell, immer froh über meine Erfolge, besorgt über meine Probleme und immer glücklich über Berichte meiner zahlreichen Erlebnisse in dieser Stadt. Auch danke ich dafür, dass sie den Ort, den ich Zuhause nenne, genau so bewahrt haben, wie ich ihn verlassen habe, damit dieses Gefühl nie verblasst.

Zuletzt danke ich all den Menschen, die eine entscheidende Rolle in dieser Erfahrung gespielt haben, indem sie sie geformt, aber vor allem ihr Farbe verliehen haben. Es gibt unzählige Menschen, die ich während dieser Phase kennengelernt habe, aber ich muss besonders denen danken, die ich aus verschiedenen Gründen als meine erweiterte Familie und meine Säulen betrachtet habe: Miguel, für das Aufkommen von Energie und Glück, das ich manchmal brauchte; Rosana, für das Erleuchten mit ihrem Lachen und Lächeln in dunklen Momenten; Josep, für das Zeigen, dass wir alle Versionen von uns selbst in anderen Regionen haben; Debora, für das Lehren, dass Sprachbarrieren für tiefe Gespräche nicht existieren; Paola, für das Zeigen, dass ein sehr einfacher Sinn für Humor eine sehr tiefe Seele verbergen kann; David, für das ständige Auge auf mich und die Bereitschaft zu helfen; Sara, für das Begleiten in meinem Mangel an Vernunft und das

Tolerieren von mir, so wie ich bin; Álvaro, für das Teilen von viel mehr als nur Humor. Abschließend danke ich ihnen allen dafür, mir gezeigt zu haben, dass es möglich ist, mit Menschen jenseits von Kultur, Sprache und Zeit eine Verbindung aufzubauen.

Diese Arbeit repräsentiert nicht nur meine Anstrengungen, sondern auch die kollektive Unterstützung und Ermutigung derer, die eine entscheidende Rolle in meinem persönlichen Wachstum gespielt haben.

Acknowledgements

In this short but intense stage, I want to thank all the people who have accompanied me directly and indirectly through this journey:

Firstly, I would like to express my immense gratitude to my supervisor, Professor Renata Raidou, for all the support, guidance, mentorship, flexibility, advice, and professionalism she has provided me during the development of my Thesis. Above all, I would like to highlight her patience with a student whose main background is far from what is necessary to tackle the thesis, always with a smile and trust placed in me about the work done. I am immensely grateful for the time and effort she has invested in me to help me resolve the complexities of the Thesis and to make all the detailed and meticulous corrections that have contributed to it ultimately being a quality work. My only complaint is not being able to stay longer under her tutelage learning much and much more about her field of work, her professionalism, and knowledge.

Secondly, to my mother, my godmother, my uncle, my aunt, Santi, Jochen, Jesús, and Borja, in short, to my family. Their support from a distance has been substantial in this experience, always rejoicing in my achievements, worrying about my problems, and enjoying my multiple experiences in this city. Also, for keep the place I call home just as the day I left it so that the sense of belonging never fades.

Lastly, to all the people who have played a crucial role in this experience, shaping it but above all giving it color. There are countless people I have met during this stage, but I must particularly thank those who I have considered my family and my pillars for different reasons: Miguel for being the burst of energy and happiness I sometimes needed; Rosana for enlightening me with her laughter and smile in dark moments; Josep for showing me that we all have versions of ourselves in other regions; Debora for teaching me that language barriers are non-existent for having deep conversations; Paola for showing me that having a very simple sense of humor can hide a very deep soul; David for always keeping an eye on me and being willing to help; Sara for accompanying me in my lack of sanity and tolerating me as I am; Álvaro for sharing much more than humor with me. In conclusion, I thank all of them for showing me that it is possible to connect with people beyond culture, language, and time.

This thesis represents not only my efforts but also the collective support and encouragement of those who have played a crucial role in my personal growth.

Kurzfassung

Radiomics bezieht sich auf das Studienfeld, das die Extraktion und Analyse einer großen Anzahl quantitativer Merkmale aus medizinischen Bildgebungsdaten umfasst und ein umfassenderes und quantitatives Verständnis von Krankheiten ermöglicht. Radiomics ist besonders nützlich in der Onkologie für die Krebsdiagnose, Behandlungsplanung und Überwachung. Andererseits ist Visual Analytics (VA) ein interdisziplinäres Feld, das interaktive Visualisierungen mit analytischen Techniken kombiniert, um komplexe Datensätze zu erkunden und zu verstehen. Die Hauptziele sind, die Exploration von Daten zu erleichtern, den Benutzern zu helfen, Einsichten zu gewinnen (Muster, Ausreißer, bedeutungsvolle Informationen usw.) und informierte Entscheidungen zu treffen. Durch die Zusammenführung der Vorteile von Radiomics und VA können Forscher und Kliniker wichtige Informationen erhalten, die das Verständnis und das Management von Krankheiten verbessern. In dieser Arbeit präsentieren wir die Entwicklung eines VA-Frameworks für die Analyse von Radiomics-Daten, das Forschern und Klinikern eine intuitive Plattform bietet, um bedeutungsvolle Einsichten aus den medizinischen Bildern von Patienten mit Leberkrebs zu gewinnen. Diese Kohorte umfasst insgesamt 19 Personen, die von der Universität San Diego stammen, einige davon haben ein hepatozelluläres Karzinom (HCC) oder Metastasen (MET). Darüber hinaus haben diese Patienten eine Strahlentherapie durchlaufen, und ihre Daten beinhalten auch Informationen darüber, ob sie irgendwelche Nebenwirkungen erlebt haben (Toxizität oder Nicht-Toxizität). Bildgebungsstudien im Kontext der Strahlentherapie konzentrieren sich häufig auf verschiedene Aspekte des Behandlungsprozesses, sodass wir in unserem Datensatz den Atemzyklus, die durchschnittliche Behandlung und die Dosisverteilung studieren können. Unser Framework nutzt all diese radiomischen Merkmale, um Tumoren zu charakterisieren, Muster oder Korrelationen zu identifizieren und die Kohortenstratifizierung vorzunehmen. Als Versuch, neue Informationen zu enthüllen, wird eine Couinaud-Segmentierung der Leber durchgeführt, die in der Radiologie für die Planung nützlich ist und bei der Lokalisierung spezifischer radiomischer Merkmale der Lebersegmente hilft. Wir bewerten die Stratifizierung der Patienten anhand bestimmter Metriken wie dem Silhouette Score, der Trägheit oder dem Davies-Bouldin-Index. Zusätzlich versuchen wir durch die Nutzung von radiomischen Daten und Dosierungsinformationen, Unterschiede zwischen den Patientengruppen (hepatozelluläres Karzinom/metastatisch oder Toxizität/Nicht-Toxizität) durch statistische Tests oder grafische Darstellungen zu finden, und auch die Toxizität vorherzusagen. Schließlich, mit genauer Kenntnis der Tumorposition dank der Couinaud-Segmente und

der Dosismessungen, identifizieren wir Bereiche der Leber, die möglicherweise über- oder unterbestrahlt sind.

Abstract

Radiomics refers to the field of study that involves the extraction and analysis of a large number of quantitative features from medical imaging data, allowing for a more comprehensive and quantitative understanding of diseases. Radiomics is particularly useful in oncology for cancer diagnosis, treatment planning, and monitoring. On the other hand, Visual Analytics (VA) is an interdisciplinary field that combines interactive visualizations with analytical techniques to explore and understand complex datasets. The main goals are to facilitate the exploration of data, to enable users to gain insights (patterns, outliers, meaningful information...etc), and to make informed decisions. By bringing together the benefits of both radiomics and VA, researchers and clinicians can get important information improving the understanding and management of diseases. In this work, we present the development of a VA framework for radiomic data analysis, providing researchers and clinicians with an intuitive platform to extract meaningful insights from the medical images of liver cancer patients. This cohort comprises a total of 19 individuals sourced from the University of San Diego, some of whom have Hepatocellular Carcinoma (HCC) or Metastasis (MET). Furthermore, these patients have undergone radiotherapy, and their data also includes information on whether they have experienced any adverse effects (toxicity or non-toxicity). Imaging studies in the context of radiotherapy often focus on various aspects related to the treatment process, so in our dataset, we are able to study the breathing cycle, treatment average, and dose distribution. Our framework uses all these radiomic features to characterize tumors, identify patterns or correlations, and make cohort stratification. As a trial to unveil new information, Couinaud segmentation of the liver is performed, useful in radiology for planning and helping in localizing specific radiomic features of liver segments. We evaluate the stratification of patients using certain metrics such as the Silhouette Score, Inertia, or the Davies-Bouldin index. Additionally, by utilizing radiomic data and dosage information, we try to find differences between the patient groups we have (hepatocarcinoma/metastatic or toxicity/non-toxicity) through statistical tests or graphical representations, and also the prediction of toxicity. Finally, knowing the precise position of the tumor thanks to Couinaud's segments and dosage measurements, we identify areas of the liver that may be over or under-irradiated.

Contents

Kurzfassung **xi**

Abstract **xiii**

Contents **xv**

1 Introduction **1**
1.1 Background and Problem Definition 1
1.2 Aim of the Work 3
1.3 Research Questions and Tasks 3
1.4 Methodology and Evaluation Approach 4
1.5 Contribution of the Thesis 5
1.6 Structure 5

2 Clinical Background **7**
2.1 Liver Cancer 7
2.2 Liver Anatomy and Couinaud Segmentation 9
2.3 Radiotherapy 11
2.4 Radiomics and Clinical Data 17

3 State of the Art **23**
3.1 Radiotherapy 23
3.2 Radiomics 24
3.3 Visual Analytics (VA) 29

4 Methodology **35**
4.1 Features Extraction 35
4.2 Preprocessing 41
4.3 Cohort Stratification 46
4.4 Comparative Analysis 52
4.5 Predictive Analysis 53
4.6 Implementation 55
4.7 Design of the LiverVis Dashboard 60

5 Results **71**
5.1 Tasks and Research Questions 71

6 Conclusion and Future Work **109**
6.1 Limitations and Future Work 109

List of Figures **111**

List of Tables **115**

Acronyms **117**

Bibliography **119**

CHAPTER 1

Introduction

In this introductory chapter, we provide a comprehensive overview of the content and key concepts that form the motivation of this thesis. We will begin establishing the context and articulating the problem at hand, its possible solutions, and the questions that will help guide our research.

1.1 Background and Problem Definition

Liver cancer is a significant global health concern, with approximately 700.000 deaths recorded each year, making it the third leading cause of cancer-related deaths worldwide [MVM+21]. The liver is a common site for the spread of cancer from other parts of the body, known as metastasic disease [RDRS+19]. Liver metastases often occur close to radiosensitive organs at risk (OARs) which limits the possibility of applying sufficiently high doses needed for optimal local control [HZTM+22].

Between the most commonly used and well-established treatments in radiotherapy, both primary liver cancer and metastatic liver lesions can be effectively treated with a non-invasive alternative called Stereotactic Body Radiation Therapy (SBRT) [MG10]. This treatment consists of delivering a high and concentrated dose of radiation over a limited number of sessions employing sophisticated and precise radiation equipment. Striking a balance between effectively targeting the tumor and avoiding complications in nearby healthy tissues is essential, so determining the right radiation dose is crucial [MVM+21].

To add difficulties, tumors exhibit significant variability both between individuals and within themselves, consisting of heterogeneous regions with distinct tissue characteristics, which should be treated with different radiation doses [RBV17]. Medical imaging modalities have progressed and are currently able to provide us with valuable tissue information. However, the generated data are extensive and require adequate solutions

for their analysis, for instance from machine learning (ML) approaches for clustering or predictions, and/or visual analytics (VA) methodologies for data visualization.

In our study, the main source of information includes radiomic features [LRVL+12] obtained from medical imaging data sets available for us. Starting with a dataset of 19 patients who have undergone radiotherapy, each patient's data includes Computed Tomography (CT) scans for the doses during the breathing cycle (one series for each section of the cycle) and CT scans for the mean radiation across this cycle. Additionally, CT scans to observe the radiation dose in isolation and structural DICOM files for the analysis of the structures of interest are also included. Considering mean dose and variations in dose during the breathing cycle is crucial in radiotherapy to optimize treatment plans, reduce side effects, and enhance the overall efficacy of targeting tumors. For this thesis, we will focus on the study of the Internal Target Volume (ITV), Planning Target Volume (PTV), Liver-ITV, and Liver Structures for simplicity. Comparisons based on the Couinaud segmentation [OFC08] method will also be conducted to discern potential insights regarding radiomic features or the dose delivered.

The current state of the art in radiomics research indicates a growing interest in the field of radiomic features, surpassing that of shape features over the last decade. In 2020 alone, over 1500 publications were referencing radiomic features, as highlighted in Shur et al. [SDK+21]. Previous studies such as Huyhn et al. or Shen et al. [HCN+16, SHD+22] have demonstrated the potential prognostic utility of radiomic features in Stereotactic Body Radiation Therapy (SBRT), showing predictive capabilities for outcomes that conventional imaging metrics cannot achieve. Furthermore, research by Shen et al. [SHD+22] has explored the prediction of toxicity occurrence using a combination of radiomic and clinical features, employing machine learning and statistical analysis. Their findings suggest that incorporating radiomic features enhances predictive accuracy compared to using clinical features alone. Various statistical methods are examined, including heatmap correlations as noted by Kumar et al. [KGB+12], dimensionality reduction algorithms and feature selection strategies as proposed by Mayerhoefer et al. [MML+20], as well as classifiers deep learning models discussed by Scapicchio et al. [SGB+21].

Visual analytics is described as the science of analytical reasoning facilitated by interactive visual interfaces [Fis05]. VA includes different kinds of goals including synthesizing information from massive, dynamic, ambiguous, and conflicting data, detecting expected and unexpected patterns, providing understandable assessments, and effectively communicating these assessments for action [KAF+08a]. VA solutions can play a significant role in addressing the complex issue of liver cancer treatment, including the use of SBRT, by handling vast and complex datasets, integrating diverse data sources, designing user-friendly interfaces, and deriving insights and knowledge from the underlying data.

Machine Learning, defined as a subset of Artificial Intelligence (AI), extracts patterns from data across various fields of knowledge, with a notable increase in the field of medicine [RYY+21]. Particularly in Radiotherapy, the abundance of annotated imaging data has enabled ML applications in tasks such as organ segmentation, decision-making, and response assessment [LKT23]. Additionally, the application of ML techniques to

Stereotactic Body Radiotherapy (SBRT) based on radiomic data enables the prediction of specific outcomes, such as mortality or toxicity [ZKW+23], or the stratification of cohorts [CJB+21].

1.2 Aim of the Work

The primary objective of this thesis is to investigate and analyze the radiomic features extracted from CT scans of patients undergoing radiotherapy for liver cancer. Specifically, we aim to develop a visual analytics tool for our cancer specialists at the University of San Diego, enabling them to extract valuable insights from their radiomic and clinical datasets. We aim to gain a comprehensive understanding of the distribution of radiation doses within the liver, identify potential areas of under- or over-radiation, and evaluate the correlation between dose distribution and radiomic features. Furthermore, we seek to explore population variances in dose-response and investigate the utility of Couinaud segmentation for stratification and prediction of potential toxicity side effects.

This tool will allow for the exploration of various radiomic features categorized by factors such as tumor type (hepatocarcinoma vs metastasis), gender, side effects, or structures of interest. Using multiple representation types, users can comprehensively understand the data through stratification or clustering of patients based on their anatomical structures across different phases of the breathing cycle, along with examining correlations between features. Additionally, users will be able to compare average and breathing cycle radiomic data, enabling them to formulate and test their hypotheses. Furthermore, the tool will provide access to dose data per patient and structure, supplemented with DICOM images for enhanced visualization.

1.3 Research Questions and Tasks

The main scientific contribution of this thesis is the development of a comparative visual analytics tool that enables cohort stratification and analysis based on radiomic features extracted from CT scans of liver cancer patients. According to the previously mentioned VA goals for addressing the complex issue of liver cancer treatment, this tool should be designed to fulfill specific requirements, such as presenting the data in a visually understandable way and helping researchers/clinicians make informed decisions. Moreover, clinical physicists at the University of San Diego want to discern differences in the population, regarding treatments and their outcomes, and if Couinaud segmentation would be useful to obtain potential insights about stratification or toxicity. As a result, we formulate our primary research questions and corresponding tasks as follows.

Research questions:

- **(R1):** Which regions of the liver are more subjected to under- or over-radiation?

- **(R2):** Which disparities can be observed in the population, concerning treatment and their outcomes?

Tasks:

- **(T0) Features Extraction:** We extract radiomic features and clinical data extraction from the patient's CT scans and structures of the dataset.
- **(T1) Preprocessing:** To ensure data quality and the efficiency of our analysis, we perform proper data cleaning, including scaling the data, reducing dimensionality, and identifying outliers.
- **(T2) Cohort Stratification:** We perform disease subtyping and stratification, contributing to personalized medicine and targeted treatment strategies, as different subtypes of liver diseases are anticipated to require distinct approaches.
- **(T3) Comparative Analysis:** We analyze and compare mean dose and breathing cycle dose. Additionally, we evaluate the potential of Couinaud segmentation, providing higher specificity in our stratification.
- **(T4) Radiation Analysis:** This phase involves investigating potential regions of under- or over-radiation within the liver. Furthermore, we utilize dose and radiomic features to predict potential toxicity outcomes.
- **(T5) Prediction of Toxicity:** This phase consists of using ML algorithms to classify patients according to their dosage or radiomic features and predict which patients might face toxicity.

1.4 Methodology and Evaluation Approach

To address the research questions and solve correctly the tasks, the methodology of the work is organized as follows:

1. **Literature Review:** We review the current state-of-the-art approaches in SBRT, visual analytics techniques, radiomics, and liver segmentation. This guides us to explore and work with the data accordingly.
2. **Extracting and Preprocessing Data:** We prepare the data in a way that is comprehensible for exploration and stratification. Data quality ensures good data processing, and reliable outputs for exploration, visualization, or predictions.
3. **Data Analysis:** We examine the radiomic and clinical features using machine learning solutions and statistical analysis to extract meaningful insights from these features. Additionally, we stratify the cohort and compare its subcohorts based on:

- Type of organ structure (Liver, Liver-ITV, ITV, or PTV).
- Couinaud classification.
- Study of the average, breathing cycle and doses.
- Clinical data (male/female, hepatocarcinoma/metastasis, toxicity/non-toxicity).

We obtain conclusions from the results of these evaluations, which are used to answer the research questions and the prediction of toxicity according to the doses received or radiomic features.

4. **Visual Analytics Tool:** We design and create an interactive visual analytics tool that enables healthcare professionals to explore, compare, test hypotheses, and derive conclusions from the collected data above.

5. **Evaluation:** We will assess and compare various strategies employed throughout the thesis, including different scaling methods, dimensionality reduction techniques, and clustering algorithms.

   These evaluations will be based on metrics such as explained variance (PCA), Silhouette Coefficient, Inertia, or Davies-Bouldin index, which measure the effectiveness of each approach in separating our different classes in clustering methods. All these methods and metrics are currently employed and cited in their respective sections. Within the tool, users will have the flexibility to select and compare different algorithms and scaling methods to make informed decisions.

1.5 Contribution of the Thesis

The main contribution of this thesis is the design and development of a tool that will allow domain experts to explore and analyze radiomic data from CT scans, helping them to draw conclusions and propose new hypotheses. The tool will have enough interactivity to explore all the possibilities of preprocessing methods, algorithms, structures, and group classification according to their clinical data. In addition to this contribution, the provision of multimodality is crucial for the understanding and stratification of patients, since we have not only the study of the average in the treatment but also the complete respiratory cycle, as well as all the Couinaud segments, resulting in an approach that is less common in the current state of the art.

1.6 Structure

Once we have a general idea of what the thesis entails, the goals we aim to achieve, and how to reach them, we continue with Chapter 2, which details all the clinical background necessary for understanding the study. This chapter starts by explaining how the liver works, from the basics to liver cancer, including its anatomy and the Couinaud segmentation procedure, highlighting its relevance to this work. In addition, the treatment of Radiotherapy is explained, emphasizing SBRT and essential elements

such as Dose Plans or DICOM-RT files. Finally, the concept of radiomics, its usefulness in the field of oncology, and the procedure for its use in our topic are introduced.

Subsequently, Chapter 3 consists of the state of the art in Radiotherapy, radiomics, and Visual Analytics, referring to recent and current works in general and focused on liver tumors. The methodology in Chapter 4 details the entire process carried out to obtain the radiomic data, its processing, its analysis, and the resolution of all the tasks or research questions explained in Chapter 1. In addition, this chapter includes the implementation as a separated section, including the libraries and computer support used for the development of the thesis, and also reviews the tool and its different parts.

Finally, Chapter 5 consist of the results, obtained in response to the questions posed at the beginning of it.

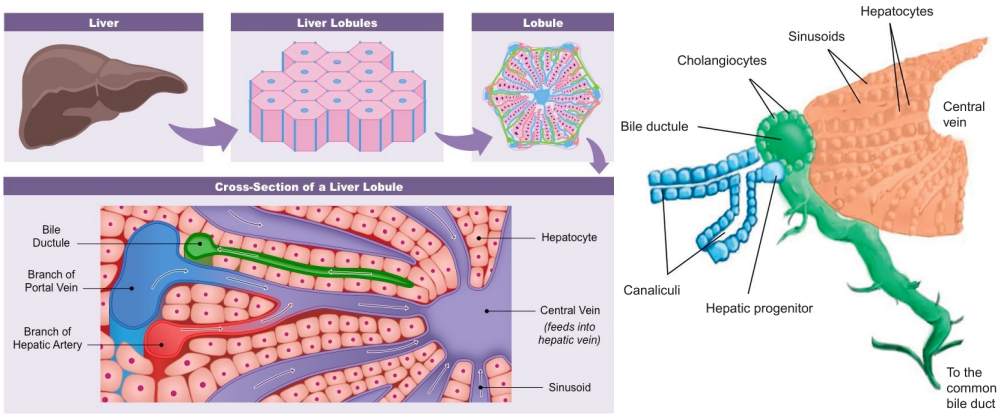
CHAPTER 2

Clinical Background

2.1 Liver Cancer

The liver, the body's largest solid organ, receives blood from two sources: 80% from the portal vein rich in bacterial products, environment toxins, and food antigens, while the remaining 20% is from vascularization by the hepatic artery [GJT08]. The liver plays a crucial role in our body, carrying out more than 500 vital functions. These include filtering waste and foreign substances from the blood, regulating blood sugar levels, and producing essential nutrients [col]. Furthermore, the liver produces and releases bile to help with food digestion. All these different tasks are carried out by **hepatocytes**, which constitute 80% of the liver's total cell population [BBW77]. Hepatocytes possess a highly advanced and intricate system for the transport of particles inside the cell membrane and their correct cellular location [SSC+19]. To better understand the biology of the hepatic cells, Figures 2.1 and 2.2 provide the structure of the Liver, from the organ as a whole down to the cellular level. Hepatocytes are arranged into plates separated by expansive capillary spaces known as sinusoids. These sinusoids are characterized by discontinuous, fenestrated endothelial cells lacking a basement membrane. Additionally, cholangiocytes, forming the epithelial lining of the bile ducts, constitute another vital component [Rud14].

One of the liver's most remarkable capabilities is its ability to regenerate and regulate its own size [Apt15]. Liver regeneration can be triggered by surgical resection or injury-induced cell death resulting from chemical exposure or viral infection. This regenerative process encompasses the replication of all liver cell types, as well as the reconstruction of blood vessels, biliary structures, and extracellular matrix components [Apt15]. Notably, hepatocytes, cholangiocytes, and endothelial cells all undergo proliferation to restore liver mass [Apt15, Rud14]. Due to this phenomenon, numerous medical professionals, scientists, biologists, and bioengineers have devoted their efforts to understanding liver



(a) Liver structure (b) Hepatocytes

Figure 2.1: Structural organisation of the liver [bio] and the hepatocyte [HJR+14]

regeneration. Their goal is to harness this regenerative capacity to develop innovative therapies for liver failure [ARZN+19].

Overall, our body is made of trillions of cells that usually grow, multiply, and replace old or damaged cells. Sometimes this process goes wrong and abnormal or damaged cells form lumps called tumors, which can be cancerous or not cancerous [cand]. According to the American Society of Cancer [cana], more than 800,000 people are diagnosed with this cancer each year worldwide.

When a cancerous growth begins in the liver, this is called primary liver cancer [canb]. The most common type of liver cancer is hepatocellular carcinoma (HCC), which represents 75–85% of liver cancer cases. Other types of liver cancer include cholangiocarcinoma (bile duct cancer) and angiosarcoma [Mas23]. Liver cancer is a leading cause of cancer-related deaths globally, accounting for over 700,000 deaths annually. The incidence of HCC is higher in mean ages from its diagnosis between 55–59 years in China and 63–65 years in North America and Europe [SFS+21], correlating with hepatitis B and C prevalence in the case of South-East Asia and Africa [LCC15]. In countries like the United States and Central Europe, the incidence is rising due to factors like obesity and alcohol-related cirrhosis [LCC15].

Hepatocellular carcinoma (HCC) is influenced by a combination of environmental and genetic factors. Chronic liver diseases like hepatitis B and C, liver cirrhosis, and excessive alcohol consumption significantly increase the risk [TCNBC+17]. Worldwide, the main causes of liver cancer include hepatitis B or C virus infections, chronic alcohol consumption, non-alcoholic fatty liver disease (NAFLD), smoking, obesity, diabetes, and iron overload. These factors play a significant role in HCC development and are crucial considerations for understanding the disease and developing effective treatment strategies [BMC+21]. Also, biomarkers are essential for diagnosing and treating hepatocellular carcinoma (HCC), such as microRNAs (miRNAs) and proteins like GP73, GPC3, and

OPN [TCNBC+17]. Challenges include standardization, but ongoing research aims to discover new biomarkers [Mas23]. Key biomarkers like Alpha-Fetoprotein (AFP), Des-Gamma-Carboxy Prothrombin (DCP), liver function tests, genetic and epigenetic markers, and Circulating Tumor Cells (CTCs) aid in early detection and personalized treatment of HCC [Mas23, TCNBC+17].

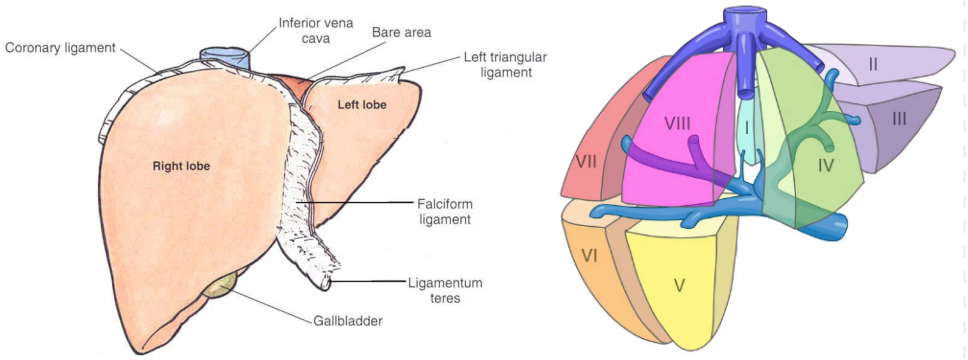
Treatment options for hepatocellular carcinoma (HCC) are diverse, ranging from curative to palliative and immunotherapy approaches, depending on disease stage and patient health. Early-stage HCC may benefit from surgical interventions like resection or transplantation, while intermediate-stage HCC can be targeted with locoregional therapies such as radiofrequency ablation or transarterial chemoembolization [LCC15, Mas23]. Emerging therapies, including targeted and immunotherapies, are under investigation [BMC+21]. A multidisciplinary approach involving various healthcare professionals is crucial for tailoring treatment plans and improving outcomes for hepatocellular carcinoma (HCC) patients. While regular monitoring is vital to detect recurrence early, late-stage HCC may necessitate palliative care and systemic therapy to enhance the quality of life [TCNBC+17]. Consulting healthcare providers is essential to determine the most suitable treatment approach based on individual patient factors and disease characteristics. However, geographic disparities and variations in management recommendations make standardization difficult [Mas23]. To improve outcomes for liver cancer patients, we need continuous research, treatment advancement, and personalized approaches. This is where **radiomics** can help, comparing the existing differences between patients' medical images, and drawing conclusions from them.

2.2 Liver Anatomy and Couinaud Segmentation

The liver, located in the upper abdomen, primarily on the right side, is the largest glandular organ in the body protected by the rib cage. It is anatomically related to structures like the abdominal esophagus, stomach, and upper duodenum, and its position and relationships are essential for understanding its function and for surgical interventions [Baz23]. Figure 2.2 (a) depicts its two main surfaces: the diaphragmatic surface, which rests against the right hemidiaphragm, and the visceral surface, covered by visceral peritoneum except at the gallbladder fossa [Baz23]. However, the liver is typically described through *morphologic* and *functional* anatomy [AMB10, Bis14].

Anatomically, the liver is divided into four lobes: the larger right and left lobes, and the smaller caudate and quadrate lobes. The falciform ligament separates the left and right lobes and attaches the liver to the abdominal wall as we can see in Figure 2.2 (a) [col]. The porta hepatis, analogous to the liver's hilum, receives essential vessels and ducts, crucial for vascular and biliary functions. Understanding these divisions is vital for medical professionals, particularly in liver surgeries and interventions [Baz23].

The liver segmentation system introduced by **Couinaud** is now the most commonly adopted classification describing the liver's functional anatomy [OFC08, GFC+14]. It is based on identifying the three hepatic veins and the plane passing through the portal



(a) (b)

Figure 2.2: Comparison between the liver's anatomy (a) [Liv] and Couinaud segmentation (b) [ima]

vein bifurcation. Dividing the liver into eight functional segments can take into account the portal and hepatic veins and bile ducts representing each a functional unit of the liver with its vascular supply and biliary drainage. Currently, Couinaud's classification is widely preferred, especially in surgical contexts, as it offers better accuracy in localizing and monitoring intra-parenchymal lesions [GFC+14, MMT+14].

As seen in Figure 2.2 (b), the liver can be divided into 8 segments:

- Segment I: Posterior segment, Caudate lobe.
- Segment II: Left posterior lateral segment.
- Segment III: Left anterior lateral segment.
- Segment IV: Left medial segment. (Usually is divided into IVa and IVb).
- Segment V: Anterior medial segment.
- Segment VI: Anterior lateral segment.
- Segment VII: Posterior lateral segment.
- Segment VIII: Posterior medial segment.

Understanding liver anatomy is crucial for surgical precision, imaging, disease management, and minimizing risks during interventions [AMB10, Bis14]. Functional assessments help evaluate these risks and allow for personalized treatment plans to improve results by meeting each patient's unique needs [Bis14]. Integrating both anatomical and functional aspects significantly improves patient care, surgical planning, and long-term outcomes in liver surgery, particularly in Hepatocellular Carcinoma (HCC) treatment. This approach

enables precise targeting of the tumor's location within the liver segments. Notably, segment 8 is identified as the most common site for HCC tumors, while segment 1 is the least common [RBP+22]. This strategic focus on segment-specific tumor location is crucial for optimizing surgical interventions and enhancing patient prognosis [Bis14].

2.3 Radiotherapy

Patients with hepatocellular carcinoma (HCC) are, ineligible for therapies such as surgery, transplantation, or radiofrequency ablation (RFA). Their treatment alternatives consist of transarterial chemoembolization (TACE), and **radiotherapy (RT)** [KWZH16a, KZK+15]. TACE has given favorable results on local control and survival, delivering highly concentrated doses of chemotherapy to tumor cells while minimizing toxicity, and radiotherapy has also achieved efficacious control of the tumor in advanced disease [KZK+15]. Radiation therapy kills cancer cells or slows their growth by damaging their DNA. When the damaged cells die, they are broken down and removed by the body [canc]. Radiotherapy was limited historically by its inability to deliver a dose capable of effectively treating tumors, but modern RT techniques have renewed their interest in the use of liver-directed RT to treat patients with primary hepatic malignancies [KWZH16b].

One fundamental equation that explains the interaction between radiation and tumor matter is the Linear-Quadratic Model, used to describe the biological effect of radiation on cells [MNR07]:

$$S = e^{-(\alpha D + \beta D^2)} \tag{2.1}$$

Where S represents the survival fraction of cells after radiation, D is the dose of radiation administered, α and β are parameters that characterize the radiosensitivity of the cells, with α associated with linear DNA breaks and β with quadratic DNA breaks.

The unit used to measure the total amount of radiation the patient is exposed to is the Gray (Gy) [MNR07], equivalent to the absorption of one joule of radiation energy per kilogram of matter. These doses are given over a period of time and divided into multiple smaller doses, usually 2 Gy but the specific dose for each patient depends on the location and severity of the tumor [YS]. The hepatic toxicity [RLC+10] has been extensively investigated, and the tolerance of the **whole liver** is in the order of approximately **30 Gy** [GCDGO04, RLC+10], which seriously restricts its clinical application. In many cases, the tumor receives a higher dose compared to the whole liver [San14] as shown in Figure 2.3, where the isodose lines (white solid lines) from inner to outer represent 40, 30, 20, and 10 Gy, respectively and the center of the tumor receives as high as 125% of the prescribed dose.

Table 2.1 provides a comprehensive comparison of various radiotherapy techniques, focusing on their efficacy, potential side effects, and specific indications. Among these, **Stereotactic Body Radiotherapy** [GAA+13] stands out due to its significant relevance

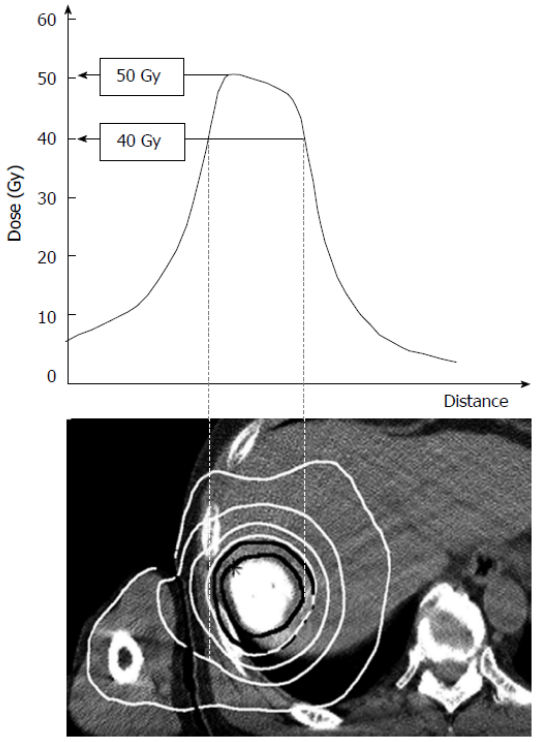


Figure 2.3: Dose distribution of SBRT for Hepatocellular Carcinoma [San14].

and effectiveness in current clinical practice. SBRT has emerged as a critical advancement in improving local control, and potential cure with high rates ranging from 70%–90% at 1–2 years for liver lesions not amenable to traditional treatments like surgery or ablation [San14]. SBRT will be discussed more deeply in Section 2.3.4.

2.3.1 Treatment Planning

Treatment planning involves determining the optimal method for delivering radiation therapy to a patient with cancer while minimizing damage to surrounding healthy tissues [MNRP08, CHM19]. It involves determining the optimal radiation dose and target volume [DPP+19], treatment technique and treatment schedule to effectively target the tumor while sparing critical organs [CHM19]. The treatment planning process in radiotherapy typically involves several key steps to ensure the safe and effective delivery of radiation. These steps may vary slightly depending on the specific treatment technique and technology used [DPP+19, CHM19, MNRP08], but the general outline of a treatment planning process includes:

1. **Patient Evaluation:** This is the assessment of the patient's medical history, physical examination, imaging studies (such as CT scans, MRI, PET scans), and

Technique	**Efficacy**	**Indications**	**Potential Side Effects**
SBRT	High precision, the potential for ablative doses, promising local control and potential cure rates [PR21, San14]	Unresectable liver lesions, lesions not amenable to surgical resection or radiofrequency ablation [San14]	Minimal due to high precision and sparing of normal tissue [PR21]
Individualized image-guided SBRT	Allows for personalized treatment plans, effective in selected patient populations [DEC06]	Patients with specific liver tumor characteristics and needs, based on individualized planning and image guidance [DEC06]	Similar to SBRT, with added emphasis on minimizing liver damage and other toxicities [DEC06]
IMRT	Enables higher doses to target lesions, sparing noninvolved liver [FBJ11]	Focal HCC not suitable for surgery, particularly with underlying liver dysfunction or in a cirrhotic liver [FBJ11]	Reduced risk of classic radiation-induced liver disease and other toxicities [FBJ11]
MRI-guided radiotherapy	Enhances the risk-benefit ratio through direct visualization, allows for adaptive treatment [WRB20]	Metastatic and primary liver cancers, especially in the context of liver mobility and changing anatomy [WRB20]	Potentially lower due to adaptive planning and the ability to customize treatments [WRB20]

Table 2.1: Comparison of Radiotherapy Techniques for Liver Cancer.

tumor characteristics to determine the extent and location of the tumor. It can be complemented with simulations.

2. **Contouring:** This step involves defining the shape and position of the tumor (i.e., the target) and nearby organs at risk (OARs).

3. **Optimal Position:** This step comprises selecting an optimal patient positioning and immobilization technique to ensure treatment reproducibility.

4. **Dose Prescription:** Based on the tumor characteristics, stage, and treatment goals, the radiation oncologist prescribes the appropriate radiation dose that needs to be delivered to the tumor.

5. **Treatment Technique Selection:** The treatment team selects the appropriate radiation therapy technique, such as intensity-modulated radiation therapy (IMRT),

volumetric modulated arc therapy (VMAT), or proton therapy, based on the tumor location and complexity.

6. **Treatment Planning System (TPS) Input:** The contoured structures, dose prescription, and treatment technique parameters are input into a Treatment Planning System (TPS) where the actual treatment plan is generated.

7. **Treatment Delivery:** Once the treatment plan is approved, the patient undergoes radiation therapy sessions according to the prescribed schedule and dose distribution.

8. **Monitoring and Follow-Up:** Throughout the course of treatment, the patient's response to therapy is monitored, and adjustments to the treatment plan may be made if necessary. After treatment completion, patients are typically followed up to assess treatment outcomes and manage any potential side effects.

For the development of this thesis, target definitions and target volumes play a crucial role, since they define the specific area or volume within the patient's body that is intended to receive the prescribed dose of radiation [MNRP08]. Delineating target volumes with precision and incorporating them into the treatment planning process, radiation oncologists can optimize treatment outcomes, enhance patient safety, and improve overall treatment efficacy [DPP+19].

Figure 2.6 shows a schematic overview of these Target Volumes. The *Gross Tumor Volume (GTV)* is defined as a visible tumor in images, palpable and demonstrable extent and location of the malignant growth [SRM+19, DPP+19]. The *Clinical Target Volume (CTV)* is a tissue volume encompassing the GTV but with an additional margin accounting for any microscopic extension of the primary tumor or regional lymph node spread [DPP+19, CHM19]. Internal Target Volume (ITV) accounts for tumor motion due to breathing or other physiological factors. It is used in cases where the tumor moves during treatment, such as in lung or liver tumors [CHM19]. The *Planning Target Volume (PTV)* is usually the final volume defined by the clinician during the planning process, expands the CTV to accommodate for uncertainties in treatment delivery, such as patient setup errors, organ motion, and variations in radiation beam delivery [DPP+19].

2.3.2 Dose Plans and Dose Volume Histograms

While a treatment plan involves considerations relating to the patient-specific factors and the treatment goals, as well as the overall treatment strategy, **dose plans** simply represent an elaborate map that explains how the concerned radiation dose would get distributed inside the body of the patient over the courses of treatment. They should include the beam angles, beam energies, dose distribution, dose constraints, and treatment parameters [DPP+19, CHM19]. A highly complex computer algorithm then develops a plan to figure out the best arrangement of the beams of radiation based on the shape and location of the tumor. This planning is very important to improve the therapeutic ratios and the preservation of the integrity of healthy tissue [Sem20].

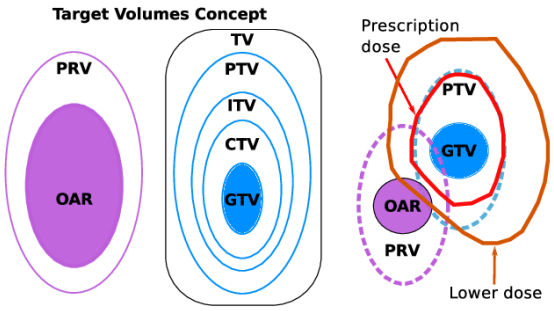


Figure 2.4: Volume concepts used in RT planning [SRM+19].

Dose-volume Histograms (DVHs) are very important in any plan evaluation and comparison of plans in radiation therapy. The graphical representation of the percentage of a target volume (tumor or organ at risk) is plotted against the dose of radiation it receives for the whole of the dose distribution within the irradiated volume ([ICR]). Figure 2.5 shows a comparison between the DVHs of different organs. A visual inspection by the clinicians of DVHs can help delineate the dose uniformity to the target, dose sparing of the critical structures, and overall conformity of the plan. Cumulative DVHs also enable an estimation of the probability of the tumor's control given the volume of tissue receiving a dose or more, and of the risk of incurring side effects, and are useful to arrive at informed decisions on adjustments of treatment by an oncologist [ICR].

DVHs are crucial in clinical decision-making as they provide a means of the calibration of optimized treatment plans. They allow for the assessment of the potential effectiveness of a plan and the risks associated, hence allowing personalized therapy adjustments. For instance, if a DVH indicates excessive radiation to a critical area, the plan may be adjusted to reduce toxicity [BCD+10]. Moreover, DVH metrics further improve the assessment of treatment outcomes, which allows for the development of better treatment protocols for radiation therapy, and consequently enhances patient care and quality of life post-treatment.

2.3.3 DICOM-RT

According to the official DICOM-RT supplement [dic], several information objects applicable to the domain of radiation oncology are defined:

- *RT Image*: specifies the semantic content of radiotherapy images that have been obtained on a conical imaging geometry, or calculated images such as digitally reconstructed radiographs (DRRs).
- *RT Structure Set*: information related to patient anatomy, identified on devices such as CT scanners, virtual simulation workstations, or treatment planning systems. The process of segmentation of the images leads to a set of structures, such as body (external) contour; target volumes (e.g. GTV, CTV, and PTV), organs at risk,

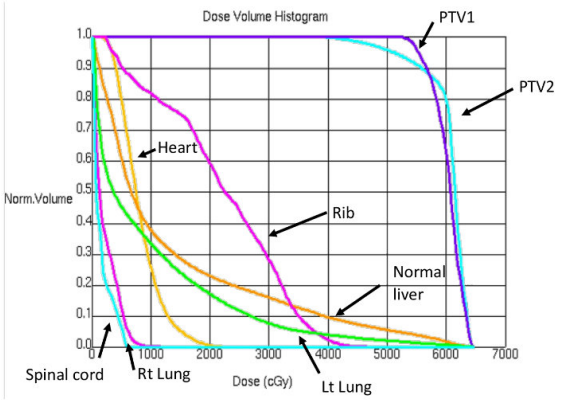


Figure 2.5: Dose-volume histograms comparisons for Liver Metastases [WCH18]

regions, and volumes of interest, points of interest (e.g. dose references). These structures have special meanings that must be maintained across the different IS having the object a link to an image.

- *RT Dose*: contains dose distributions generated by a treatment planning system in one or more formats, such as 2D/3D dose data, isodose curves, dose-volume histograms (DVHs), or dose points.
- *RT Plan*: geometric and dosimetric data specifying a course of external beam and/or brachytherapy treatment, for example, beam angles. The RT Plan can originate from a simulation workstation and then be further refined by a treatment planning system before being transferred to a record and verify system or treatment device. Typically, an instance of the RT Plan object links to an RT Structure Set instance, which establishes a coordinate system and defines a set of patient structures.

2.3.4 Stereotactic Body Radiotherapy

Stereotactic Body Radiation Therapy (SBRT) is an advanced form of external beam radiation therapy designed to precisely target and deliver high doses of radiation to tumors while minimizing damage to surrounding healthy tissues [San14, Mat23], and typically in fewer treatment sessions compared to conventional radiation therapy [MHGS+14, Mat23]. It is also known as Stereotactic Ablative Radiotherapy (SABR) or Stereotactic Radiosurgery (SRS) when used for certain types of tumors [Mat23]. It uses stereotactic, non-coplanar conformal radiation therapy techniques to administer larger radiation doses per fraction, typically between 8–12 Gy [San14].

SBRT is used to treat tumors in various locations, including the liver, lungs, spine, and prostate [MHGS+14]. However, the treatment is particularly effective for treating liver malignancies such as hepatocellular carcinoma (HCC) [Mat23], and is suitable for patients who cannot undergo surgery, liver transplantation, or other traditional treatments. Key

advantages include its non-invasive nature, which provides high precision in tumor targeting and effective local control. This approach can potentially improve tumor response and patient survival outcomes [San14]. Additionally, it eliminates the need for surgery, making it ideal for patients who are not surgical candidates [Mat23].

Discussing side effects, SBRT is known to cause significant toxicity of grade 2 or higher, manifesting as fatigue, dyspnea, and pain [Har18]. The liver, being a vital organ with a limited tolerance for radiation as told in the last section, is particularly susceptible to adverse effects from the high-dose radiation delivered in a few fractions characteristic of SBRT. The most concerning toxicities associated with liver SBRT encompass radiation-induced liver disease (RILD) [MHGS$^+$14, San14], liver dysfunction, and gastrointestinal complications, highlighting the need for careful dose management and patient monitoring during treatment [San14].

The safety profiles for SBRT are generally positive, showing only a rare occurrence of severe side effects [Mat23]. To guarantee SBRT's safety and handle its possible toxic effects properly, careful choice of patients, accurate planning of treatment, and detailed monitoring are essential. This approach helps in minimizing risks and efficiently managing complications [San14].

In SBRT, using the average tumor position and information throughout the Breathing Cycle enables a comprehensive strategy for locating the target, optimizing dosage, adapting margins, monitoring in real-time, and ensuring quality. This significantly improves treatment accuracy and effectiveness for patients receiving radiation therapy [BAC$^+$20]. While the breathing cycle refers to the sequence of inhalation and exhalation that occurs during breathing (with 0 representing end inspiration and 50 representing end exhalation), the average cycle refers to a calculated mean or typical value derived from this sequence [KOW$^+$21]. Also, the Breath Hold technique [KNM$^+$23] exists, where patients are instructed to hold their breath at a specific point in the respiratory cycle while radiation is delivered.

2.4 Radiomics and Clinical Data

Recent advancements in medical imaging have facilitated the conversion of medical images into meaningful and mineable data through a quantification process. Extracted quantitative features, although not easily interpretable by humans, reflect the underlying pathophysiology and can be analyzed using complex algorithms [SGB$^+$21, MML$^+$20]. This systematic process of extraction and analysis of the data is named **radiomics** [MML$^+$20]. The goal is to use these extracted features to better understand and characterize various aspects of medical conditions and potentially improve diagnostic and prognostic capabilities [MML$^+$20, KGB$^+$12].

Radiomics can be employed to capture tissue and lesion properties, including shape and heterogeneity, as well as their changes over time, particularly during treatment or surveillance [HCN$^+$16]. In oncology, assessing tissue heterogeneity is crucial, as genomic

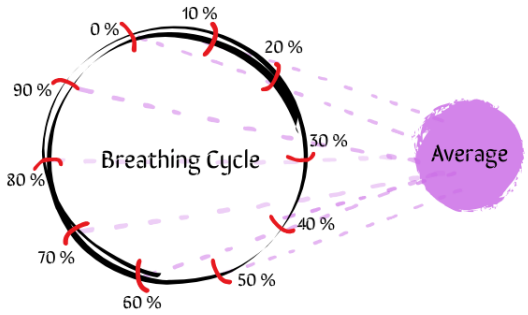


Figure 2.6: Breathing Cycle and Average, representative diagram.

analyses have revealed its significance as a prognostic factor for survival and a challenge in cancer management. Radiomic features are strongly correlated with heterogeneity indices at the cellular level. Unlike biopsies, which only capture heterogeneity within a small portion of a tumor and at a single anatomic site, radiomics encompasses heterogeneity across the entire tumor volume [KGB+12]. Consequently, radiomic features are also indicative of tumor aggressiveness. Radiomics presents a non-invasive strategy for precision medicine, extracting numerous advanced quantitative features from medical images to evaluate the tumor phenotype. [HCN+16, MML+20].

A well-defined workflow with radiomic data as shown in Figure 2.7 is important to assure standardization, quality control, efficiency, interdisciplinary collaboration, and data management [KGB+12, SGB+21]:

1. **Image Acquisition:** biomedical standardized protocols for acquiring medical images, requiring parameter settings based on the imaging modality, intended purpose (diagnosis or treatment planning), and tissue identification.
2. **Image Preprocessing:** pre-process to enhance the quality of images before segmentation, feature extraction, and analysis. The most common techniques are noise reduction, intensity normalization, image registration, or removal of artifacts.
3. **ROI Segmentation:** Definition and segmentation of the region of interest, whether lesion or normal tissue, achievable manually or automatically.
4. **Features Extraction:** Radiomics feature extraction follows, generating numerous features from statistical, filtering, and morphological analyses, forming a high-dimensional feature space.
5. **Features Analysis:** Subsequent analysis identifies informative features based on user-defined criteria. Machine learning is then employed to automatically extract

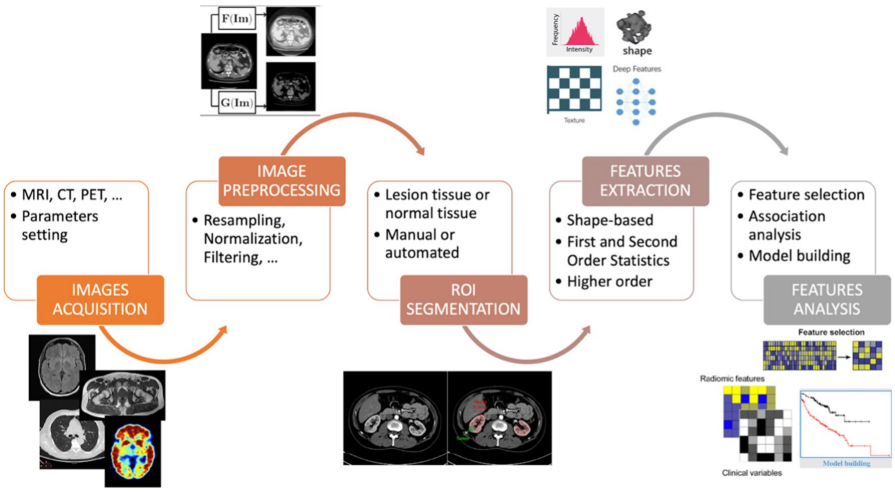


Figure 2.7: Workflow of radiomic data, a process that starts with acquiring images and extends to analyzing the extracted features [SGB+21].

and select relevant features, building predictive models. These models are trained on analyzed features to learn decision functions for tasks like group characterization, tumor classification, disease prognosis, therapy response assessment, etc.

Each of these steps presents specific challenges that must be overcome [KGB+12]. For instance: identifying and harmonizing optimal protocols (standardization) for *image acquisition*, robust *segmentations*, and effective *preprocessing* techniques due to image qualities and variability, since both have an enormous impact on the quality of the feature extraction [MML+20]. Also *features* should capture volume complexity without being redundant, i.e selecting relevant features due to high dimensionality [SDK+21], statistical approaches for *data analysis* must be optimized, being aware of overfitting and complex models that may lack interpretability [SDK+21].

2.4.1 Classification of Radiomics Features

Feature extraction is the next step after a region of interest is segmented. It is the selection of useful information to assist in the characterization of normal and abnormal radiological images [SGB+21, MML+20]. These features describe characteristics of the tumor intensity histogram (e.g., high or low contrast), tumor shape (e.g., round or spiculated), texture patterns (e.g., homogeneous or heterogeneous, irregularity and compactness), as well as descriptors of tumor location and relations with the surrounding tissues (e.g., near the heart) [KGB+12, MML+20]. According to this description, we can classify the radiomic features into three classes: Histogram-based, shape-based, and texture-based features.

Histogram-based features, also known as *First-order features* are statistical descriptors derived from the global gray-level histogram of an image or ROI, calculated based on

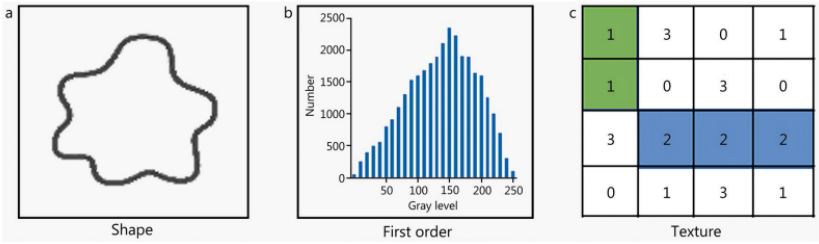


Figure 2.8: Categories of radiomic features: shape, first order, and texture features [ZZC+23].

the frequency distribution of pixel intensities [MML+20, KGB+12]. Examples of first-order features include gray-level mean, maximum, minimum, variance, and percentiles. Other important examples are skewness, which describes the asymmetry of the intensity distribution curve, and kurtosis, which describes the peakedness or flatness of this intensity distribution.

Shape-based features describe geometric properties of the ROI, providing information about the spatial configuration and morphology of the region under analysis [MML+20]. These features can provide insights about growth patterns or invasiveness [SDK+21]. Examples of shape-based features include diameter, area, sphericity, and compactness.

Texture-based features, also known as *Second-order features*, describe spatial patterns and variations in pixel intensities, capture spatial relationships between pairs of pixels or voxels in the ROI [MML+20, SDK+21]; Matrices such as the gray-level co-occurrence matrix (GLCM) or the run-length matrix (GRLM) are often used for underlying tissue characteristics [SDK+21].

Figure 2.8 shows an example of the three categories. By combining these different types of radiomic features, researchers and clinicians can create comprehensive profiles of tumors, enabling better characterization, classification, prediction of tumor behaviour, treatment response, and patient outcomes [MML+20]. For instance, kurtosis and skewness are combined with high-order texture features and shape-based features to capture quantitative descriptors for predictive modelling and outcome assessment [KGB+12].

2.4.2 Clinical Data

The medical record is essential in medical care, teaching, and research, serving as a detailed record of the patient's health status, including history, diagnoses, treatments, and responses to these treatments [edu]. Clinical data obtained from this record, along with biobanks, provides an essential database for diagnosis and treatment, as well as for research and the study of specific cases [DC14].

Rates of liver cancer are typically higher in men compared to women due to several factors, including higher prevalence of risk factors, biological differences, differences in immune response, and genetic and epigenetic factors [BRDC04]. Risk factors include

chronic infection with hepatitis B or C viruses, excessive alcohol consumption, tobacco use, obesity, and exposure to certain chemicals or toxins [BRDC04].

The relationship between obesity and liver cancer appeared to be stronger in men than in women, compared to individuals of normal weight, those who were overweight had a 17% increased risk of liver cancer, while those who were obese had an 89% increased risk [LW07]. A higher mean hepatic dose was associated with an increased risk of RILD in patients undergoing 3D-CRT for hepatocellular carcinoma [CWH$^+$02]. As explained in section 2.1, biomarkers such as microRNAs are essential for diagnosing and treating hepatocellular carcinoma, and Barshack et al. [BMR$^+$10] were able to specify various miRNAs with differing expression patterns in Hepatocarcinoma and Metastatic.

CHAPTER 3

State of the Art

In this chapter, we will discuss the current state of the art of the main branches of this Thesis, radiotherapy (focusing on SBRT), radiomics and their applications, and the concept of Visual Analytics. This will help us to have a basic understanding of what the current techniques, processes, and solutions are so that we can study possibilities in our thesis, and compare the results with those already existing.

3.1 Radiotherapy

One of the therapeutic approaches to treat small hepatic malignancies includes Stereotactic Body Radiation Therapy (SBRT), providing good local control for selected patients [San14, Mat23]. However, it carries the risk of affecting nearby normal tissues and causing different kinds of toxicities, among which are changes in liver enzymes and changes in the gastrointestinal tract when facing hepatocarcinoma [PKD+10].

The requirements for the high confidence of the location of the tumor are that SBRT provides near-real-time imaging of the physiologic motion of the tumor, considering it may vary, and access diagnostic studies of the best quality, offering an accurate treatment delivery [CT07]. Saw et al. [SBL12] highlight the need for such special considerations in avoiding or reducing radiation toxicities in SBRT.

While local control of the tumor has been the basis of the treatment aim with SBRT, the interesting observations of abscopal (or out-of-field) effects pointed toward the exciting possibility that high-dose radiation could activate anti-tumor immunity. Popp et al. [PGND16] summarize currently available evidence of immune modulation by SBRT alone or potentially combined with immunotherapy.

One challenge in SBRT is the variability in image segmentation, which can affect the dose-volume/outcome complication relationships between the liver and other abdominal normal tissues i.e., esophagus, stomach, duodenum, and small intestine. The technique

of breath-hold computed tomography (CT) scanning, oral, and intravenous contrast media has been recommended to assist with accurate and reproducible segmentation of some of the upper abdominal normal tissues, whose shape is irregular and is not well demarcated at cross-sectional images [MVM+21]. Also, Katano et al. [KNM+23] in their dosimetric comparison study, suggest that the breath hold technique is associated with lower radiation exposure to the bowel and normal liver tissues.

Studies of the difference in toxicities among patients with various liver conditions gain importance, due to a need for any future result reports to include achieved dose-volume metrics and completed absolute dose-volume histograms for organs at risk (OARs), in order to further improve the dose-response analyses. In another study, liver toxicity prediction with SBRT and the influence of fraction size is discussed [MHGS+14]. This report suggests that when the biological effect of the fraction size is taken into account, it results in a very important reduction of the predicted risk of hepatic toxicity, which would evidently call for biological normalization, with a view of clinical practice, to avoid the overestimation of toxicity in the liver.

The total liver volume receiving specific doses of radiation has been identified as a dose-limiting factor in SBRT. In studies, it has been shown that certain thresholds of radiation dose to the total liver volume are associated with an increased risk of hepatic toxicity and deterioration of hepatic function [SCR+10].

3.2 Radiomics

Radiomics is an emerging and highly promising field in medical research with an intention to extract a set of quantitative features from a wide variety of visible radiographic images of interest for a large spectrum of medical specializations. This is in the interdisciplinary area that demands combined work from radiologists and imaging scientists to data specialists. The derived features, obtained at single or multiple time points, cover a wide range of imaging data in oncology [SDK+21, SGB+21].

Time-to-event analyses in radiomics are generally designed to divide populations or predict clinical outcomes using radiomic features. The categorization into populations is generally for classification tasks, such as the nature of malignancy of a tumor, genomic status, stage, or presence of metastasis, by the prediction of survival rate based on dividing the patients into the risk group of their clinical outcomes [SDK+21]. With CT-based radiomics showing significant potential, particularly in developing imaging biomarkers for non-small cell lung cancer (NSCLC) patients undergoing SBRT, it remains a widely utilized imaging modality for RT treatment planning, guidance, and follow-up [HCN+16].

Radiomics can extract quantitative features from medical imaging to provide valuable insights into hepatocellular carcinoma (HCC). Radiomic models, as well as the ones mentioned for NSCLC, can predict various aspects of HCC, including histology, treatment response, recurrence, and survival. Compared to qualitative characteristics, radiomics

offers a more objective assessment of HCC. Notably, radiomic features derived from CT images obtained before radiation therapy may provide insights into baseline liver function and the risk of Radiation-Induced Liver Disease (RILD) [SHD+22].

Radiomics aids also in predicting HCC grade, differentiating pathological types, identifying lesions, and predicting microvascular invasion. Overall, radiomics has the potential to revolutionize HCC diagnosis, treatment, and prognosis, offering a personalized and precise approach to patient care [MHF+23].

However, despite favorable results, the authors mentioned the following challenges for broad clinical use. Chetan et Gleeson [CG20] have outlined some of the barriers to the use of radiomics for:

- **Heterogeneity between studies:** studies are heterogeneous in terms of the kind of populations of patients, stages of cancer, kinds of treatment modalities, follow-up times, and methodological differences within the radiomics workflow. This variability makes the comparison of results across studies and the replication of findings difficult, highlighting the necessity for uniform standards in research to improve both reproducibility and reliability [HTVR17].

- **Lack of Reproducibility:** ensuring reproducibility in radiomics faces several obstacles, including the very technical complexity, suboptimal study designs, and data overfitting issues, along with an extraordinarily large number of confounding variables, particularly in retrospective data [SMM+17]. This issue is critical for the clinical translation of radiomics research, as reproducible results are essential for developing reliable predictive models [CG20].

- **Low Scientific and Reporting Quality:** Hatt et al. [HTVR17] find evidence that a high proportion of studies is of quality considered too low, insufficient, or incomplete to provide information with an appreciable probability of validity, and more specifically, they quote at least 50% of the studies facing one such issue in one of the analyses. This reflects a need for improved methodological rigor and reporting transparency in radiomics research.

- **Need for Standardization and Collaboration:** to address the issues of heterogeneity and reproducibility, there is a need for standardization in imaging protocols, including image acquisition, post-processing, and segmentation algorithms, to ensure comparability and accuracy across studies [SMM+17]. Additionally, collaboration across institutions is essential to validate promising radiomic models and to accumulate a large enough dataset for meaningful analysis [CG20].

- **External Validation and Clinical Utility:** for radiomics models to be implemented as a clinical decision-making tool, they must be externally validated and their impact evaluated within the clinical pathway [CG20]. This is validated against independent, prospective populated cohorts, ideally within clinical trials, in

order to be able to statistically verify associations or biomarkers identified through radiomics studies [SMM+17].

Future research should focus on implementing standardized radiomics features and software for prospective external validation to identify reliable and reproducible radiomic predictors of response that can be employed in practice. This would help identify a set of reproducible radiomic predictors of response that will be used in clinical practice reliably. [CG20].

3.2.1 Radiomics: Methodology of Work

In numerous articles that make use of radiomic data in medical imaging, a similar workflow is mentioned. Figure 3.1 shows the procedure proposed by Huyhn et al. [HCN+16], divided into five steps: *Imaging*, *Tumor segmentation*, *Feature Extraction*, *Feature selection*, and *Analysis*. That is, as developed in section 2, the first step is to take the patient's images that correspond to the case, then segment the tumor or regions of interest, and with this, extract their radiomic features.

Kumar et al. [KGB+12] also advocate for following a line of work identical to that defined by Huyhn et al. [HCN+16], emphasizing the challenges posed by some of its phases, such as image acquisition and segmentation of the ROIs. Specifically, it highlights that there is a wide variation in imaging parameters such as resolution, patient position, and reconstruction algorithms across different scanners and institutions, making it difficult to compare results obtained from different sources. Additionally, segmentations need to be robust and involve minimal operator input to ensure accurate delineation of regions of interest.

Shur et al. [SDK+21] and Scapicchio et al. [SGB+21], unlike the two previously cited, add an image preprocessing stage before the extraction of radiomic features.

Structures Extraction

In our study, the ROIs can be segmented manually, semi-automatically, or automatically. Kumar et al. [KGB+12] point out that manual segmentations are typically utilized as the gold standard or ground truth for assessing the accuracy of automated segmentation. However, these hand-drawn boundaries typically show marked inter-reader differences, and their reproducibility is low.

According to Hojjat and Kitler [HK98], and Dehmeshki et al. [DAVY08], the Region-Growing method is employed, which selects a seed point within the volume of interest (VOI) and iteratively expands the region according to specific criteria. Despite its speed, region-growing methods may yield unintended regions in noisy environments. Conversely, Boykov et al. [BVZ01] apply Graph Cut techniques for image segmentation, transforming the image into a graph to identify the optimal cut that delineates the intended regions. This approach is notably effective for segmenting structures with distinct boundaries.

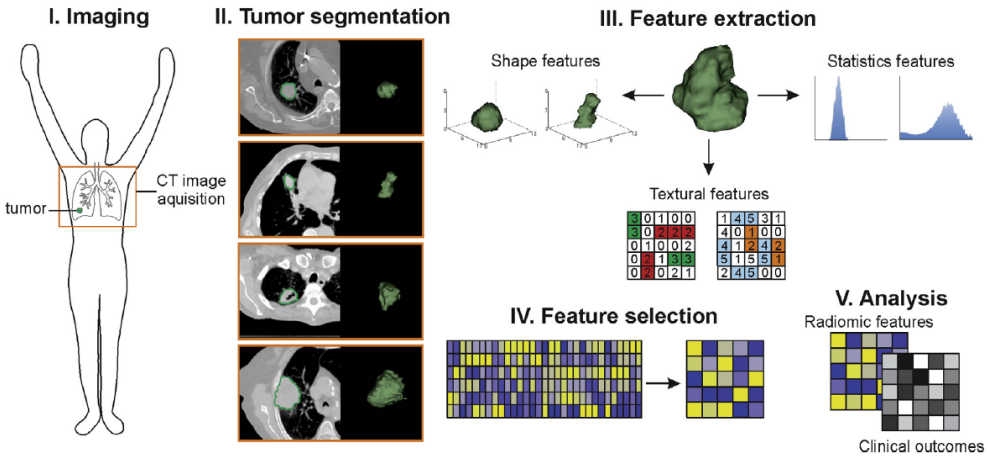
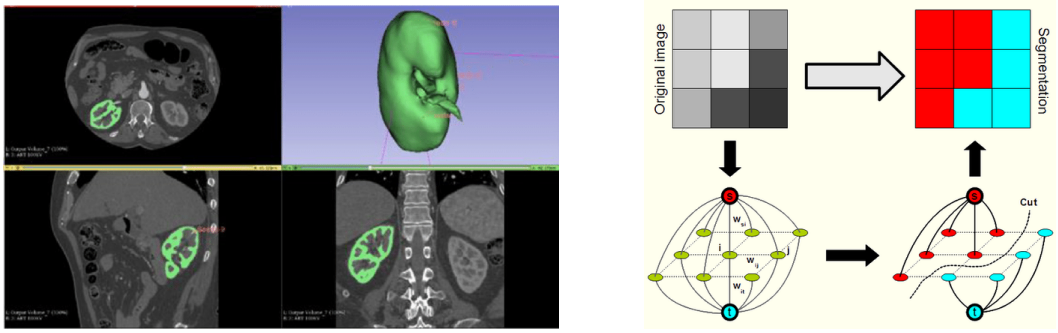


Figure 3.1: Radiomics Workflow proposed by Huyhn et al. [HCN+16].



(a) (b)

Figure 3.2: (a) Segmentation with Region Growing method [LPS+20] and Segmentation with Graph Cut method (b) [gra]

The approach used by Kumar et al. [KGB+12], is also supported by authors such as Kogkuo et al. [LH06]: the semi-automatic segmentation, where methods like livewires involve user interaction to guide the process of segmentation, being a combination of manual input with automated algorithms to improve accuracy and efficiency.

Since within the work of our thesis, we consider the Couinaud segmentation as a part of the same, we will mention some articles related to the Couinaud segmentation. Neural network architectures are used throughout the articles to segment the liver in the 8 Couinaud segments: Wang et al. [WJL+22] describe their method with an attentive residual hourglass-based cascaded network (ARH-CNet), while Arya et al. [ARJA21] use a U-Net. Both achieved accuracy of 90 and 80% without using complex neural networks. Finally, Tian et al. [TLSX19] applied a GLC-Unet and reported an accuracy of 92%.

Radiomics Extraction and Preprocess

The next step, once we have performed the segmentation of the ROIs, is to extract the characteristics of the radiomic features from our medical images, for which different methods are suggested. Shen et al. [SHD+22] suggest the use of the 3D Slicer software, as it is capable of performing the segmentation of the region of interest, which in this case is a tumor, and subsequently using a plugin that allows for the extraction of radiomic features. This plugin works with a Python library called PyRadiomics, which authors such as Griethuysen et al. [vGFP+17] or Sack et al. [SNM+22] work on directly in Python. Similarly, Tang et al. [TWX+23] and Wang et al. [WJL+22] use the PyRadiomics library for extracting radiomic features, as having the data and being able to work on it in Python makes the use of subsequent Machine Learning very efficient.

Subsequently, it is necessary to preprocess the data once obtained, focusing on the detection of outliers. Boukerche et al. [BZA20] discuss in their article the different categories into which outlier detectors can be classified, highlighting supervised, semi-supervised, and unsupervised techniques, as well as global or local-level outlier detectors. El-Sherbiny et al. [ESNH+23] employ both local and global methods, specifically the *Local Outlier Factor* for global outliers, and the *Isolation Forest* at a global level. Furmanová et al. [FGM+20], on the other hand, use dimensionality reduction techniques, such as PCA, to detect outliers in a low-dimensional space.

Lastly, when working with a high-dimensional space, it is necessary to employ dimensionality reduction techniques. Scapicchio et al. [SGB+21] reduce their radiomic data dimensional space using techniques such as PCA and t-SNE, similar to Huyhn et al. [HCN+16]. The latter article mentioned, along with Lambin et al. [LRVL+12], takes advantage of the explained variance feature of the PCA method, as it allows for optimizing the number of components to use. On the other hand, Xiang et al. [XWY+21] utilize the U-MAP method since it is capable of retaining more global structure, and Borg et al. [BG05] advocate for the use of MDS.

Cohort Stratification and Predictive Analysis

With the data prepared for analysis, we discuss the methodologies employed for cohort stratification and predictive analysis using Machine Learning models.

One technique for cohort stratification from radiomic data is Hierarchical Clustering, as advocated by Sing et al. [SJY23] and Tsujikawa et al. [TRY+17] to categorize genetic subtypes of autism or cluster PET image features, respectively. Furthermore, leveraging the previously proposed dimensionality reduction techniques, clustering can be performed as a next step using algorithms such as K-Means or Spectral Clustering. El-Sherbiny et al. [ESNH+23] use algorithms like K-Means, Mean-shift, Hierarchical Clustering, OPTICS, and Gaussian Mixture Models for clustering their radiomic and genomic data. Furmanová et al. [FGM+20] utilize Hierarchical Clustering exclusively, while on the other hand, Raidou et al. [RCMA+18] employ the Gaussian Mean Shift again.

In addition to clustering, authors such as Raiter et al. [RBGR18] suggest reducing the high-dimensional space through two consecutive dimensionality reduction techniques such as PCA and t-SNE, so that the advantages of both can be combined.

Finally, to conduct predictive analysis, many articles utilize radiomic data for classification tasks or predicting a response variable. Tang et al. [TWX+23] use radiomic features to predict steatosis in liver cancer patients, employing the Support Vector Machine (SVM), and also comparing the utility of these radiomic data against the use of only clinical data. Shen et al. [SHD+22] attempt to predict the previously mentioned RILD in patients who have been treated with SBRT, using models such as Random Forest (RF) or Logistic Regression (LR). Finally, Shur et al. [SDK+21] provide a general review of the advantages and disadvantages of using models such as Gaussian Naive Bayes, Logistic Regression, Random Forest, and SVM, applied to radiomics in oncology.

3.3 Visual Analytics (VA)

While technologies like visualization, statistics-based data analysis, and machine learning have been explored, they may hide some relevant conclusions in the studies. The concept of visual analytics, introduced by James Joseph Thomas, combines algorithmic data analysis with interactive visual interfaces to integrate human judgment into the analytical process [Cui19]. This approach facilitates decision-making by leveraging human flexibility, creativity, and background knowledge. Also, VA is used in a large variety of areas, like physics and astronomy, business, environmental monitoring, disaster and emergency management, security, software analytics, biology, medicine and health, engineering analytics, personal information management, and mobile graphics and traffic [Ham10].

Visual analytics bring also challenges and future directions, characterized by scalability, interaction, infrastructure, and evaluation [Cui19]:

- **Scalability:** As data volumes continue to explode, visual analytics systems must be capable of handling large and complex datasets effectively [REE+09]. The challenge involves creating tools and techniques that can scale with both the size and the dimensionality of the data, ensuring that visual analytics applications can manage massive datasets without compromising performance or usability [Cui19].
- **Interaction:** Interactivity in visual analytics systems is defined as the quality of interactions among components, influenced by cognitive psychology, human-information interaction, visualization sciences, and interaction design [SPB12]. Think about and implement interactions that are meaningful, intuitive, and effectively support the users' analytical processes. This includes careful consideration of the interaction patterns, their cognitive utilities, and how they fit within the broader context of the user's tasks and goals [PSDK15].
- **Infrastructure:** The lack of a common framework or infrastructure makes difficult the development and deployment of visual analytics tools [Cui19]. Significant

investments are required to develop infrastructures that support the unique needs of visual analytics, including scalable storage, fast queries, efficient distribution, and exploration-friendly database technologies [Fek13].

- **Evaluation:** Evaluating the effectiveness, utility, and trustworthiness of visual analytics applications is complex due to the integral role of human cognition, perception, and reasoning [Cui19]. Developing theoretically grounded evaluation frameworks that can assess both the visualization and data analysis components of visual analytics applications is crucial [Wij13].

Among other application domains, VA has also been employed as a supporting methodology for the dields of radiotherapy. Previous work includes tools as the one Furmanová et al. [FGM+20] describe in their article. This tool was developed to explore and predict changes in patient pelvic organ variability throughout RT treatment, thereby assisting in the assessment of different treatment strategies considering anticipated anatomical changes. Features include an intuitive representation of past patient data, enabling analysis of anatomical similarities and variations. Additionally, the framework supports matching incoming patients with past cases to predict anatomical changes and optimize RT plans accordingly.

Raidou et al. [RCMA+18] present their tool *Bladder Runner*, designed to explore and analyze the impact of bladder shape variation on the accuracy of dose delivery during prostate cancer radiotherapy (RT). This tool is aimed at aiding clinical researchers in understanding RT-induced complications through detailed visual exploration and analysis of individual patients and cohorts throughout the entire treatment process. Also, Raidou et al. [RBV17] present a comprehensive approach to enhancing radiotherapy planning and analysis through the use of visual analytics. This work is significant in the context of prostate cancer treatment, where radiation therapy plays a critical role. The paper outlines the development of a series of visual analytics solutions designed to facilitate the interactive exploration and analysis of complex radiotherapy planning data and processes.

In the context of this thesis, previous work includes El-Sherbiny et al.'s approach [ESNH+23] for radiogenomic and clinical data analysis and exploration. This innovative framework aims to assist cancer experts and biomedical data scientists in navigating the complexities of radiogenomic data, characterized by its size, heterogeneity, and intricate relationships between different data types.

Yu et al. [YJY+17] introduce a visual analytics system designed to assist researchers and practitioners in exploring and analyzing large-scale medical imaging data, particularly focusing on radiomics. This system, named iVAR, addresses the challenges posed by the massive volume of imaging data and the high dimensionality of derived radiomic features, which quantify tumor phenotypes in detail by analyzing image intensity, shape, texture, and other clinical features.

Within the field of Visual Analytics, there are different ways to represent data. Some articles detail the methods they use in their tools for data visualization.

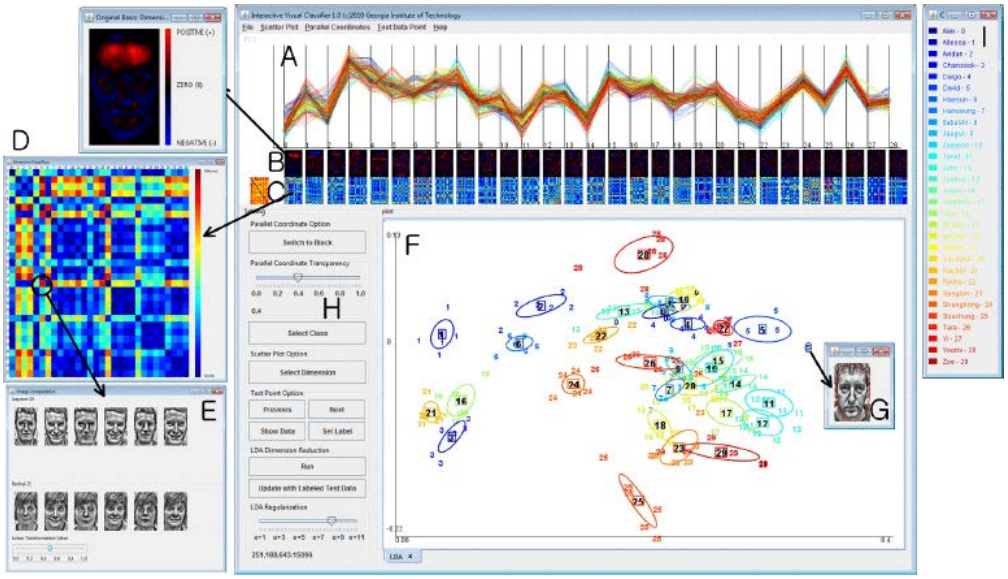


Figure 3.3: TripVista: visual analytics tool for traffic trajectory data [GWY+11].

Guo et al. [GWY+11] developed *TripVista*, a tool for exploring and analyzing complex traffic trajectory data, which was mainly based on a parallel coordinate plot and coordinated multiple views, as shown in Figure 3.3. Choo et al. [CLKP10] presents a visual-analytics system tool (iVisClassifier) for classifications based on a supervised dimension-reduction, shown in Figure 3.4, where parallel coordinates plots are again highlighted, as well as the use of heatmaps.

With application in radiomics, the VA tool by Yu et al. [YJY+17] is shown in Figure 3.5, where various forms of visualization are combined. In (a), the 2D visualizations of the CT scans and the 3D reconstructions of the tumor are shown. In (b), all the features are displayed as small multiple-frequency plots, where one can select and view an enlarged plot. Additionally, heat maps, correlation matrices, and a plot showing the correlation between two features are used. Lastly, a multivariate panel with Parallel Coordinates and a plot with PCA are included.

Another example is the previously explained VA tool by El-Sherbiny et al. [ESNH+23] and shown in Figure 3.6. This tool provides some of the visualization techniques from the previous case, such as heat maps (C) and scatter plots (A), though in this case, they are used to represent other types of data such as patient stratification. Additionally, bar charts (B) and (D) are used, along with various buttons for interactivity and selection options of the linked charts.

Our work will aim to incorporate multimodality into the use of radiomic data, meaning the ability to compare different types of data simultaneously. We will apply patient exploration and stratification techniques, and display the results on our dashboard so that users can interact with and draw conclusions from the data.

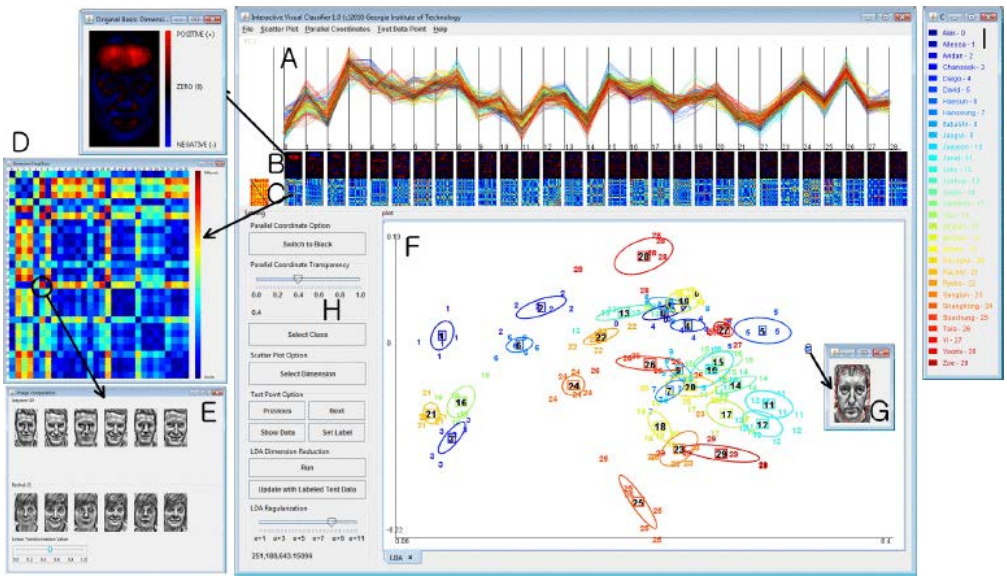


Figure 3.4: iVisClassifier: Visual Analytics tool based on supervised dimension-reduction [CLKP10].

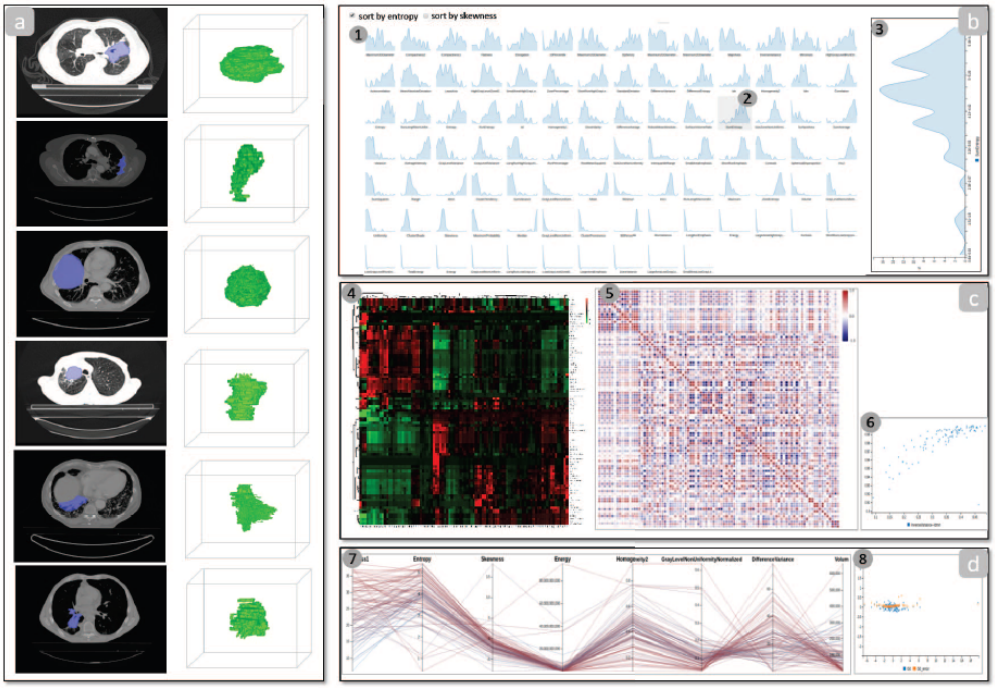


Figure 3.5: iVAR: Visual Analytics tool for radiomic features [YJY+17].

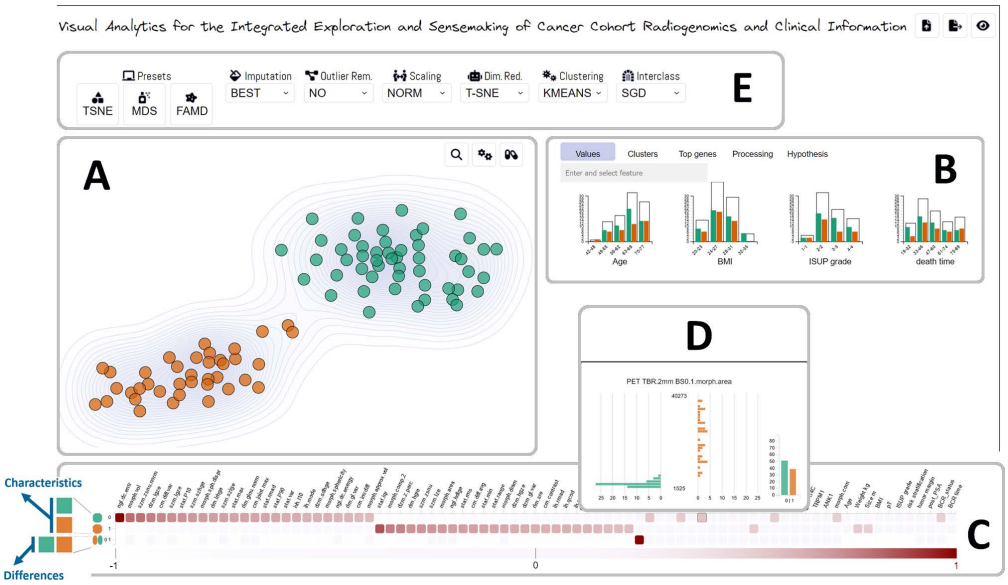


Figure 3.6: Visual Analytics tool for radiogenomics and Clincal Information [ESNH+23].

CHAPTER 4

Methodology

As detailed in previous sections, we are going to work with radiomic features obtained from the CT scans of our dataset. We will follow the pipeline established according to Figure 4.1. This pipeline distributes the different tasks to be followed throughout this thesis, starting with the extraction of our structures of interest (Liver, Liver-ITV, ITV, and PTV), followed by the extraction and computation of the respective radiomic features of these structures, continuing with their exploratory analysis, patient stratification, toxicity prediction, and finishing with the design of an interactive tool for the management of these data.

In the following sections, each of the steps designed in the workflow will be developed in detail. This workflow is defined by numerous articles that have similar ways of working. Huynh et al. [HCN+16] do not include in their procedure a preprocessing of the images once they are acquired, but place more emphasis on the way of analyzing the data and reducing the dimensionality of the radiomic features. This is also done by Shur et al. [SDK+21] and Scapicchio et al. [SGB+21] who share a structure much more in line with ours. Despite not preprocessing the acquired images, we do process the extracted radiomic data, cleaning it for the posterior analysis.

4.1 Features Extraction

4.1.1 Structures Extraction

In order to perform feature extraction, we need a mask that segments or establishes a Region of Interest (ROI) in our images. These ROIs can be delineated manually, semi-automatically, or automatically. In the case of manual segmentations, the presence of a second expert to corroborate is necessary [SDK+21]. Software such as 3D Slicer or Eclipse Software is commonly employed for this task if it is to be performed entirely manually [SGB+21, HCN+16]. Nonetheless, it is also advocated that a semi-automatic

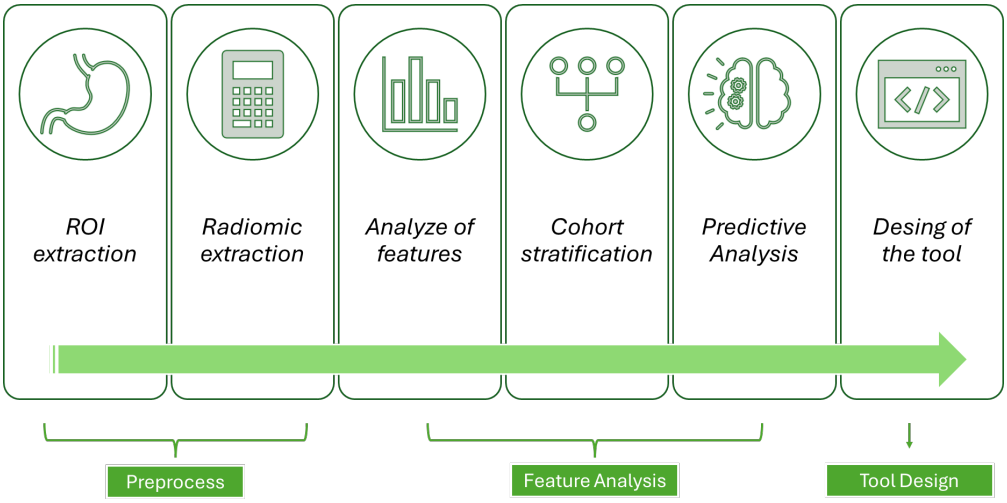


Figure 4.1: Processes and subprocesses of the implementation pipeline.

segmentation followed by an expert's curation of the results is the most time-efficient and accurate method [SGB+21].

In our case, as we have not been directly provided with masks along with the CT scans, we will make use of the DICOM RT-Structure, where the various target volumes and OARs have been previously segmented, allowing us to export them as masks from which to subsequently obtain the radiomic features. Additionally, as part of the objective of this thesis is to study what impact the use of Couinaud's Segmentation could have on the analysis, this task will be carried out to add the 8 segments to each patient and analyze their radiomic features separately. For both extraction tasks, the 3D Slicer software will be used [sli].

RT-Structures

In Section 2, reference is made to the different most common Target Volumes in Radiotherapy, such as *GTV*, *IV*, *ITV*, and *PTV*, among others. In some cases of Liver Cancer treatment, target volumes include *Liver-ITV*, as in the case of this thesis. The difference between *ITV* and *Liver-ITV* lies in the specific area each covers. While *ITV* can refer to any internal target volume for any part of the body, *Liver-ITV* specifically focuses on the liver area. This distinction is crucial for treatment planning, allowing physicians and dosimetrists to precisely adjust radiation doses to cover the tumor while minimizing exposure to surrounding healthy tissues, especially in sensitive organs like the liver [ZKW+23].

Our dataset is composed of anonymized patients that come from studies carried out on different dates, which is why not all of them have all the target volumes explained above. The structures *Liver*, *Liver-ITV*, *ITV*, and *PTV* are the most common in their studies,

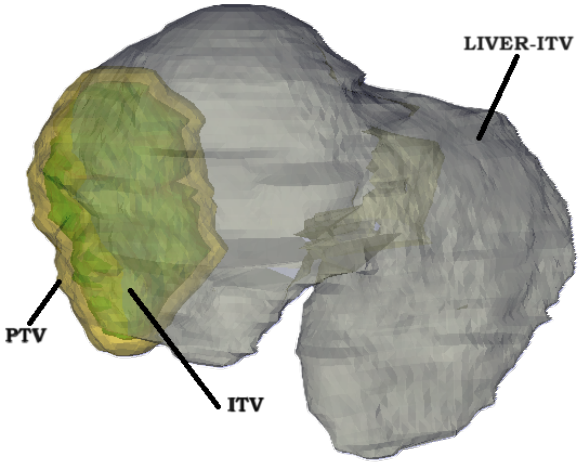


Figure 4.2: Structures of Interest: Liver-ITV (gray), ITV (green), and PTV (yellow) visualization in Slicer 3D.

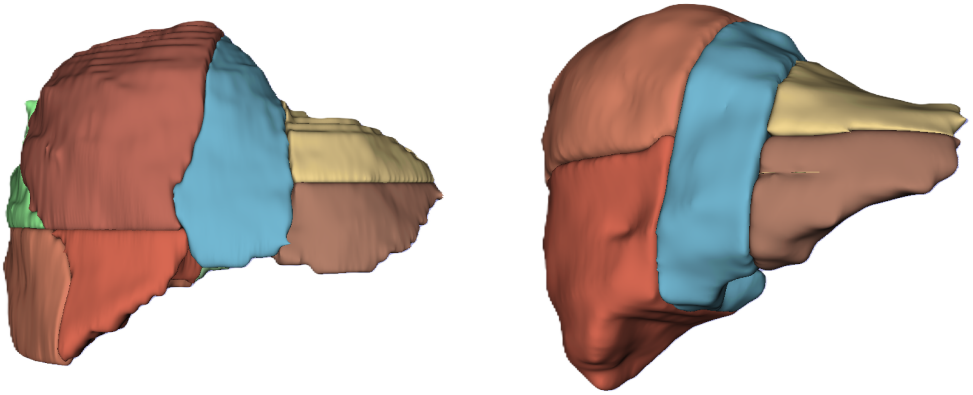
which is why we will primarily work with them. In this way, we can ensure that we have features related to the **liver** (Liver, Liver-ITV), and the **tumor** (ITV, PTV).

Despite being the most common in all patients, it is necessary to point out that not all patients in the dataset have all these structures in their RT-Structure, and others have more than one *ITV* or *PTV*. For the latter case, and given that *Liver-ITV* is a structure common to all patients, we selected the *ITV* that is part of the *Liver-ITV* and its corresponding *PTV*.

In the first instance, an attempt was made to automate the structure extraction process. However, as the structures are not common to all patients and may have different names assigned to them, the option of using the 3D slicer is preferable. In this way, all the structures can be observed in detail, and possible incidences such as the existence of more than one *ITV* or PTV can be detected. Since the total number of patients is 19, and a maximum of four structures per patient, it is efficient to perform these manual observations. An example of the visualization is shown in Figure 4.2, where the *Liver-ITV*, *ITV*, and *PTV* structures are shown simultaneously, observing that the *PTV* surrounds the *ITV*, and both denote the position of the tumor in the liver. Once we have located the structures of interest, we export them as a separate .nrrd file and assign the same name for each patient (i.e., *ITV*, *PTV*...etc) in order to optimize the process of extracting radiomic features.

Couinaud Segmentation

We additionally perform a Couinaud segmentation of the livers according to the protocol explained in Section 2. There are numerous articles detailing neural network architectures



(a) Annotated Couinaud segmentation (b) Manual Couinaud segmentation

Figure 4.3: Comparison of our manual Couinaud segmentation using 3D Slicer (b) with the annotated reference from the GitHub repository (a) [git].

capable of automatically or semi-automatically segmenting these livers, such as Wang et al. [WJL+22]. This work details how an attentive residual hourglass-based cascaded network (ARH-CNet) is used for landmark detection on the hepatic vascular system, thus achieving an average accuracy of 90%. Arya et al. [ARJA21] utilize a U-Net with an accuracy of 80%, from CT images. However, in none of the articles where deep learning segmentation methods are explained, are the datasets with DICOM images used for training provided, or explanatory code of the process.

A dataset obtained from GitHub [git] with 193 NIfTI files of livers segmented according to Couinaud, although not from CT images that include the liver. An attempt was made to follow the model of Tian et al. [TLSX19] based on an Unet (GLC-Unet) that used the same database and achieved 92% accuracy, but our results were not satisfactory given the high computational requirements for training, and the complexity of the problem for segmentation with 9 classes. Nevertheless, this dataset has been used as a reference, taking into account the intra- and inter-patient variability necessary for the manual segmentation of our livers in the dataset.

The result and comparison are shown in Figure 4.3, where we can see the annotated Couinaud Segmentation (a) and our manual Couinaud Segmentation (b). Another advantage provided by the use of Couinaud segmentation is the ability to identify the **location of the tumor** in the liver, that is, approximately in which segments it is found. An example of this is shown in Figure 4.4, where it can be seen that the tumor is located between segments 2 and 3. This will allow us to know which areas of the liver may be overradiated or underradiated, as detailed in the research question (R2).

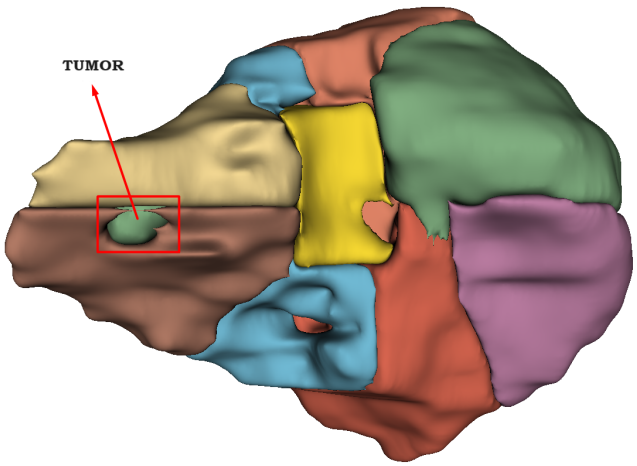


Figure 4.4: Location of the tumor in the liver, between segments 2 and 3. Visualization in Slicer 3D.

4.1.2 Radiomics Extraction

For the extraction of radiomic features, both an **image** and a **mask** are necessary. We have images according to the average and breathing cycle of each patient, and masks as structures discussed in the previous section. Once the image and the mask are imported and all the radiomic features of each study and patient are processed, we obtain a dataframe as output. This dataframe consists of 112 columns, one for each feature that allows the extraction of Pyradiomics, together with additional columns that refer to the patient's ID, gender, whether they present toxicity Yes/No, and type of cancer. They will also include, depending on the case, the time of the respiratory cycle, the structure, or the Couinaud segment.

Once the processing is finished, we will have **four resulting dataframes** with similar structures that we will work with for their analysis, stratification of patients, prediction of toxicity, and implementation in the tool to be designed in the most efficient way possible. Figure 4.5 shows a pipeline of how dataframes are constructed from images, masks, and clinical data.

The resulting dataframes are called **AVG**, for the case where the reference images are the average, **AVG Couinaud** for when the structures are not general but the liver segmented according to the Couinaud method, **B.C** when each part of the respiratory cycle is used as images, and **B.C Couinaud** when we again use the Couinaud segments instead of the general structures.

4.1.3 Doses Extraction

Using the 3D-Slicer tool and leveraging our Eclipse Doses files, we are able to obtain different dose measures for our patients, and for their structures of interest. Among

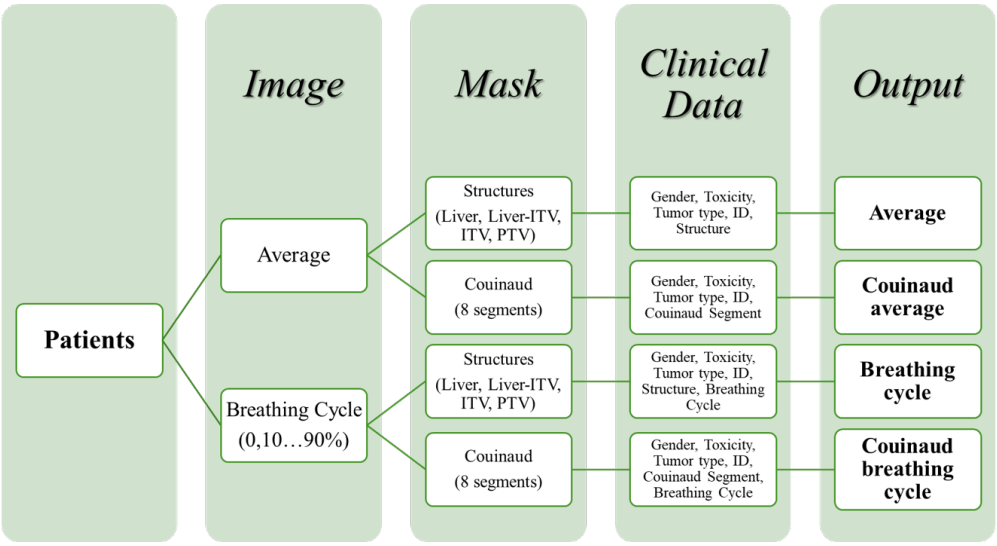


Figure 4.5: Pipeline of Dataframes building, a combination of radiomic features and clinical data.

Structure	Volume (cc)	Mean dose (Gy)	Proportion (Gy/cc)	Min dose (Gy)	Max dose (Gy)	Toxicity	Disease	ID	Tumor
Segment 7	195.54	1.77	0.009	0.127	7.269	0	1	1	2,3
Segment 4	156.62	3.15	0.02	0.189	14.44	0	1	1	2,3
...	...	...	...	...	...	...	...	...	...
ITV	2.34	25.37	10.83	7.31	31.87	0	1	19	4
Liver-ITV	1239.78	2.24	0.002	0.021	31.94	0	1	19	4

Table 4.1: Doses data frame: structures, measures, clinical data, and tumor location.

these values are the minimum, average, and maximum dose in Gy, and the volume of the structure itself (measured in cc). Additionally, an extra value resulting from the ratio between the average dose value and the volume of the structure has been calculated, which is called 'Proportion', and measured in Gy/cc. Finally, we thus have information about these dose values for each patient, for each of their structures including the Couinaud segments, and also our clinical data. Moreover, since the segment or the specific liver segments where the tumor is located have been visually verified, this information has been added to the last column.

An example of how this dataframe is structured is shown in Table 4.1, where the variables are displayed, with examples of the different types of structures that are possible to explore and how the location of the tumor is expressed in the last column. In this way, *2,3* implies that the tumor contains segments two and three of the liver, while *4* indicates that the tumor completely encompasses only this segment.

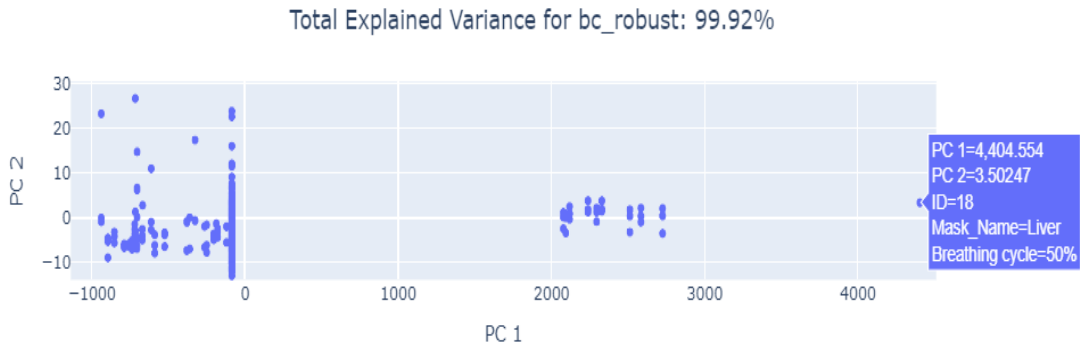


Figure 4.6: PCA (2 components) with RobustScaler for outliers detection, Breathing Cycle dataframe.

4.2 Preprocessing

As detailed in Section 3, different articles and previous work suggest approaches to the preprocessing of radiomic features. This includes the detection of outliers, scaling of the data, and dimensionality reduction.

4.2.1 Outliers Detection

Once we have completed the feature extraction process and shaped our dataframes, before scaling the data, we will perform a detection of possible outliers, to try to mitigate errors in the measurements due to the segmentation of structures or Couinaud segments. In some cases, outliers may indicate errors in data collection or processing, though they can also signal the existence of subpopulations within a larger dataset. Instead of being errors, these values may represent real and significant variability [BB19]. In data-driven decision-making, having an accurate representation of the data is crucial. Outliers can lead to decisions based on a misunderstanding of the data [BZA20].

In Figure 4.6, an example of the visual detection of these outliers is shown for the breathing cycle data frame, in this case after applying RobustScaler and a Principal Component Analysis (PCA), discussed in the following sections). The figure clearly highlights that the liver data of patient 18, during 50% of the respiratory cycle, differ from the rest. In this way, and for the rest of the dataframes, we can find out which structures (or which structure and cycle) are disturbing the analysis of our data. However, visual identification of outliers is not the best option, especially considering that we have four dataframes for the study.

There are several methods for detecting outliers, which are divided into categories based on the type of data they use. Boukerche et al. [BZA20] discuss these categories: *Supervised outlier detection* involves using data that has been labeled to create models that can predict outliers, treating the process as a type of binary classification where the classes are unbalanced, while *semi-supervised outlier detection* employs a combination

of a small amount of labeled data and a larger portion of unlabeled data to construct models. *Unsupervised outlier detection*, on the other hand, operates completely with unlabeled data, using it either to build models that can score outliers or to directly assess outlier scores without creating a model [BZA20, FC23]. The unsupervised approach is the most extensively studied due to the challenges associated with acquiring labeled data for outlier detection and the interest in identifying unknown anomalous patterns [BB19]. Our data may contain outliers, as will be observed at the end of this section, and with the use of scalers, and given the uncertainty of which data are correct and which are not, we perform unsupervised methods for outlier detection.

There are other ways to classify outlier detection, being treated **locally** or **globally**, depending on the characteristic difference between the outlier and its neighbors, or compared to the entire dataframe [HMI+22]. Therefore, unsupervised methods such as K-Nearest neighbors and local outlier factor (LOF) as examples of local algorithms, and isolation forest (IF) and one-class support vector machine (OSVM) as global methods have been used for outlier detection.

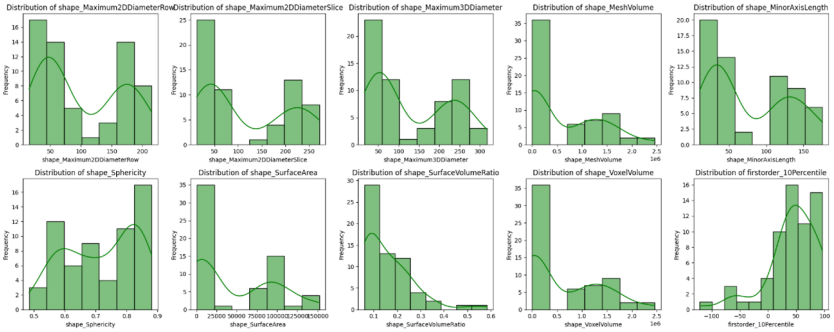
The *local* approaches identify outliers based on their relations with nearby data points [Smi20]. The nearest-neighbor-based outlier detection approaches measure the degree of abnormality based on a data point's relation to its nearest neighbors while clustering groups data into multiple clusters with similar data instances ending up in the same cluster [BZA20]. Outlier detection algorithms based on clustering usually take a two-step procedure: grouping the data with clustering algorithms and analyzing the degree of deviation based on the clustering results. The Local Outlier Factor (LOF) [Smi20] algorithm is designed to quantify how much of an outlier each data point is within a multidimensional dataset. It computes the LOF of a data point by comparing its local density to the densities of its k-nearest neighbors.

On the other hand, *global* methods like isolation forest [BZA20] consists of an unsupervised tree ensemble that intuitively resembles the random forest for classification problems, and One-Class SVM [HMI+22] which, unlike traditional SVMs that separate two or more classes with a hyperplane, is designed to identify a single class over all data points.

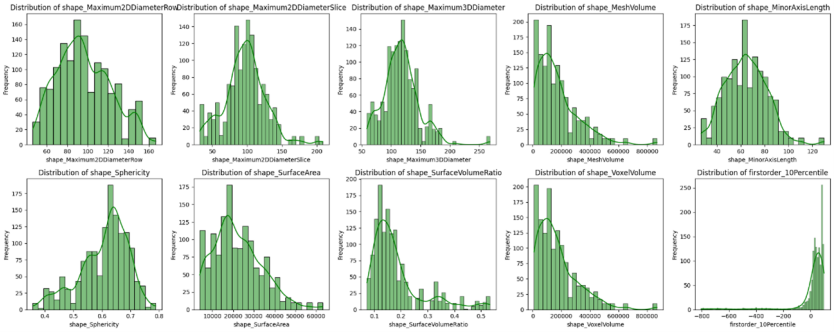
The impact of outlier removal and the differences it introduces compared to the original dataframe is explored in the patient stratification.

4.2.2 Data Scaling

When comparing different radiomic features during exploratory analysis, we often encounter variations in the measurement scales and units used. These discrepancies can make it challenging to interpret and compare the features accurately. By applying scaling techniques, such as Min-Max scaling or standardization [dCC23], we can transform the features onto a common scale, ensuring that they are directly comparable. This helps to clarify the representations of the features and facilitates more meaningful comparisons between them.



(a) Average Dataframe



(b) Couinaud Breathing Cycle Dataframe

Figure 4.7: Distribution of 10 random radiomic features, comparison between dataframes.

Before selecting a scaling method for our research, it is essential to conduct a normality test to determine whether our data follows a normal distribution. Figure 4.7 displays histograms illustrating the distributions of 10 random features extracted from the average and Couinaud breathing cycle data frames, respectively. These histograms facilitate a comparison between the dataframe with fewer rows or patient-structure combinations and the one with the most. In both cases, the presence of outliers complicates the determination of whether the data follows a normal distribution.

To address this issue, we employ the Normal Test [FF21], which returns a p-value. If the p-value is less than 0.05, this indicates a significant deviation from the normal distribution, suggesting that the data are not normally distributed. Alternatively, if the p-value exceeds 0.05, we cannot reject the null hypothesis of normality.

In the four cases (data frames), the number of columns that do not follow a normal distribution exceeds that do, this will make us establish that our data frames, in general, exhibit a non-normal distribution. Because of this, we will not be able to use a Standard Scaler. Two possibilities can be selected for scaling the data, the *MinMaxScaler* or the *Robust Scaler*, each has its advantages and disadvantages, and their effectiveness can vary depending on the nature of the data.

MinMaxScaler [dCC23] scales and translates each feature individually so that it is

in the given range on the training set, typically between zero and one [0,1]. It is best used when you know the distribution of the data does not have outliers and can be well approximated by a bounded interval. It is particularly useful when algorithms are sensitive to the scale of data, such as neural networks or k-nearest neighbors. The formula of the MinMaxScaler is given by:

$$X_{\text{scaled}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{4.1}$$

Where X is the original value, X_{min} is the minimum value in the feature column, X_{max} is the maximum value in the feature column, and X_{scaled} is the resulting scaled value.

RobustScaler [dCC23] scales features using statistics that are robust to outliers. Instead of using the minimum and maximum values, it uses the median and the interquartile range (IQR). It is useful when the data contains many outliers or you expect to have outliers and you want to mitigate their effects on the scaling of your dataset. It is effective for models that are sensitive to the scale of data but can be negatively impacted by outliers, such as linear regression, logistic regression, and other linear models. The formula of the RobustScaler is given by:

$$X_{\text{scaled}} = \frac{X - X_{\text{median}}}{IQR} \tag{4.2}$$

Where X is the original value, X_{median} is the median of the feature column, IQR is the interquartile range (calculated as $Q_3 - Q_1$, where Q_1 and Q_3 are the 1st and 3rd quartiles, respectively), and X_{scaled} is the resulting scaled value.

RobustScaler is a good option for capturing information from all the data, and considering outliers, but, on the other hand, MinMaxScaler brings all the features to the same scale, which is beneficial for the visualization of the data. De Amorim et al. [dCC23] explore the impact of different data scaling techniques on the effectiveness of classification models. They highlight that choosing an inappropriate scaling technique can significantly degrade model performance, potentially even more than if the data were not scaled at all. This is the reason why we perform both for cohort stratification, data exploration, and prediction of toxicity, so the advantages and disadvantages are discussed in these sections.

4.2.3 Dimensionality Reduction

The extraction methods can generate from dozens to thousands of features, producing a high-dimensional feature space and making more complex every task we want to perform [SGB+21, MML+20]. This is why after the scaling of data is necessary to perform a dimensionality reduction, where the most common algorithms are Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE), Uniform Manifold Approximation and Projection (U-MAP), and Multidimensional Scaling (MDS). Scapicchio et al. [SGB+21] and Huynh et al. [HCN+16] discuss PCA and t-SNE

performances, and how they use them for their radiomic data. PCA analysis is performed and compared between the two scalers mentioned before, and used to complete Task 4 (T4) and Task 5 (T5).

PCA (Principal Component Analysis) [Jol02] is a statistical technique that transforms data into a new coordinate system, where the greatest variance by any projection of the data comes to be the first principal component, the second greatest variance becomes the second principal component, and the process is repeated until the new component is almost ineffective or reaches the threshold set by users [XWY+21]. The utility of PCA also includes its ability to reduce the dimensionality of the data and to study how much variance it can explain or retain when using multiple components, as demonstrated by Huynh et al. [HCN+16] and Lambin et al. [LLD+17a]. The formula (Equation 4.3) for calculating the principal components is the eigen decomposition of the covariance matrix (C) of the data:

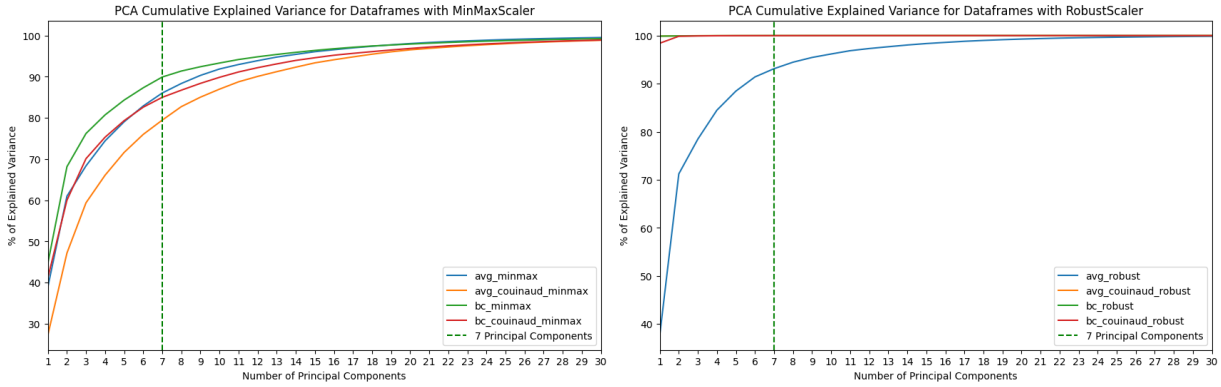
$$C = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^T \qquad (4.3) \qquad X = V^T U \qquad (4.4)$$

where x_i are the samples, $\bar{x}$ is the calculated mean of our data and n the number of observations.

While the linear method PCA is based on the Covariance Matrix, the Non-Linear method **t-SNE** [vdMH08] starts by converting the high dimensional Euclidean distances between data points into conditional probabilities that represent similarities. **U-MAP** [MHM20] is a dimension reduction technique that can be used not only for visualization similar to t-SNE but also for general non-linear dimension reduction. Compared with t-SNE, UMAP retains more global structure with superior run-time performance [XWY+21]. **MDS** (Multidimensional Scaling) [BG05] which aims to preserve a measure of similarity (dissimilarity or distance) between pairs of data points has been used for exploratory analysis, multivariate analysis, and visualization.

PCA in its different versions (standard PCA, robust PCA, sparse PCA, kernel PCA, ... etc) is still one of the preferred techniques due to its simplicity and intuitiveness [AHT20]. A comparison between both scalers and the number of necessary components to explain variance is shown in Figure 4.8. While MinMaxScaler needs a minimum number of 6 components (green vertical line) to explain the total variance of our data, RobustScaler needs for almost all the dataframes only 1 component.

By performing PCA with two components on our dataframes, we can identify structures (or structures within a specific respiratory cycle) that significantly differ from the rest of our data, as shown before in Figure 4.6, which could be problematic for subsequent tasks such as cohort stratification or accurate visualization in our exploratory analysis. Nevertheless, the method used for detecting these outliers in this thesis has been previously carried out and explained.



(a) Cumulative Explained Variance, MinMaxScaler (b) Cumulative Explained Variance, RobustScaler

Figure 4.8: Comparison of the explained variance between MinMaxScaler and RobustScaler.

The PCA components are linear combinations of the original variables, and "loadings" describe the weight or importance of each original variable in forming the principal components [Loh]. The original data matrix X can be decomposed as two orthogonal matrices U and V, which are the loadings matrix and score matrix respectively, as described in Equation 4.4. Loadings help us to interpret the principal components by showing how each original variable contributes to them and can reveal patterns and relationships among variables that might not be apparent in the high-dimensional space.

In addition to PCA, other techniques like t-SNE, U-MAP, and MDS are applied to both scalers, and their efficiency will be assessed in Section 4.3, and compared.

4.3 Cohort Stratification

For the cohort stratification, we have already explained the dimensionality reduction techniques employed. Next, once we have reduced our high-dimensional feature space, we will perform unsupervised machine learning techniques, specifically clustering, to try to separate our patients into the different combinations allowed by our dataframes. The methods used will be combinations of dimensionality reduction with clustering, hierarchical clustering, and chained dimensionality reduction.

4.3.1 Hierarchical Clustering

Hierarchical clustering [TRY+17] is a technique that arranges items in a hierarchy based on the distance or similarity between them. This method helps in grouping similar features based on their characteristics, allowing for a visual representation of associations between the features. The method we are going to use is Agglomerative Clustering [Gir], where each data point is assigned to an individual cluster. At each iteration, the

clusters are merged based on their similarity and the process repeats until one cluster or K clusters are formed. Articles using radiomic data such as Sing et al. [SJY23] performed hierarchical clustering as the primary method employed in this study to analyze and categorize genetic subtypes of autism. In the same way, Tsujikawa et al. [TRY+17] used it for approaches in cervical cancer and clustering the PET-derived features.

Figure 4.9 represents the stratification of the cohort, comparing the different types of parameters that can be selected during the study. The upper subfigure represents the breathing cycle, and the lower subfigure represents the average, based on the following selected options:

CASE 1

1. **Breathing Cycle:** (50%).
2. **Structures:** All.
3. **Scaler:** MinMaxScaler.
4. **Outliers:** No removed.

CASE 2

1. **Average**.
2. **Structures:** Segment 6.
3. **Scaler:** RobustScaler.
4. **Outliers:** Global outliers removed.

These two contrasting examples reflect the differences that using one scaler over another can entail, as well as the inherent differences between patients classified or filtered based on the same characteristics. For all cases, the RobustScaler tends to identify one or more patients who differ from the rest, as has been shown in Figure 4.6, since it is based on the percentile method. However, the use of the MinMaxScaler distributes the clusters much better and facilitates the visualization of the results. In both subfigures, three clusters are established, and the difference between them is much greater when studying the Couinaud segments, reflected in the height of the y-axis.

4.3.2 Dimensionality Reduction and Clustering

To combine with the mentioned dimensionality reduction techniques, clustering methods such as *K-Means*, *Mean-Shift*, and *Spectral Clustering* will be performed.

K-Means [TSKK18] is a clustering algorithm that groups data by attempting to separate samples into groups with the same variance, minimizing a criterion known as inertia (WCSS) or the sum of squares within the cluster. It works by randomly assigning initial points as the centers of the clusters and then reassigning each point to the nearest cluster center, updating the cluster centers after each assignment. This requires setting the number of Clusters in advance, and their optimal value can be calculated using the Elbow Method [Saj].

The elbow method is shown in Figure 4.10. It displays the sum of squared distances within clusters (WCSS) on the vertical axis for various K values on the horizontal axis. The ideal K is identified at the point where the graph forms an elbow. For example in this figure, the optimum number of clusters for the average Couinaud, the breathing

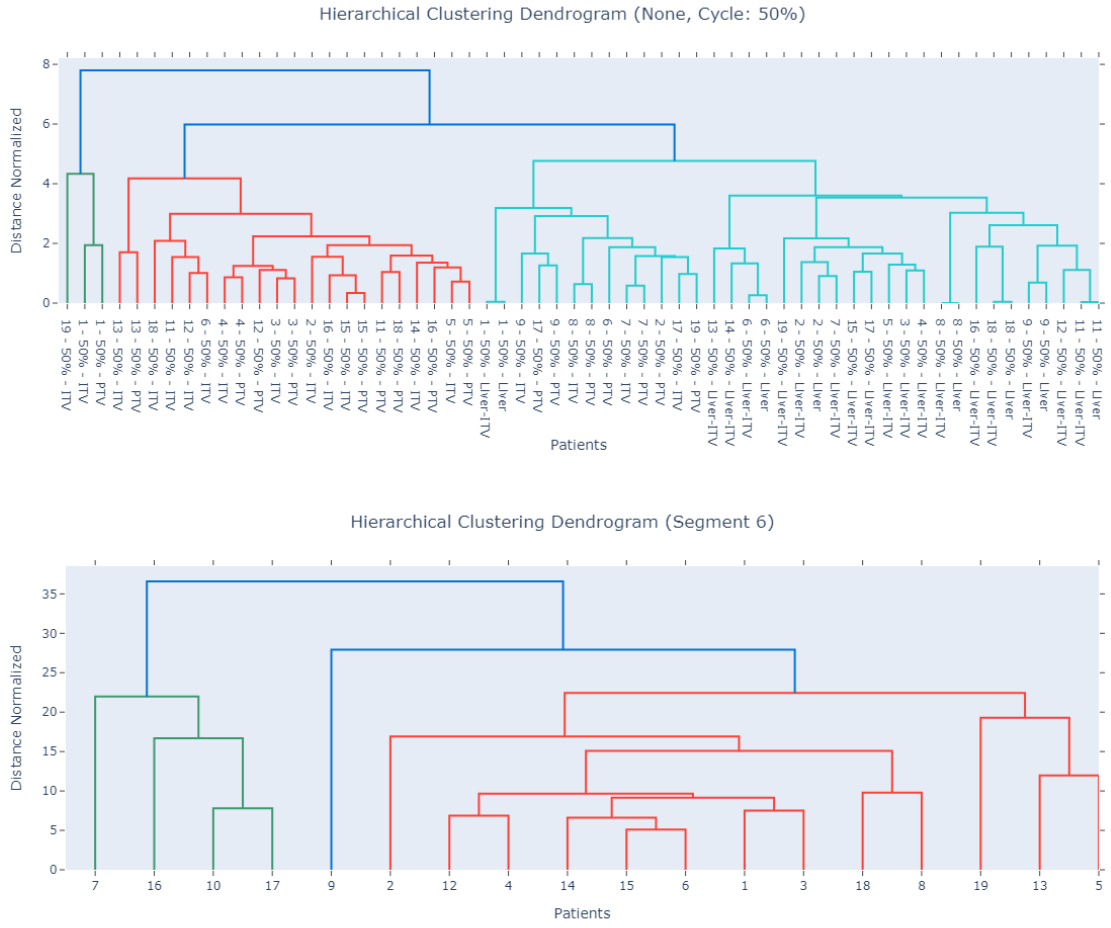


Figure 4.9: Comparison between Hierarchical Clustering for Breathing Cycle, and Average with Couinaud Segments.

cycle, and the breathing cycle Couinaud are 2, while the average is more difficult to establish.

Spectral Clustering [Lux04], unlike K-Means, does not assume a priori a particular shape for the distribution of points in each cluster, therefore, it is a method especially useful for identifying clusters that have an irregular shape. Spectral Clustering transforms the data space into a lower-dimensional space where it is easier to separate the clusters, so it can be combined with other Clustering techniques.

Lastly, Mean Shift [CM02] is a clustering algorithm that also does not require specifying the number of clusters in advance. It works by finding cluster centers as the points for which the "mean shift" of points within a local window converges to zero. That is,

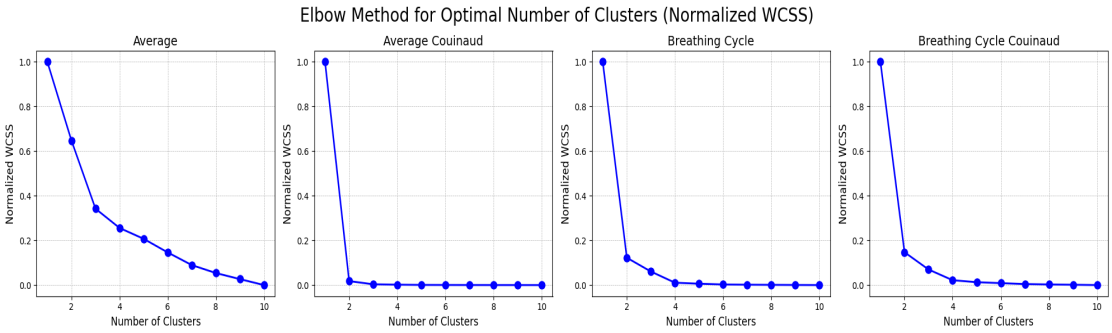


Figure 4.10: Elbow Method for the optimal number of Clusters (K).

it iteratively updates the location of the cluster centers to the region of the highest density of points, similarly to what K-Means does. Mean Shift is particularly useful for identifying clusters of arbitrary shapes and variable sizes.

We are dealing with different methods and potential parameter combinations, this makes the optimal parameters for generating the "best" cluster setup prove challenging and typically involves a "trial and error" approach performed by the researcher [CAC10]. To evaluate the quality of the Clustering, three measures will be taken as reference: the *Silhouette Score*, *Davies-Bouldin Score*, and *Inertia*.

The Silhouette Score [SK21] measures how well each point is assigned to its cluster by calculating the difference between its similarity (or closeness) to its cluster and its similarity to the closest cluster to which it does not belong. On the other hand, Davies-Bouldin index (dbi) [ITS+22] measures the quality of the separation between clusters by calculating the ratio between the average distance of all points in the cluster to its center and the distance between this center and the center of the nearest cluster.

Inertia [BC09], previously mentioned in the elbow method as WCSS, measures how well a dataset has been clustered by calculating the sum of the squared distances of each point in the dataset to the nearest center of the cluster to which it belongs. In other words, it tries to measure how much the clusters extend around their centers.

It is important to note that clustering is better with a higher value of Silhouette Score, and a lower value of Inertia and Davies-Bouldin. It is also worth mentioning that for the Mean-Shift and Spectral Clustering techniques, it is not possible to calculate inertia since, as explained, the number of clusters is not pre-established. An example of this clustering is observed in Figure 4.11, where a comparison is made between the use of two different dimensionality reduction methods and the type of parameters selected for them, but with the optimal number of Clusters in both cases:

CASE 1	CASE 2
1. **Average**.	1. **Breathing Cycle Couinaud:** All.
2. **UMAP and K-Means**.	2. **PCA and K-Means**.
3. **Structures:** Liver-ITV.	3. **Structures:** All.
4. **Scaler:** MinMaxScaler.	4. **Scaler:** RobustScaler.
5. **Clusters:** 5.	5. **Clusters:** 2.
6. **Outliers:** No removed.	6. **Outliers:** Local and Global outliers removed.

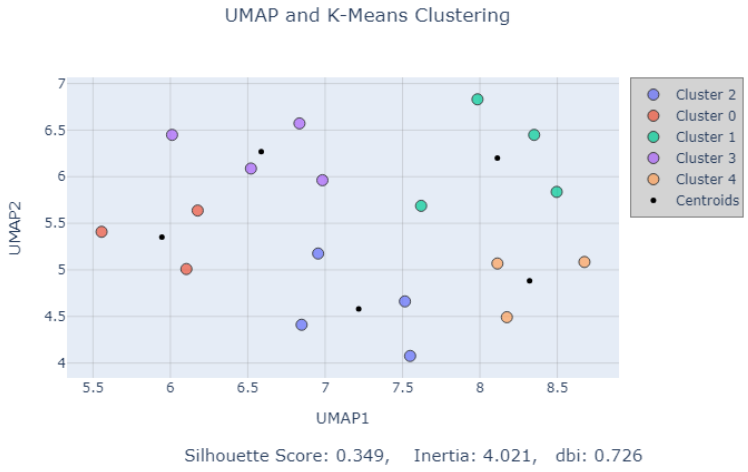
It is necessary to highlight that for case (a), UMAP is used as a dimensionality reduction technique, and K-Means as a Clustering algorithm. The value of the Silhouette Score gives us information that the data are well associated with each cluster, although considering that this metric varies between -1 and 1, the data-cluster similarity is not entirely precise. In addition, it can be observed that the clusters are quite extended from their centroids, as measured by the value of Inertia or the dbi.

In case (b), PCA and K-Means are used, obtaining a very good Silhouette Score value, but a very high Inertia value. This indicates that the data are clustered accurately, but there are some data very distant from their own centroids. Likewise, the dbi indicates that the two clusters are well separated from each other. Note that for this case, filtering by structure or respiratory cycle has not been performed, therefore all data are being handled simultaneously. Moreover, the RobustScaler has been used, and local and global outliers have been removed to slightly increase the precision of the metrics.

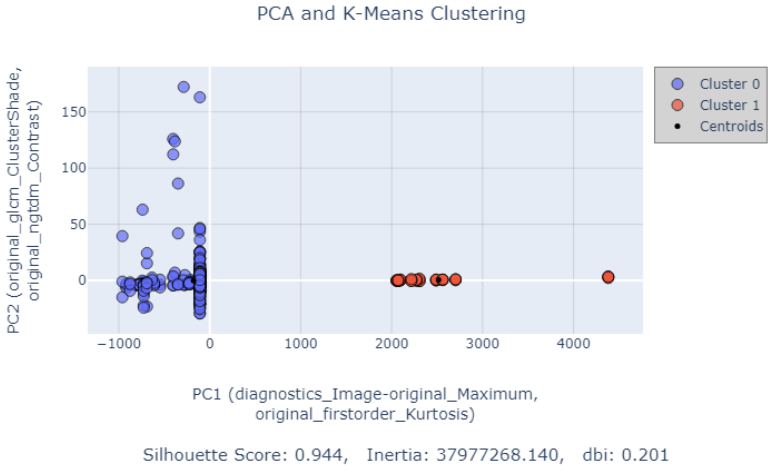
4.3.3 Other Dimensionality Reduction methods

Next, an attempt will be made to perform Dimensionality Reduction in two steps, one for intraclass and another for interclass. Reiter et al. [RBGR18], propose to reduce the high-dimensional space by using two consecutive Dimensionality Reduction techniques. By integrating PCA and t-SNE, the authors can combine the strengths of both techniques. PCA is effective for analyzing variance within organ classes, while t-SNE works better in distinguishing between classes.

Figure 4.12 shows a workflow of how this type of technique works. Initially, there is no filtering by structure or respiratory cycle; instead, all data are pooled together for the first reduction (t-SNE). With this reduction, one of the classes is selected, returning to the main dimensional space, and performing another reduction (PCA), and finally the visualization of the data. Since in previous sections the representation was in 2D, and more than two components can be used for dimensionality reduction, as is the case with 3D representations, these have been used for visualization.



(a) UMAP and K-Means



(b) PCA and K-Means

Figure 4.11: Dimensionality Reduction and Clustering for Average and Breathing Cycle Couinaud datasets.

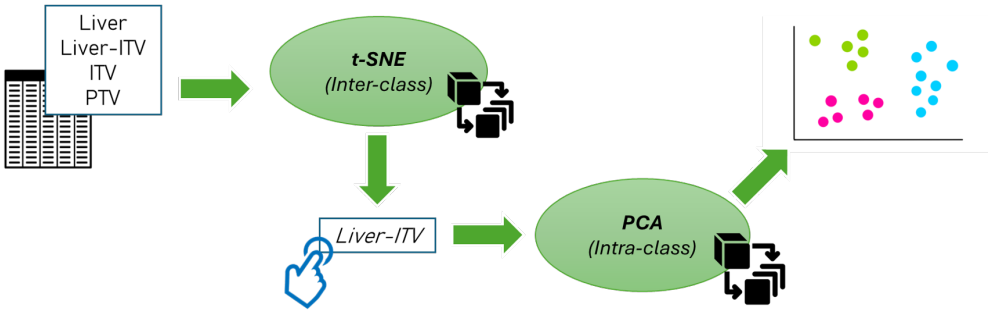


Figure 4.12: Pipeline of combined dimensionality reduction techniques.



(a) Reduction for average data frame

(b) Reduction for B.C. Cou. data frame

Figure 4.13: Comparison between two examples of combined dimensionality reduction, for average data frame (a) and breathing cycle Couinaud data frame (b).

In Figure 4.13, a comparative example of two Dimensionality Reduction technique combinations is shown. In this case, Figure (a) shows an example of how the technique is applied to the average data frame, after having used t-SNE to obtain the Liver-ITV class. On the other hand, Figure (b) shows exactly the same method but for the breathing cycle Couinaud data frame, filtering as the structure Segment 1. This figure also highlights the presence of outliers, as can be observed at the extremes of the three dimensions, for the recognition of these outliers, all are labeled and will be shown in the tool design chapter.

4.4 Comparative Analysis

To perform the comparison between our various types of data, both for the mean and for the breathing cycle, we carry out the different types of stratification described previously and display them on two panels simultaneously through the LiverVis Dashboard. This setup allows us to study the results and compare them based on the chosen modality type and desired parameters. However, to conduct the Radiation Analysis (T4) and

identify differences among the population (R2), we will use various statistical tests, which will show us whether there are significant differences between groups (HCC/MET or TOX/NT) in terms of their radiomic features or doses.

Considering that our data, both radiomic and dose, follow a non-normal distribution, and given that we are always working with a variety of structures or breathing cycle-structure combinations, we cannot use contrast tests such as ANOVA or t-test. Instead, we must apply their non-parametric equivalents such as the Multivariate Kruskal-Wallis (MKW), Mann-Whitney U (MW), and Kolmogorov-Smirnov (K-S) tests.

The **Multivariate Kruskal-Wallis test** [FHI17] is a non-parametric statistical method that compares multiple groups across various outcome variables simultaneously, without the assumption of normally distributed data. This test ranks the data within each group, calculates a test statistic from these ranks, and evaluates hypotheses to detect significant differences among the groups. A p-value below a specified threshold (typically 0.05) indicates significant differences between the groups. Likewise, the **Mann-Whitney U test** [uzh] is a test used for comparing two groups that do not have normal distributions, can be used for ordinal data, and also when one of the two groups is larger than the other, as in our case (12 patients without toxicity and 7 with toxicity). Lastly, the **Kolmogorov-Smirnov test** [Asl20] is a non-parametric test used to compare cumulative distribution functions of two datasets or to assess if a sample matches a specific distribution.

Despite conducting these three tests simultaneously to find truly significant differences between groups, in the case of our radiomic data, we may encounter a high rate of false positives. Therefore, we will apply the **Bonferroni correction** [VOA19], which is a method used to adjust for multiple comparisons in statistical hypothesis testing. The correction is used to adjust the significance level (alpha) by dividing it by the number of comparisons made. This helps keep the overall chance of a Type I error (false significance) under control when conducting multiple tests at the same time. This method reduces the risk of incorrectly rejecting the null hypothesis, thus keeping the error rate within acceptable bounds across all tests.

4.5 Predictive Analysis

In many articles that work with radiomic data, models are constructed for the prediction or classification of certain variables. For example, Shen et al. [SHD+22] attempt to predict Radiation-induced liver disease (RILD) in patients who have undergone SBRT, using models such as random forest (RF) or logistic regression (LR), and demonstrating that higher accuracies can be achieved by using radiomic features instead of ignoring them. On the other hand, Tang et al. [TWX+23] utilize radiomic characteristics to predict steatosis in liver cancer patients, through models based on the support vector machine (SVM), obtaining AUC of 0.8 and 0.9 when using radiomic data and when combining radiomic and clinical data, respectively. Finally, Shur et al. [SDK+21] in their guide on the use of radiomics in Oncology, discusses the use of models such as Gaussian

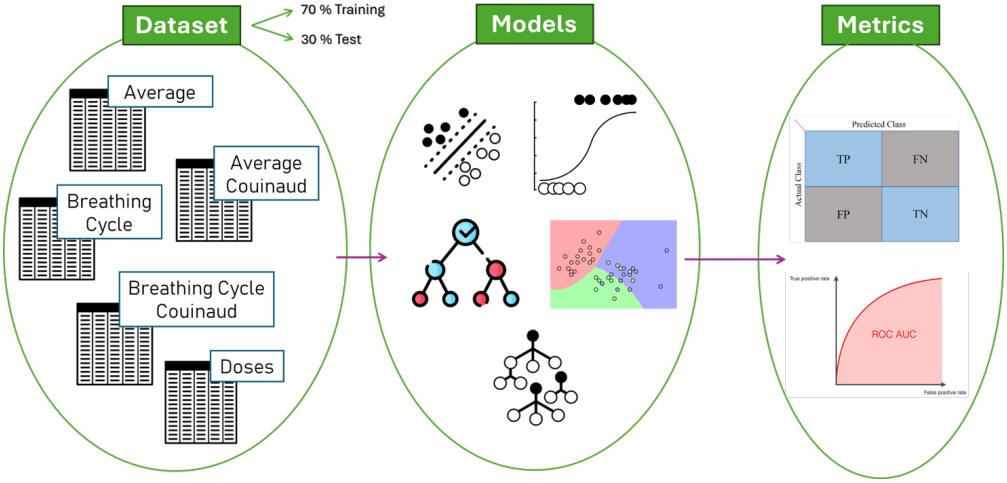


Figure 4.14: Predictive analysis workflow: training datasets, ML models and evaluation.

Naive Bayes, logistic regression, random forest, and SVM, explaining their functionality, advantages, and disadvantages.

For our analysis, we will try to make a prediction about our patients, based on radiomic features and the data we have obtained about the doses. The response variable or characteristic to predict will be both the toxicity and the type of cancer of our patients (Hepatocellular carcinoma or Metastatic). Figure 4.14 shows a workflow of the procedure to follow, where five separate dataframes are used (average, average Couinaud, breathing cycle, breathing cycle Couinaud, doses), employing several of the models mentioned previously and subsequently measuring them through a confusion matrix and AUC. Additionally, a comparison will be made between using RobustScaler or MinMaxScaler, whether to use all radiomic features or to perform prior dimensionality reduction, evaluating in such comparisons the accuracy and training times.

In each prediction process, the corresponding Dataset is divided into 70-30% for training and testing, respectively. We will employ all the models mentioned above: *Decision Tree*, *Random Forest*, *SVM*, *Gaussian Naive Bayes*, and *Logistic Regression* since they are mentioned in numerous articles dealing with radiomic data. Additionally, GridSearch [Jos] is used, which builds a model for every combination of hyperparameters specified and evaluates each model, selecting the best one.

A *Decision Tree* [May16] is a predictive model that uses a tree structure to represent a series of decisions and their possible consequences, including outcomes, costs, and utility. It starts with a root node and splits into interior nodes or decision nodes, which represent tests on some attribute, and leaf nodes, which represent the value of the prediction or classification. Assembling several decision trees, we get the *Random Forest* [Bre01] model, which operates from randomly selected subsets of the training set, and the decisions from all the trees are averaged to get the final prediction.

The *Naive Bayes* classifier [Ris01] is a probabilistic model based on Bayes' theorem along with a strong assumption of independence among the features. Bayes' theorem here is used to compute the posterior probability of a data point belonging to a class, given its features. The model's "naivety" comes from assuming that all features are independent of each other, given the class variable value. The *Support Vector Machine (SVM)* [May16] aims to find the hyperplane that best separates the classes in the feature space, transforming the data to a higher dimension space where it can be linearly separated.

Finally, *Logistic Regression* [Har21] models the probability of an input belonging to one of two categories, as it is a binary classification model. The decision function is the logistic function, which maps any real input to a value between 0 and 1, interpreted as the probability of belonging to one of the classes.

To evaluate the quality of our models, we will measure their *accuracy*, *sensitivity*, and *specificity*, calculated as detailed in equations 4.5 and 4.6. These last two parameters are often represented through a confusion matrix and ROC Curve.

$$Sensitivity = \frac{TP}{TP + FN} \qquad (4.5) \qquad\qquad Specificity = \frac{TN}{TN + FP} \qquad (4.6)$$

where TP and FP are the number of True Positive and True Negative predictions respectively, while TN and FN are True Negative and False Negative predictions.

The confusion matrix [ED23] organizes the predictions of a classification model by comparing them with the actual values, providing a clear view of the number of correct and incorrect predictions. The columns of the matrix represent the predictions made by the classifier, while the rows show the actual instances. One of the main benefits of the confusion matrix is that it makes it easier to identify if the system is confusing two specific classes. The ROC curve [ED23] displays the performance of our models by representing sensitivity (true positive rate) versus specificity (1 - false positive rate) at different discrimination thresholds.

4.6 Implementation

In this section, a general explanation will be provided regarding the software used for the development of this thesis. As has been discussed throughout the methodology, two stages can be distinguished: one for the visualization and understanding of our structures before the extraction of radiomic data, and another for their acquisition and analysis in all forms. For the visualization of our structures, as well as the Couinaud segmentation, 3D Slicer has been used, while for the rest, the entire implementation is done in Python. Figure 4.17 shows a schema of the implementation.

4.6.1 3D Slicer

3D Slicer [sli] is a free, open-source software application designed for the visualization and analysis of medical image computing datasets. It supports a wide range of data

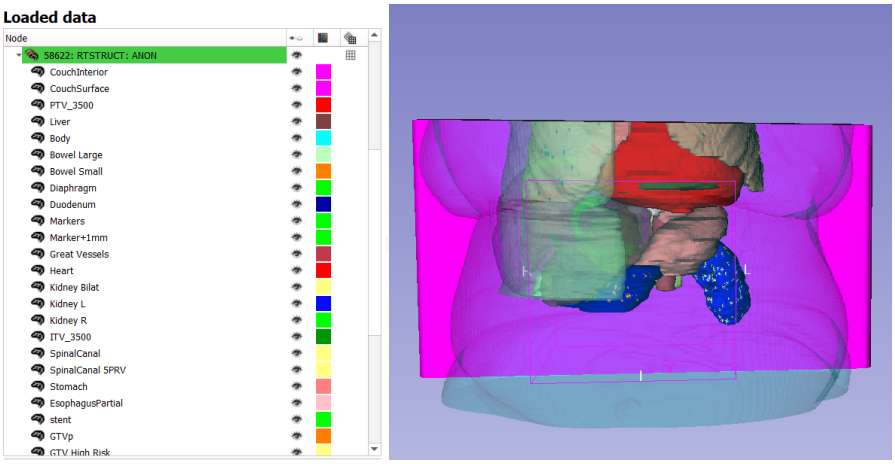


Figure 4.15: Visualization of DICOM RT Structures in 3D Slicer, example with one patient.

types, including images, segmentations, surfaces, annotations, and transformations across 2D, 3D, and 4D formats.

It works as a research platform that allows for fast development and testing of new methods. All its features can be extended in Python and C++. 3D Slicer offers a complete Python environment, works with Python packages, and has a Python console. It can also work as a Jupyter Notebook kernel, offering the ability to render 3D images remotely.

Another important feature of 3D Slicer for our thesis is its support for multi-organ and multi-modality imaging, real-time medical device interfaces, high extensibility through modules, scripts, and custom development, and a large, active user community. Specifically, we have made use of application modules for the context of radiotherapy, which allows working with DICOM RT files, enabling the visualization of structures, images of the breathing cycle, and the average, as well as the doses.

Figure 4.15 shows an example of visualization of all structures, from which we can extract those of interest and export them as files in various formats, such as .nrrd. Moreover, not only can we visualize radiotherapy doses with these plugins, but we can also obtain their measurements in the structures of interest, and export these data in .csv files.

Other specific plugins related to the liver are available, such as blood vessel detection or automatic segmentations of the entire organ. However, for the Couinaud segmentation, the tools provided by 3D Slicer were used, which flexibly allowed the liver to be sectioned into the eight segments of interest, making regular or irregular cuts as the case may be. In this way, these segmentations are exported in .nrrd files as with the previous structures of interest. The segments have been exported separately as individual and unique structures, and the entire liver with the segmentations performed on it, in such a way that we can work on the entire liver or the separate structures.

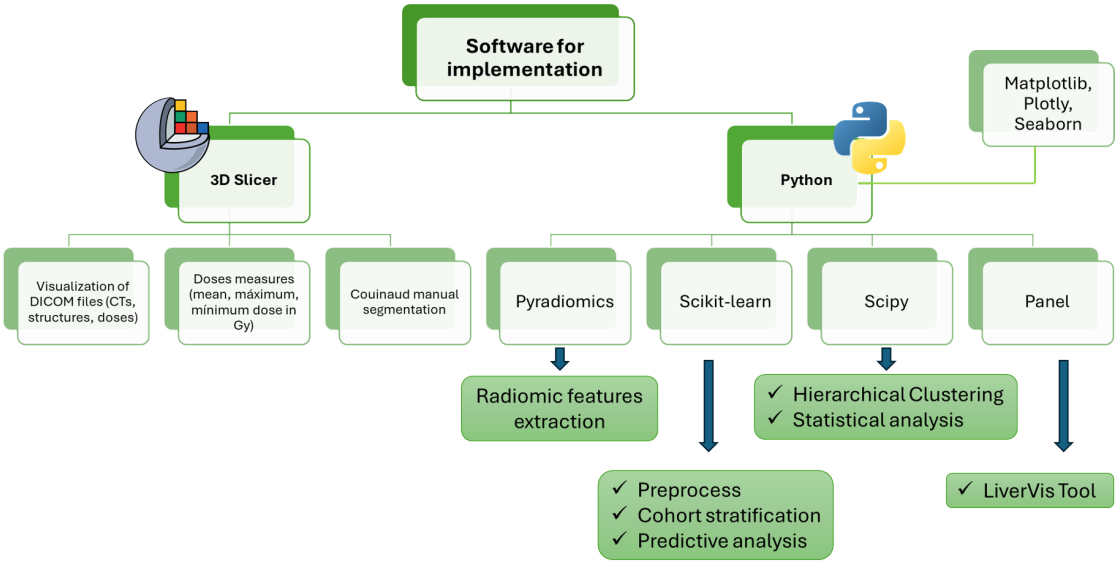


Figure 4.16: Software used for implementation, 3D Slicer and Python.

4.6.2 Python

With the structures exported and our DICOM files with images of the average and the breathing cycle, the next step is to obtain the corresponding radiomic data. The work of Shen et al. [SHD+22] the use of 3D Slicer extension for radiomics extraction is discussed, where the tumor segmentation is also performed, on the other hand, Huynh et al. [HCN+16] combines a MATLAB Toolbox with the same 3D Slicer plugin. 3D Slicer is indeed among the most used methods due to its simplicity and interactivity on the frontend. However, its backend operates thanks to the Python library PyRadiomics [pyr].

Van Griethuysen et al. [vGFP+17] highlight the ease and efficiency of working with this library, given the flexibility for processing or choosing features. Similarly, Sack et al. [SNM+22], Tang et al. [TWX+23], and Wu et al. [WLW+24] utilize PyRadiomics in their specific case of liver image analysis for subsequent statistical analysis and model building in Python. This is motivation enough for it to be the chosen option when working with radiomics.

Pyradiomics [pyr] is a Python-based open-source package designed for extracting Radiomics features from medical imaging, to set a standard for radiomics analysis. It requires an image and a mask to work, accepting a wide range of file types. We have DICOM files for the average and breathing cycle images and .nrrd files for the masks. From both, the radiomic features of the image defined by the mask are extracted and ultimately exported into a *Pandas* dataframe. By automating the extraction process for all our patients and structures, the four resulting dataframes are obtained, to which we have added the clinical data of interest (also using Pandas).

For preprocessing our dataframes, we will primarily use Scikit-learn [scia], an open-source

machine learning library for data preprocessing, supervised and unsupervised learning, and many other utilities used in this thesis. With this library, algorithms for outlier detection techniques such as LOF or IF can be used, our data can be scaled with MinMaxScaler or RobustScaler, and all dimensionality reduction techniques (excluding UMAP, for which its own library is used) can be applied.

For the stratification of the cohort, since we use algorithms for dimensionality reduction and clustering, this is also carried out with Scikit-learn, which additionally includes evaluation metrics such as the Silhouette Score or Inertia. The only exception is Hierarchical Clustering, performed through the Scipy library [scib], which provides algorithms for optimization, algebraic equations, differential equations, statistics, and many other classes of problems. With this library, normality tests and statistical tests like the Mann-Whitney are also conducted, which are necessary for identifying differences in our groups.

The tasks of predicting toxicity are also performed with Scikit-learn, since all the models, the GridSearch for hyperparameter optimization, and metrics to evaluate the accuracy or the confusion matrix are included in this library.

For data visualization and various plots, libraries such as *Seaborn*, *Matplotlib*, or *Plotly* are used. As support libraries for any of the processes mentioned above, *numpy*, *pandas*, *os*, and *counter* among others are used.

Panel

With Python being one of the most widely used programming languages for data analysis and Machine Learning, it has been attempted to use this as the main backend of the Dashboard to be designed. Specifically, the *Panel* [hol] library has been used, which is an open-source library that allows the design of custom interactive web apps by connecting the widgets we decide to different plots, tables, images, or text.

Panel belongs to the HoloViz ecosystem, and is based on Bokeh, a Python library for creating JavaScript-powered visualizations, without writing JavaScript ourselves. Panel has been chosen as the option for designing our tool since it makes use of all the common Python libraries for data analysis work, such as Plotly, Pandas, and Matplotlib among others, therefore it is possible to work with the entire volume of Python's own libraries and through functions and widgets establish interactions with the user. Additionally, some platforms such as HuggingFace or GithubPages allow for the deployment of applications for free based on Panel.

Lastly, Panel app code is very light and high-level, expressing complex functionalities with few lines of modular code, and making it possible to add, rearrange, and test components easily. A native JS/HTML/CSS application of similar complexity at the user level requires more complex code, which makes it very difficult to adapt to high-level needs.

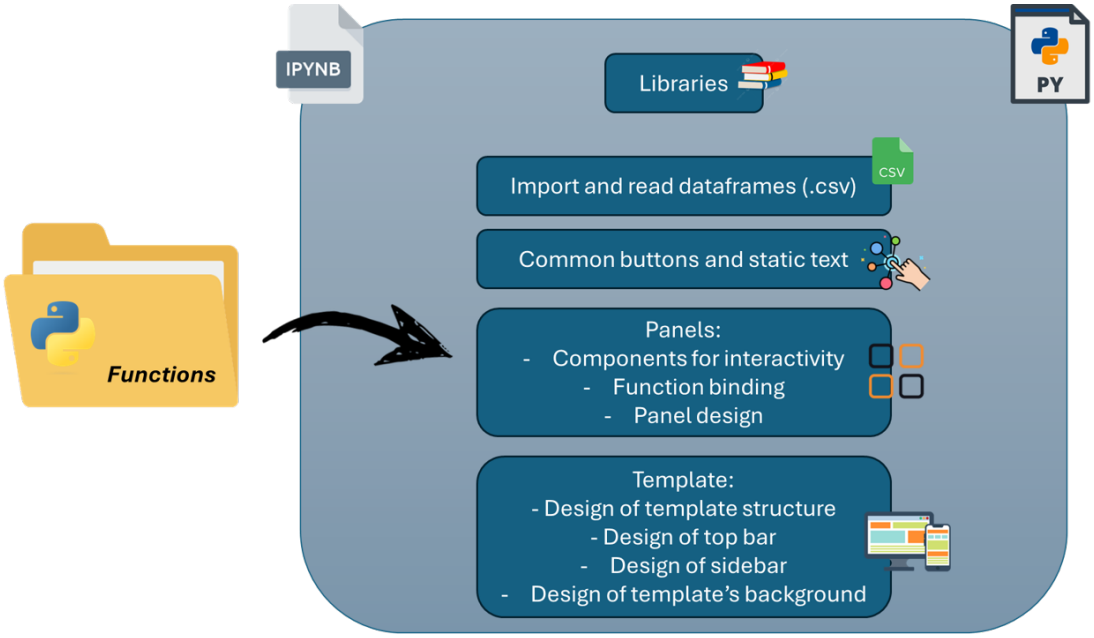


Figure 4.17: Pipeline of how to structure Panel's coding.

An example of the code structure is shown in Figure 4.17, where a folder with our created functions in .py (for example, functions for patient clustering, violin diagrams, etc.) is displayed, along with a Python file that acts as the base or "main", which is executed to launch the application. This last Python file can be a .py as well, or a .ipynb, and consists of the different stages mentioned in the Figure 4.17 scheme:

- Importing the necessary libraries, such as Panel itself or auxiliary libraries like numpy or pandas. Also, call the functions we have in the folder.
- We import the five dataframes using pandas, to work with them with the panels.
- We set some of the components that are common throughout the tool, such as the Floatpanel.
- We start to define each of the panels one by one, for which we must introduce their own interactivity components, and link them to the previously imported functions. We also design the aesthetics of the panel itself, where the buttons are located, etc.
- We establish the general structure of the application, that is, the appearance of the side and top bars, background colors, order of the panels in tabs, etc.

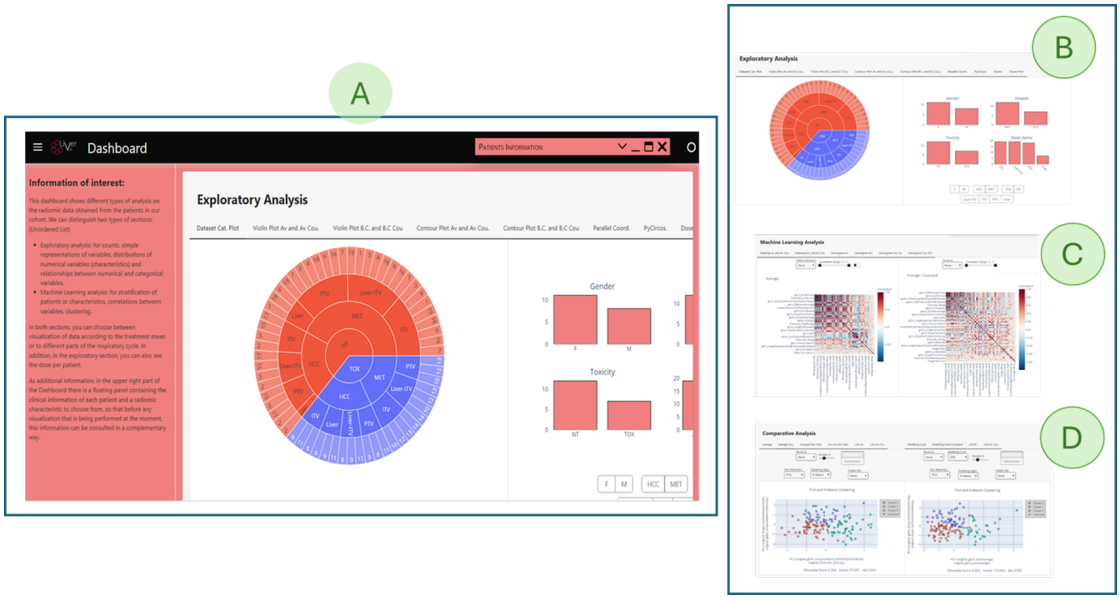


Figure 4.18: General overview of the LiverVis Tool: Main screen (A), exploratory analysis (B), machine learning analysis (C), comparative analysis (D).

4.7 Design of the LiverVis Dashboard

Given that the main goal of this thesis is the design of a tool that allows clinicians and researchers to flexibly explore and analyze the provided radiomic and clinical data, we need to find a work environment that meets the needs and requirements for its study.

4.7.1 Desing Criteria

To be able to design a Visual Analytics tool, it is essential to consider some aspects that ensure effectiveness and usability. The LiverVis Tool has been designed with a focus on the end-users, and understanding their needs and tasks. Also, as Keim et al. [KAF+08b] define, a VA tool should represent data quality, and scalability and provide visualizations that offer both detailed analysis and a global overview.

The general structure of the dashboard is shown in Figure 4.18. The tool must address the needs raised for solving tasks such as the comparative analysis of our dataframes, or the stratification of patients. In this way, the tool has a section for the exploratory analysis of our dataframes based on radiomic and clinical characteristics **(B)**, a Machine Learning part that includes correlations between features, and hierarchical clustering of patients **(C)**, and finally a more complex unsupervised Machine Learning-based stratification **(D)**. Each of the sections is designed with the necessary interactivity so that users can test different combinations of data and methods. The main panel of the LiverVis Tool is shown in **(A)**, where we can see that is provided with a collapsible sidebar, a float panel, and a busy indicator.



(a) Logo of the LiverVis Tool (b) Favicon

Figure 4.19: Desing of Logo and Favicon for the LiverVis Tool, based on Hepatocytes shape.

4.7.2 (A) Main Screen and supplementary tools

Based on the concept of the liver as the main subject of the Thesis, a logo, and a favicon have been designed as shown in Figure 4.19. Both maintain the structure of hepatocytes, with their characteristic hexagonal shape (see Figure 2.1 (a)), and a soft pink color is used, also present throughout the tool.

The tool has a collapsible Sidebar, where each section is detailed with information, and some additional features, utilities, or tips are mentioned. In the top black bar, there is a small FloatPanel, which allows at all times the consultation of specific data of the patients from our Dataset, such as values of their radiomic and clinical data, in a way that can help us at any stage of the study. Lastly, in the top right corner, a Busy Indicator is included, which serves as a guide especially when interacting in section (D), where changing Clustering methods or Dimensionality Reduction can be a computationally costly process.

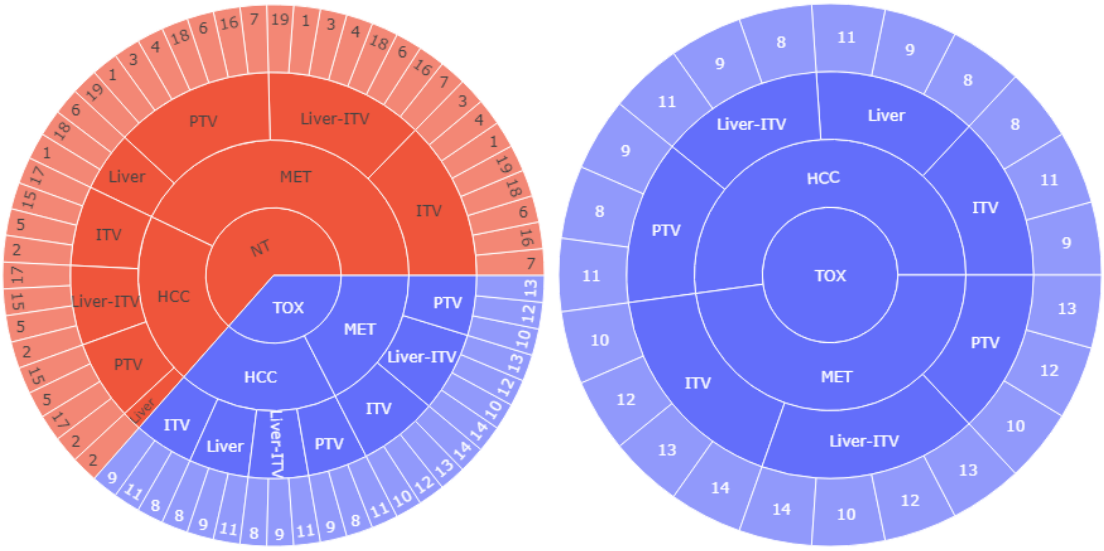
4.7.3 (B) Exploration of the Dataset

As previously described, the first section of our Dashboard focuses on the exploratory analysis of our data. For this purpose, we have used Sunbursts, simple Bar and Line charts, Violin Plots, and Contour Diagrams, among others. To explain this section of the tool, we will divide it into three subsections: one for the visualization of radiomic data, another one for clinical data, and a last for radiation therapy doses.

Clinical data

Firstly, we encounter the distribution of our dataset concerning the clinical data available to us: toxicity, cancer type, patient gender, their identification, and how many structures (Liver, ITV, etc.) we have from them.

The **sunburst plot** [Rus16] allows for immediate and interactive visualization of these data, splitting the circle into two major sections, those with toxicity and those without



(a) Closed Sunburst plot

(b) Open Sunburst plot

Figure 4.20: Comparison between Sunburst plot of the whole dataset (a), and classified by toxicity (b).

since this classification is crucial for our patients. Subsequently, the branch bifurcates into patients with hepatocellular carcinoma and metastasis, continuing with their classification in the structures each group possesses, and finally, the patients belonging to each group and subgroup. In short, what we have is a tree of clinical features based on the groups of toxicity and non-toxicity, where if we click on any of the branches, the tree narrows down only to that section we have pressed. This way, it is possible to observe the relationships between our different variables, and also provide a total count of how many structures we have available for each.

For example, Figure 4.20 shows a comparison between the sunburst plot in its two states. Figure 4.20 (b) splits the dataset by patients experiencing toxicity, revealing that the group includes 12 structures from patients with hepatocellular carcinoma and 11 from metastatic patients. Notably, the latter group does not have any *Liver* structure. Furthermore, within the entire group, there are only 4 unique patients.

To complement this chart and with the aim of providing a more direct visualization of these clinical data, a **bar chart** with patient counts from our dataset is shown in Figure 4.21, classified in the same manner as with the sunburst. At the bottom of the panel, different buttons are available that allow for combinations in filtering, such that if we want to know how many patients are male, and metastatic, all charts will rearrange to provide us with the corresponding counts regarding these two characteristics, how many present toxicity/no toxicity, and how many structures we have of them.

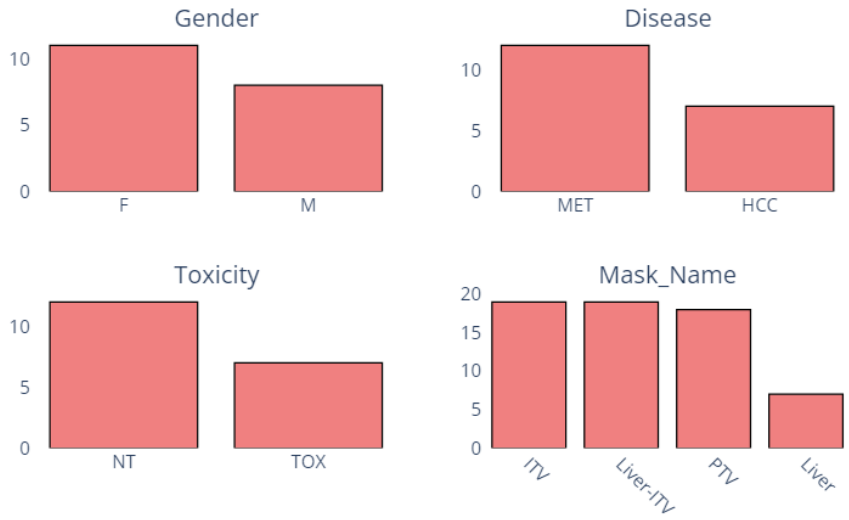


Figure 4.21: Number of patients according to clinical data (gender, disease, and toxicity) and number of available structures in the dataset.

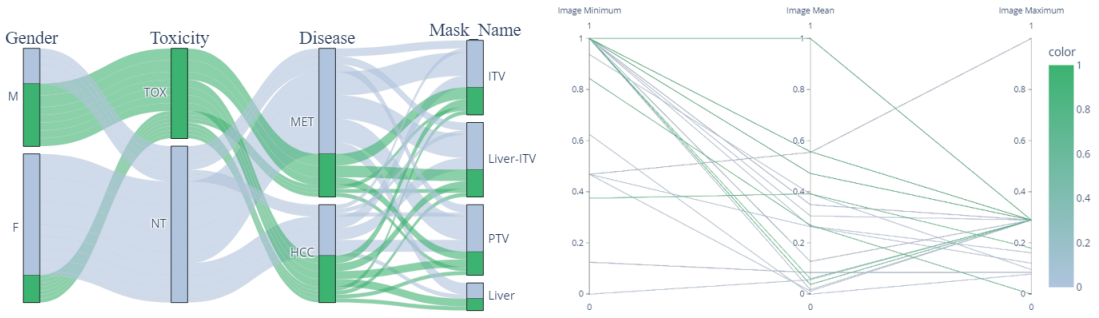
To conclude with the exploration of clinical data, a section based on the use of parallel sets and parallel coordinates is designed. **Paralell sets** [BR12] are sections that allow the study of relationships between variables, displaying our four variables of interest as columns and the relationships between them according to the classification of interest.

Figure 4.22 (a) shows how having selected the patients with toxicity, the relationships with the other columns are highlighted, directly observing that a large part of the patients with toxicity are male, with hepatocellular carcinoma, and that we have structures of all types from them. This type of chart allows the swapping of column positions for visual convenience, and moreover, hovering over them allows observing the probability between two variables, for example, that a patient has developed toxicity and is male.

Radiomic data

This type of parallel chart is not only useful for categorical variables but also for numerical variables, called **parallel Coordinates** [JWF15]. Thus, we move on to study the radiomic variables we have extracted. By clicking on the "switch" button in the same previous section, the representation changes from categorical variables (Clinical Data) to numerical (Radiomic Features). Similarly, the relationships between numerical variables are established, observing the differences by the classification method we have chosen. Both the type of classification and the radiomic features we wish to observe can be selected.

Figure 4.22 (b) shows how the average of the image, minimum, and maximum of the structures (each line represents a structure) vary, and additionally, the color indicates which of them have Hepatocellular Carcinoma and which are metastatic patients. These



(a) Categorical Coordinates (b) Parallel Coordinates

Figure 4.22: Parallel charts: categorical coordinates (a) and parallel coordinates (b).

characteristics are scaled between 0 and 1 in absolute value, and by selecting with the mouse on the columns, one can observe the path of one or more lines along the variables, studying their variability in the radiomic characteristics if we are interested in some specific patients/structures.

Violin and contour diagrams [HN98, Max14] are also available. These diagrams allow for simultaneous representation (similarly to the coordinates plot) of the distribution and value of a radiomic variable, classified by Gender/Toxicity/Cancer Type. In Figure 4.23 the values of the *shape elongation* are simultaneously represented for patients with toxicity and those without toxicity, for both *Liver-ITV* and *ITV* structures, belonging to the dataframe of the average, and exactly the same but with Couinaud segments 3, 5, and 7.

Having both representations allows us to observe how the value of this magnitude varies for each structure in the study and the differences that may exist between groups. In addition, another panel identical to the previous one is added but for the case of the breathing cycle, where, in addition to being able to select structure and classification, a specific stage of the breathing cycle can be selected. Among the advantages offered by a violin plot over a normal scatter plot is that by hovering over the distribution representation, one can observe typical statistical values such as mean, median, and quartiles, among others.

This same procedure is repeated in an additional panel, but with a **contour plot**, which allows analyzing the distributions or joint probability density of two continuous variables. Unlike the previous case, here we must select two radiomic variables of interest. For the violin plot and contour plot, we can consider the differences that choosing one scaler over another implies.

Krzywinski et al. [KSB+09] designed a visualization tool called Circos, to facilitate the identification and analysis of similarities and differences between genomes. This tool has been used for the visualization of biological data in many studies, although it does not provide interactivity with the user. This particularity has been remedied by authors

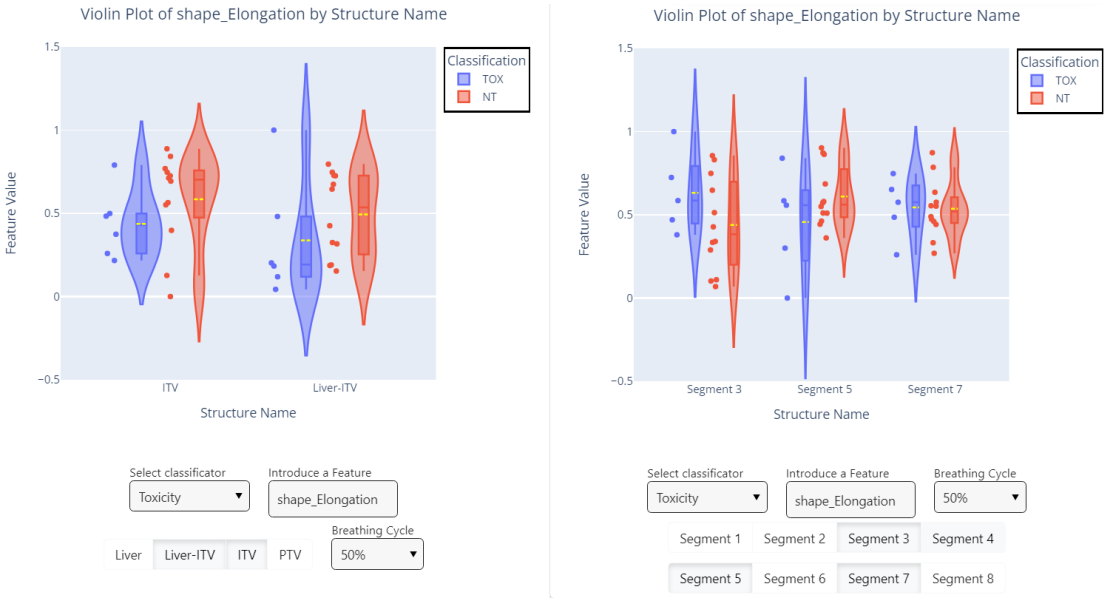


Figure 4.23: Violin plot for *shape elongation* feature, patients classified by toxicity and specific structures.

like Cui et al. [CCX+20], and in our case, we will also provide interactivity in the data handling.

By using the Python library PyCircos, we are able to display our radiomic features in several of the patients, and from our four dataframes simultaneously, as observed in Figure 4.24. In this Figure, the values of five radiomic features are represented, in each of the dataframes and for the first six patients. In this way, it is possible to study the variations of these values for each patient, in the structures, and in the different breathing cycles. Furthermore, the second plot of the figure represents the relationships that exist between those patients and their respective radiomic features through lines, where thinner lines represent smaller values and thicker lines, larger values, allowing a greater understanding.

Doses

Given that with both the Python library Scikit-RT and Slicer-3D we can obtain some values of the applied radiation dose, as well as the DVHs and the treatment images themselves, they are made available in the dashboard in an exploratory and informative manner. In Figure 4.25 the CTs of our patients that have been used as a reference in the radiation therapy dose are shown in cross-sectional view the image on the left represents the area where the dose is applied, and in the image on the right, the values in Gy of said dose with its legend to the right. Through the upper component, we can move forward or backward in the management of the Slices, and also select the patient we want to study.

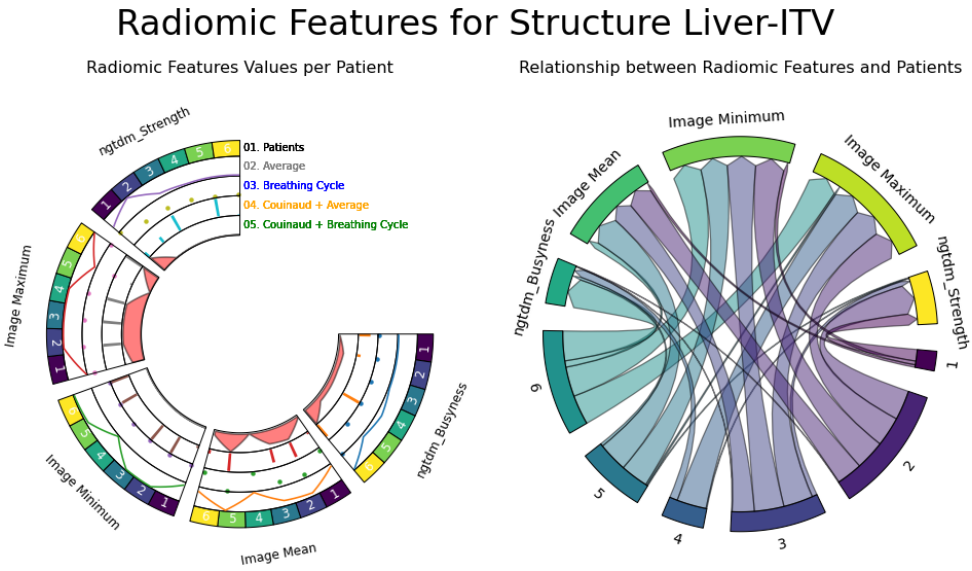


Figure 4.24: Circos Plot, values of 5 radiomic features for six patients.

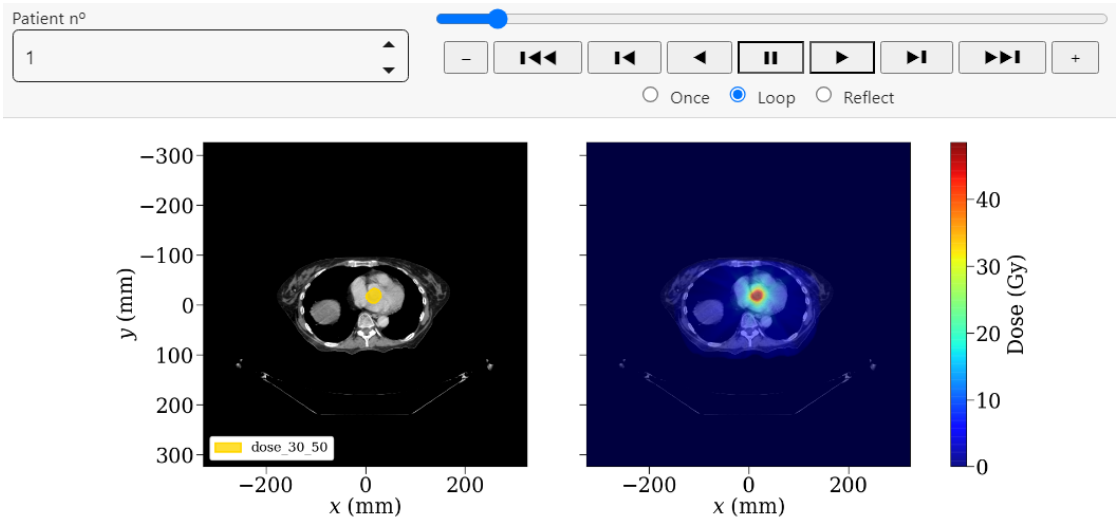


Figure 4.25: Visualization of the doses across CT images, interactive selection of patients and slices.

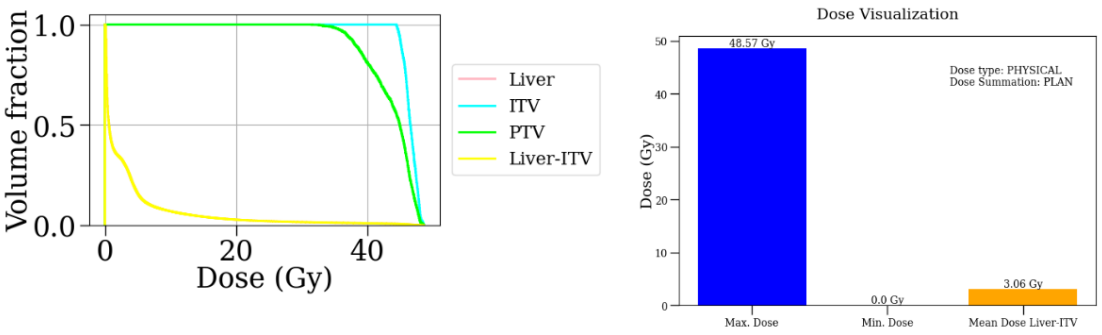


Figure 4.26: Visualization of the DVHs of patient's structures and measures of the Liver-ITV doses.

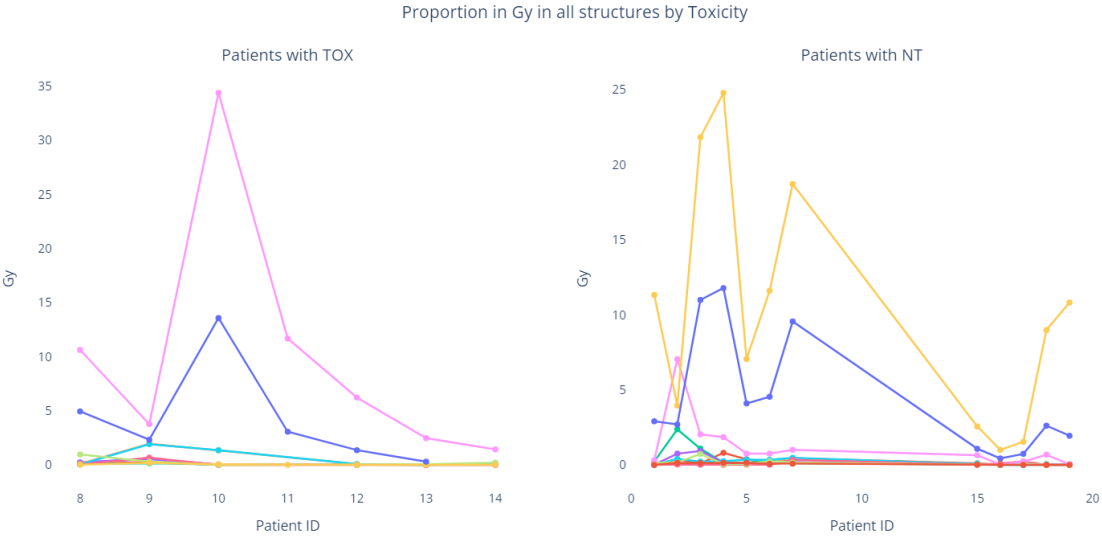


Figure 4.27: Visualization of the variable 'Proportion', patients classified by toxicity (left) and non-toxicity (right).

Below this representation, the DVHs of the patient's structures of interest are observed in Figure 4.26, as well as the values of the maximum, average, and minimum doses during the treatment. Finally, the image on the right shows how the average and maximum dose values vary according to toxicity and cancer type.

If a more exhaustive analysis of the dose values is desired, a line diagram is provided where one can see the variation of the dose dataframe measures (i.e, mean, minimum, maximum dose in Gy, and proportion), by patient group, being able to observe not only the four main structures but also the eight Couinaud segments. An example of this plot is shown in Figure 4.27 where the variable proportion is plotted, and the patients are split in toxicity and non-toxicity.

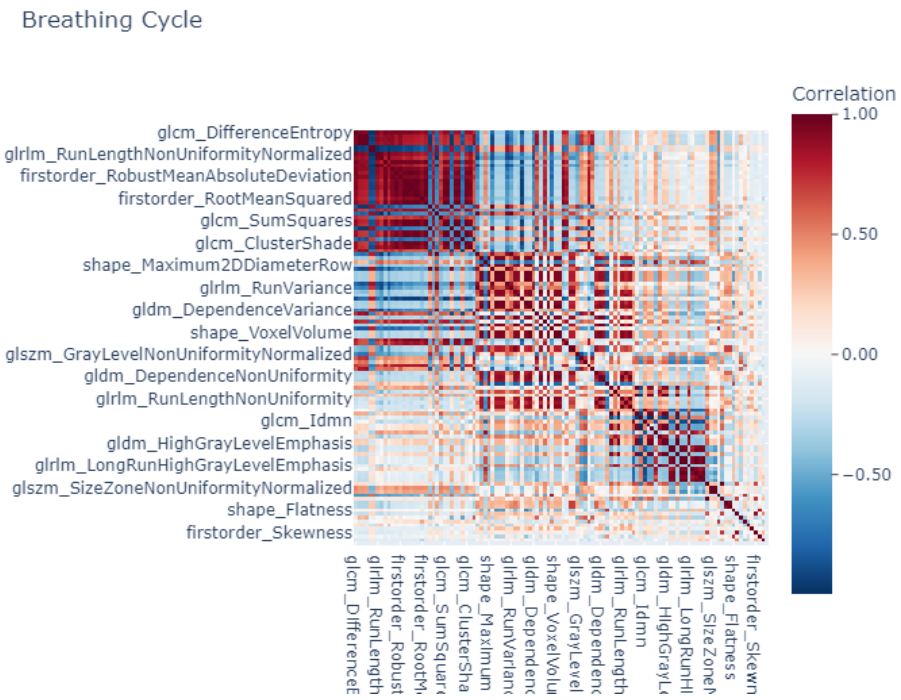


Figure 4.28: Heatmap of correlation between radiomic Features, breathing cycle data frame, and no correlation filtering.

4.7.4 (C) Machine Learning Analysis

In this section of the LiverVis Tool, relationships are established among all the radiomic features of our four dataframes through the calculation of the Pearson correlation coefficient. This coefficient is a statistical measure that evaluates the linear dependence between two quantitative variables, ranging from -1 to 1, where values close to 1 or -1 indicate a strong positive or negative correlation, respectively, and a value of 0 indicates the absence of linear correlation [Sed12]. Figure 4.28 shows an example of a heatmap, where this coefficient is reflected among all the radiomic features. A reddish color will indicate a strong correlation between features, while more blue color indicates a negative correlation, and the closer to white it is, the correlation is closer to 0.

The heatmap is arranged in such a way that it helps to visualize the strongest correlations first, whether positive or negative, as shown in the same Figure 4.28, noting that the most correlated features are at the top left. The tool allows the selection of the structure or respiratory cycle, to observe which features are more related to each other at different stages, or how they vary between Liver-Tumor structures. Additionally, it can filter by more correlated features, so that the heatmap only displays the features that are correlated between a and b, with a and b being values between -1 and 1.

The rest of the options in this section have been discussed in the patient stratification,

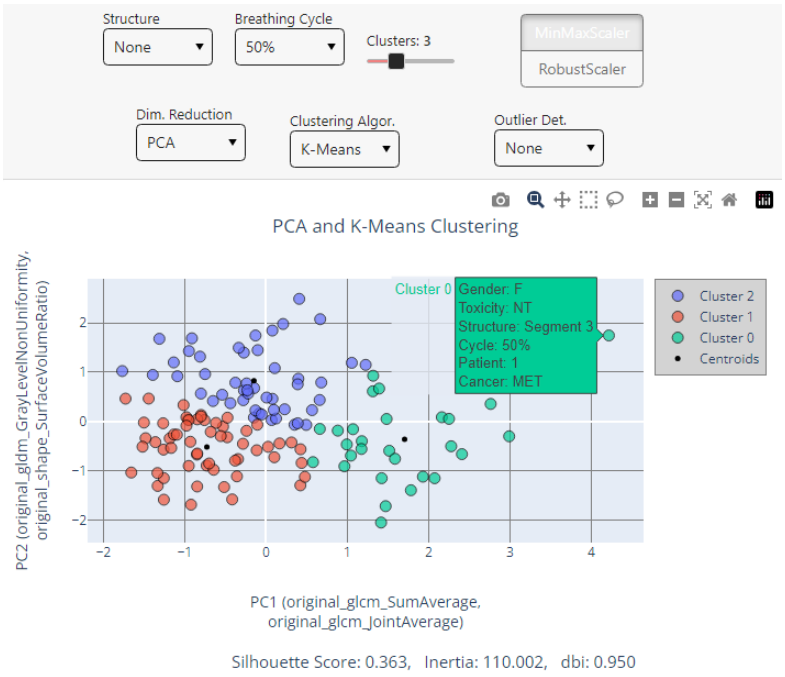
as here are the dendrograms of Hierarchical Clustering for the four dataframes. The interface allows filtering by structures, and respiratory cycle, specifying the type of outlier detection method to remove them, or changing the scaler.

4.7.5 (D) Comparative Analysis

This last section consists of the comparison between dataframes related to the mean and with the respiratory cycle simultaneously. Both subsections have the same types of visualization, which are those described earlier in the patient stratification, that is, dimensionality reduction combined with clustering, and methods of dimensionality reduction combined with each other. We can interact with the visualization in such a way that we can modify the analysis structure, the scaler, and the methods of dimensionality reduction, eliminate outliers in different ways, and select the number of Clusters, as can be seen in Figure 4.29. In addition, the data are labeled, so by hovering the mouse over it we can see which patient it is with all its characteristics (Gender, Cancer type, Toxicity, ID, Structure, Cycle of the Breathing cycle).



(a) Combined Dimensionality Reduction method for Average



(b) Dimensionality Reduction and Clustering for Breathing Cycle Dataframe

Figure 4.29: Cohort stratification for average (a) and breathing cycle data frame (b), comparative analysis in LiverVis Dashboard

CHAPTER 5

Results

5.1 Tasks and Research Questions

In this last section, we will present the results obtained throughout the thesis. Specifically, we will focus on tasks T2 and T3, aimed at patient stratification and comparative analysis, making use of the tool previously explained. These two tasks will be explained as one since the comparative analysis of our dataframes in the tool is carried out with the help of the different ways it has for patient stratification.

Conversely, we also address task T4 on radiation analysis, that is, observing the advantages of being able to plot the different variables we have obtained in the dose dataframe, as well as the differences between groups in terms of toxicity. Finally, T5 is a presentation of the results obtained with the models for the prevention of toxicity.

With all these tasks, we have all the necessary resources to respond to the research questions, perform statistical analysis to observe differences between groups (HCC and MET) and see which areas are under or over-irradiated.

5.1.1 (T1) Preprocessing

Previously, we discussed the methods used for preprocessing our dataframes, i.e., scaling the data, detecting potential outliers, and reducing dimensionality through various techniques. For outlier detection, as each of the four methods employed produces different results (i.e., K-Nearest Neighbors and LOF do not detect exactly the same outliers), those that coincide between both algorithms are established as local outliers, and the same is performed with the global outliers method. The LiverVis Dashboard enables the simultaneous testing of local, global, and a combination of both types of analyses.

The results of how many outliers are detected on each dataframe and the method used are shown in Table 5.1, which can be compared with the original size of the dataframe,

Dataframe	Local Detection	Global Detection	Local & Global Detection
Average (63)	2	4	0
Average Couinaud (144)	2	3	0
Breathing Cycle (609)	5	38	5
Breathing Cycle Couinaud (1392)	7	79	7

Table 5.1: Detection of outliers with local, global, and combination of methods.

given in parentheses in the first column. As can be seen, global methods detect more outliers than local methods, particularly for the Breathing Cycle dataframes where there are more combinations with the structures and cycles. Likewise, in these two dataframes, both local and global outlier detection methods agree on some of the outliers identified.

Subsequently, the effect of using scalers is compared after performing dimensionality reduction with PCA. For the cases of MinMaxScaler and RobustScaler, dimensionality reduction of 6 components has been performed, in such a way that the four most important loadings of each component have been calculated. In essence, we try to find out, in general terms, which are the radiomic features of our four dataframes that are repeated the most (left column MinMaxScaler, right column RobustScaler):

CASE 1

1. **GLRLM** Run Percentage (5).
2. **Shape 3D** Surface Volume Ratio (4).
3. **First Order** Image Mean (4).
4. **GLDM** Small Dependence Low Gray Level Emphasis (3)

CASE 2

1. **First Order** Kurtosis (8).
2. **GLCM** Cluster Shade (6).
3. **NGTDM** Busyness (4).
4. **GLSZM** Large Area Low Gray Level Emphasis (4).

Taking into account the number of principal components which is necessary for each scaler, the four dimensionality reduction algorithms are shown in Figure 5.1, where we can see that for the case of the Average Dataframe, UMAP tries to make a good separation between two groups using MinMaxScaler, and PCA/MDS detects some outliers when using the Robust Scaler.

Oppositely, Figure 5.2 shows the same comparison applied to the Breathing Cycle Couinaud dataframe, where, due to the large amount of data, dimensionality reduction

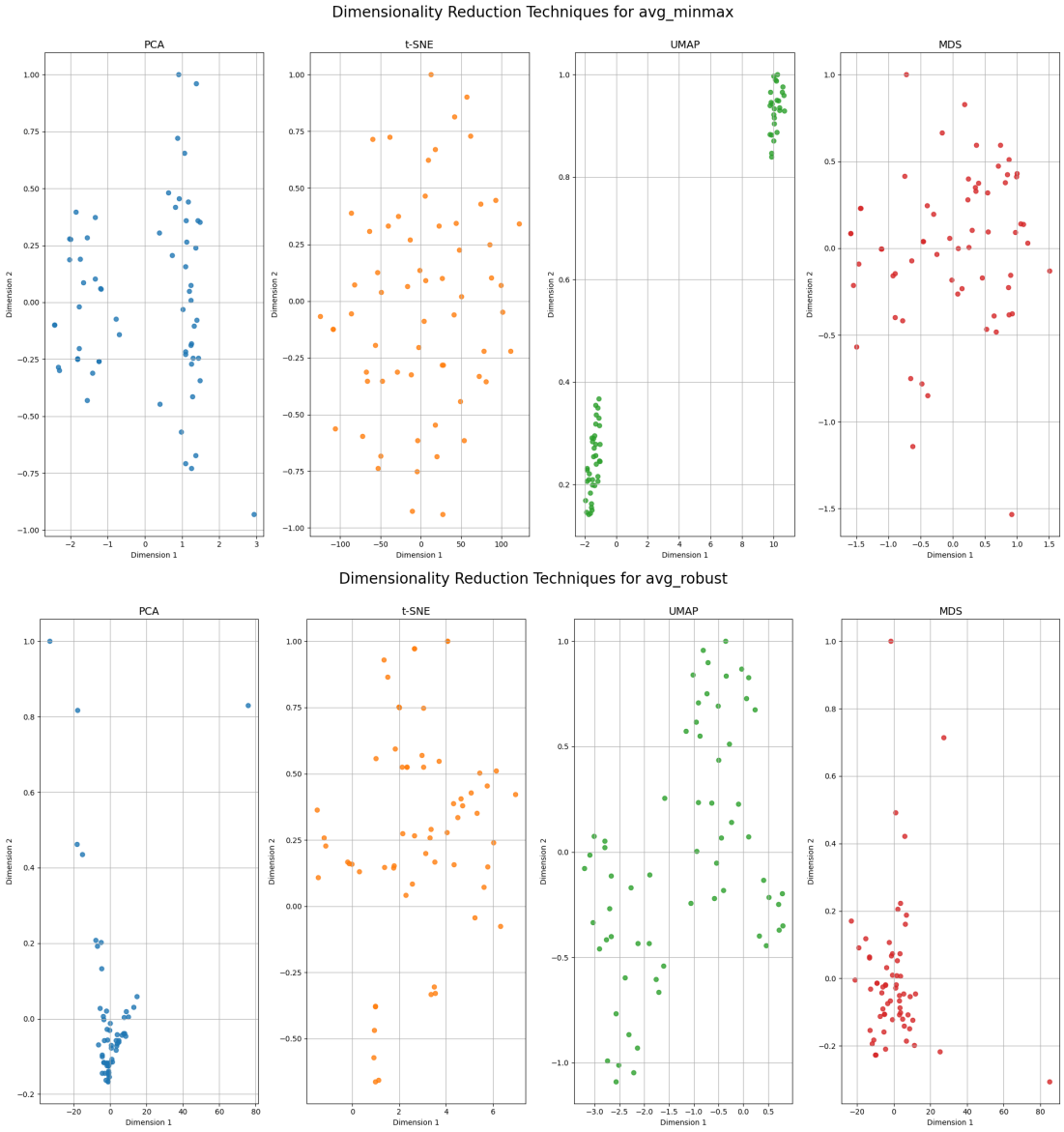


Figure 5.1: Comparison between dimensionality reduction methods, using MinmaxScaler and RobustScaler in Average dataframe. Being the data frame with the fewest data points, the PCA and UMAP algorithms are capable of clearly separating into two distinct groups when using MinMaxScaler, whereas RobustScaler is better at detecting outliers.

algorithms are unable to effectively separate the data, although in some cases, such as with PCA or MDS using RobustScaler, we can distinguish some groups and outliers.

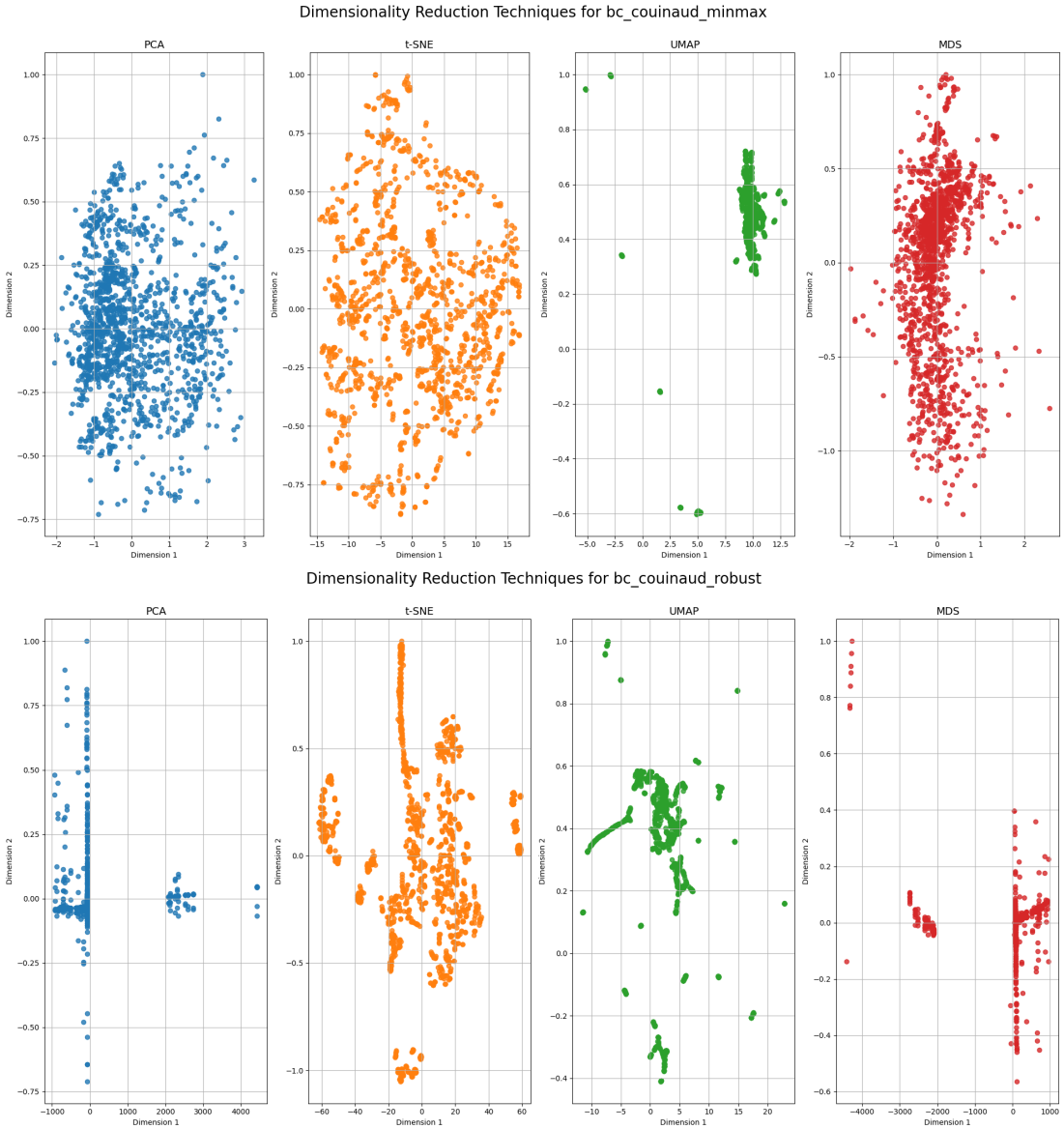


Figure 5.2: Comparison between dimensionality reduction methods, using MinmaxScaler and RobustScaler in Breathing Cycle Couinaud dataframe. Being the data frame with the most data points, the algorithms are not capable of separating when using MinMaxScaler, whereas RobustScaler can detect outliers.

5.1.2 (T2) Cohort Stratification and (T3) Comparative Analysis

In this section, we will perform the stratification of patients using the LiverVis Dashboard, employing the techniques explained in the methodology. Figure 5.3 shows the stratification of the cohort from the radiomic data of the average dataframe. Figure 5.3 (a) uses all

structures, while Figure 5.3 (b) filters by the ITV with the following settings in the LiverVis Dashboard options.

CASE 1

1. **Structures:** All.
2. **Scaler:** MinMaxScaler.
3. **Outliers:** Global outliers removed.
4. **Dim. Reduction:** UMAP.
5. **Clustering:** K-Means.
6. **Clusters:** 3.

CASE 2

1. **Structures:** ITV.
2. **Scaler:** RobustScaler.
3. **Outliers:** Local and global removed.
4. **Dim. Reduction:** PCA.
5. **Clustering:** K-Means.
6. **Clusters:** 3.

In this way, it is observed that with all structures, UMAP is capable of effectively reducing dimensionality, as was also shown previously in Figure 5.1. By applying K-Means with three clusters, we obtain a Silhouette score of 0.74 and a DBI (Davies-Bouldin Index) of 0.396, indicating that the clusters are dense or have little internal dispersion, and are also well-separated from each other. Moreover, by only stratifying patients according to their ITV, we obtain a lower Silhouette Score but a better DBI, meaning the clusters, although separated from each other, have points that are a bit farther from their respective centroids.

Figure 5.4 shows the stratification of the average Couinaud dataframe, again with an example using all the segments, and subsequently filtering by one of these structures. The configuration is as follows:

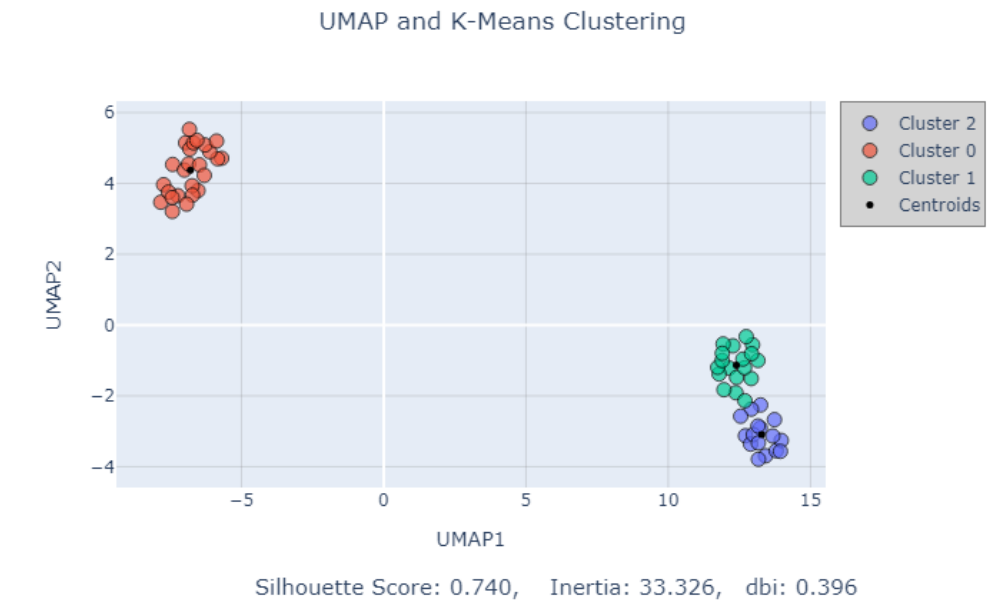
CASE 1

1. **Structures:** All.
2. **Scaler:** MinMaxScaler.
3. **Outliers:** Local outliers removed.
4. **Dim. Reduction:** T-SNE.
5. **Clustering:** K-Means.
6. **Clusters:** 3.

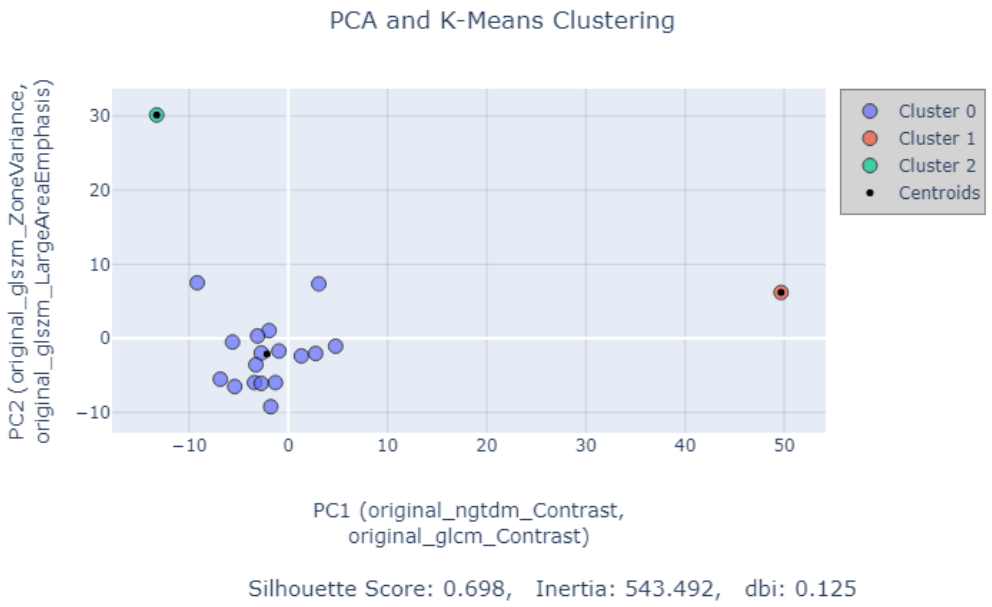
CASE 2

1. **Structures:** Segment 8 .
2. **Scaler:** MinMaxScaler.
3. **Outliers:** Local outliers removed.
4. **Dim. Reduction:** UMAP.
5. **Clustering:** Spectral Clustering.
6. **Clusters:** 3.

As can be observed, when attempting to cluster all the segments of the patients using t-SNE and K-Means clustering, the results are not good. Although the Silhouette Score is



(a) Stratification with UMAP and K-Means, all structures



(b) Stratification with PCA and K-Means, filtering by ITV

Figure 5.3: Stratification of patients in the average data frame, different configurations. In (a) UMAP reduces the dimensionality in two differentiated groups and metrics indicate clusters are dense and well-separated from each other. In (b) PCA separates into groups, but clusters have internal dispersion.

ordinary, inefficient values of DBI and inertia are obtained, indicating that the separation between clusters is not good. Moreover, when clustering segment 8 of the patients, we again obtain suboptimal results.

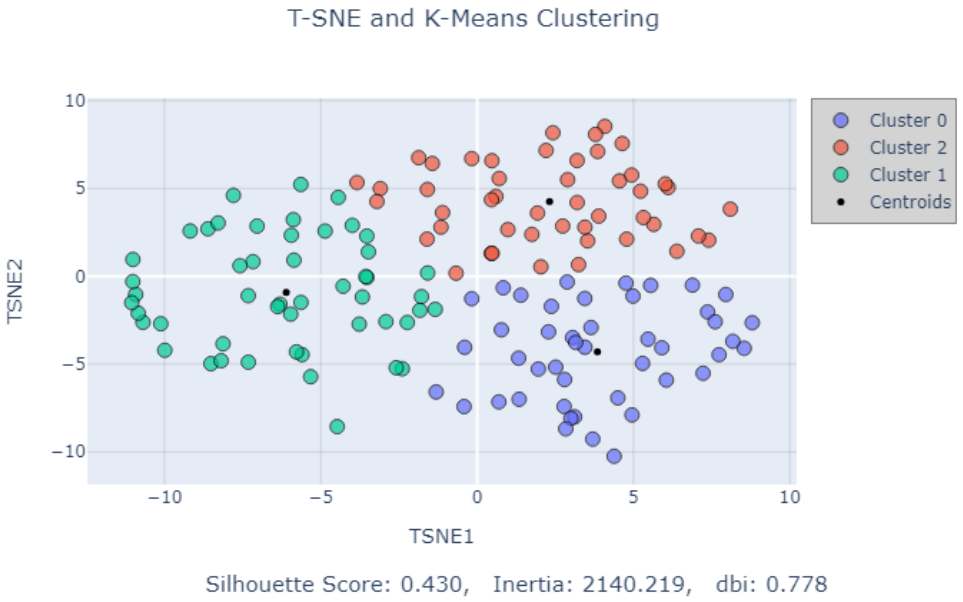
Figure 5.5 shows the stratification of the radiomic features from the Breathing Cycle data frame. Figure 5.5 (a) displays a clustering of all possible structure-cycle combinations in the patients, while Figure 5.5 (b) shows a particular case for the respiratory cycle at 30% and the Liver-ITV structure. The configuration is as follows:

CASE 1	**CASE 2**
1. **Structures:** All.	1. **Structures:** Liver-ITV (30%).
2. **Scaler:** MinMaxScaler.	2. **Scaler:** RobustScaler.
3. **Outliers:** Local and Global removed.	3. **Outliers:** Local outliers removed.
4. **Dim. Reduction:** UMAP.	4. **Dim. Reduction:** PCA.
5. **Clustering:** Mean-Shift.	5. **Clustering:** K-Means.
6. **Clusters:** 4.	6. **Clusters:** 3.

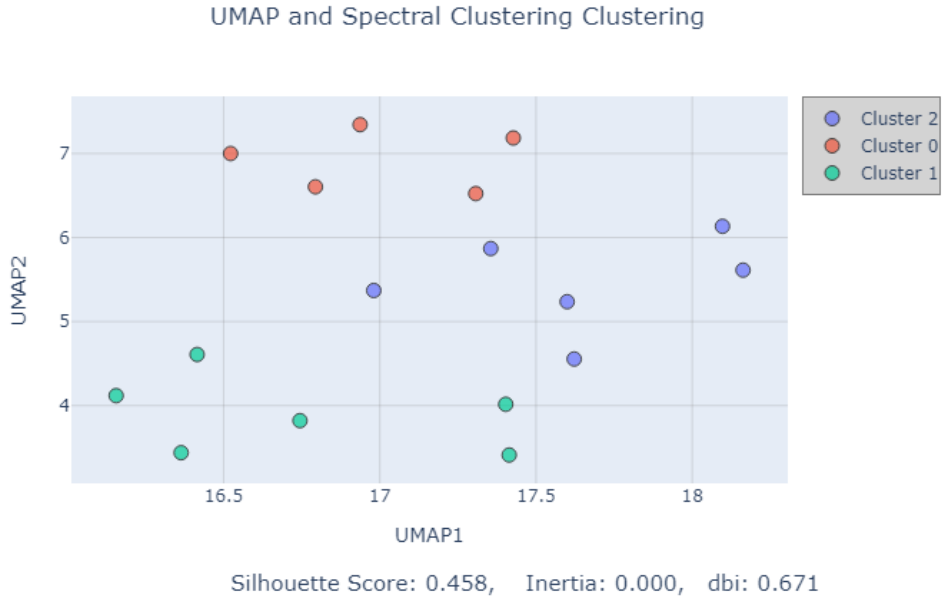
It is observed that through UMAP and Mean-Shift, the structure-cycle combinations can be efficiently separated into four clusters, which, according to the metrics, are well-separated from each other, and each is sufficiently compact concerning its centroids. Additionally, when filtering by structure and cycle, good results are also obtained in the metrics, but with a very high value of inertia due to the clusters not being compact.

Figure 5.6 displays the stratification of the Breathing Cycle Couinaud data frame, again with an example using all cycle-segment combinations, and subsequently filtering by one of these combinations. The configuration is as follows:

CASE 1	**CASE 2**
1. **Structures:** All.	1. **Structures:** Segment 3 (70%).
2. **Scaler:** RobustScaler.	2. **Scaler:** RobustScaler.
3. **Outliers:** Global outliers removed.	3. **Outliers:** Local outliers removed.
4. **Dim. Reduction:** MDS.	4. **Dim. Reduction:** PCA.
5. **Clustering:** K-Means.	5. **Clustering:** Mean-Shift.
6. **Clusters:** 3.	6. **Clusters:** 4.

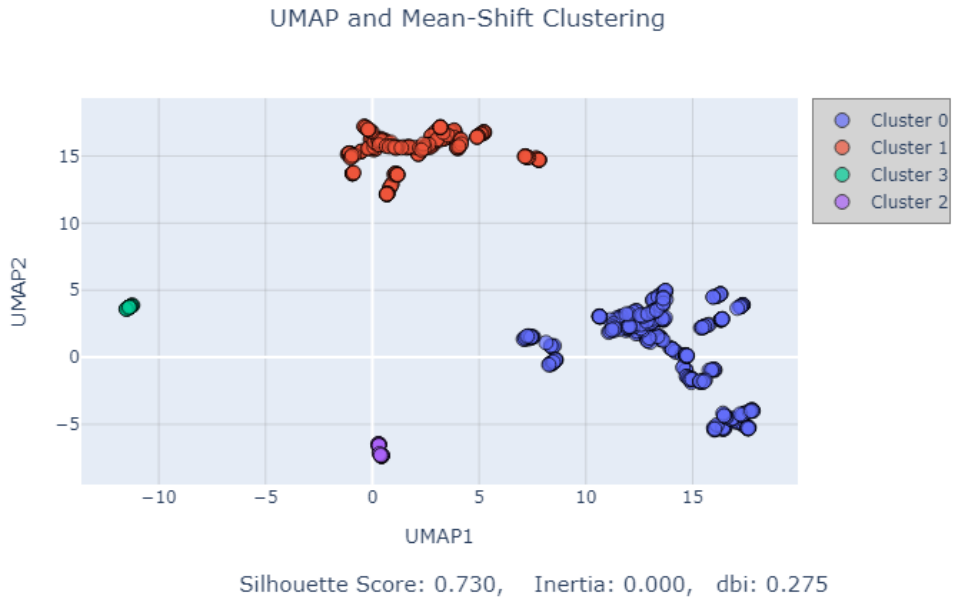


(a) Stratification with T-SNE and K-Means, all structures

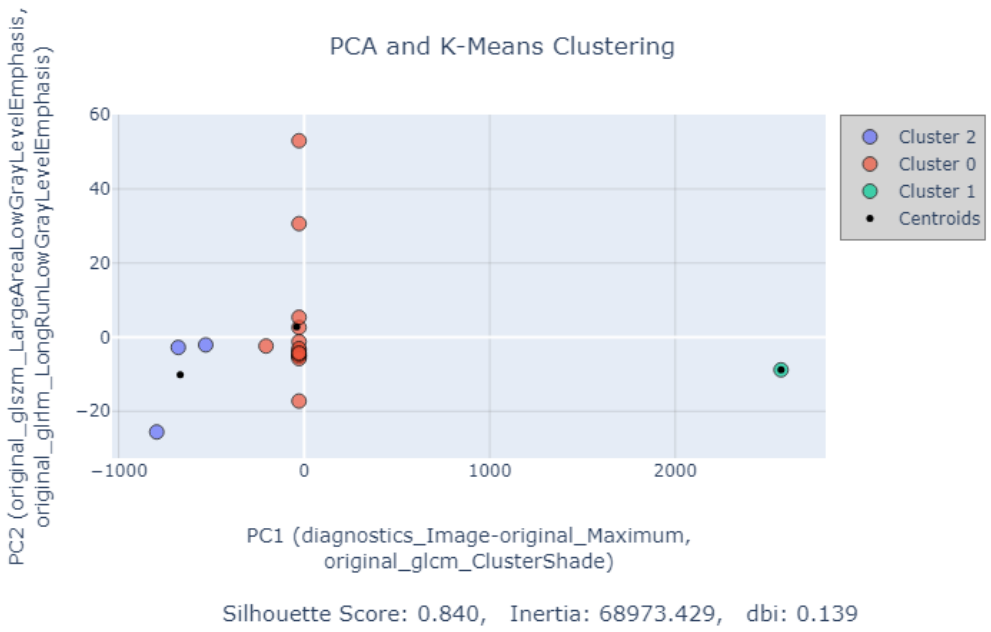


(b) Stratification with UMAP and Spectral Clustering, filtering by Segment 8

Figure 5.4: Stratification of patients in the average Couinaud data frame, different configurations. Both methods provide inefficient values of metrics, indicating that the separation of clusters is not optimal.



(a) Stratification with UMAP and Mean-Shift, all structures



(b) Stratification with PCA and K-Means, filtering by Liver-ITV and 30% of the Breathing Cycle

Figure 5.5: Stratification of patients in the Breathing Cycle data frame, different configurations. UMAP and Mean-Shift (a) separate our data into four consistent clusters, while PCA and K-Means (b) separate into three clusters but are not compact.

Lastly, in this figure, similarly to the previous case, good results are seen in the metrics of the Silhouette Score and the DBI, both for the case with all structures and for the case with segment 3. In this last example, it is observed that two of the clusters contain only one point each (and are also significantly separated from the rest), indicating that both patients differ distinctly from the cohort.

5.1.3 (T4) Radiation Analysis

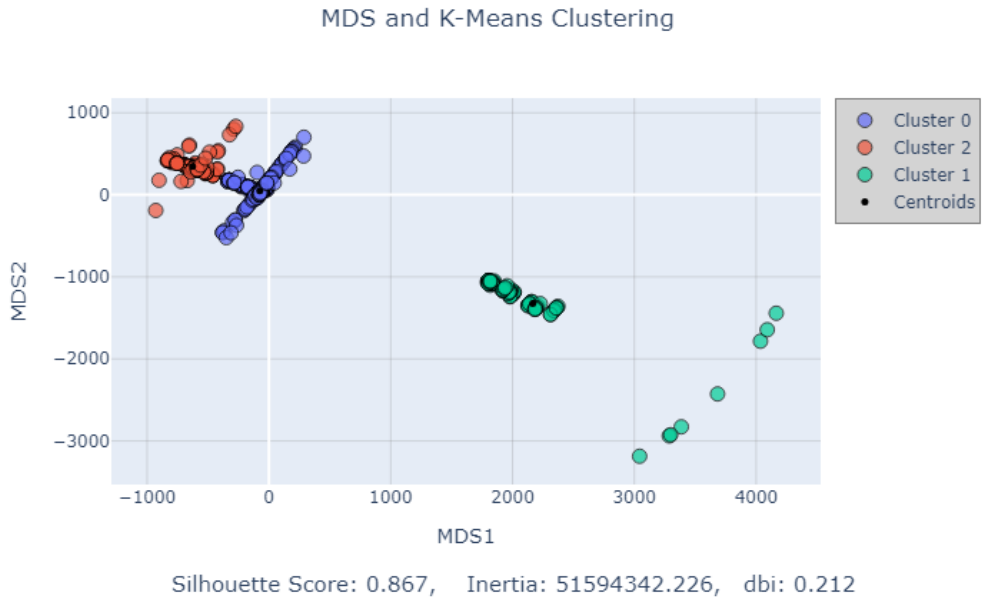
Starting from the data frame that we have obtained with the dose measurements, we will use this information to conduct a study on the treatment of our patients. Specifically, we will try to observe through graphs of these measures, how the magnitudes vary between patients with and without toxicity.

Figure 5.7 (a) explains the distribution of the average dose across all structures of interest, plotted simultaneously for all patients. As expected, the ITV and PTV structures have the highest values for almost all patients, since they are the target volumes for radiotherapy sessions to eliminate the tumor. Subsequently, the liver represented by Liver-ITV and its corresponding Couinaud segments, remains almost between 0 and 2.5 Gy/cc, except for patient number 2 who presents a higher proportion of radiation in segment 1, even higher than that of his ITV.

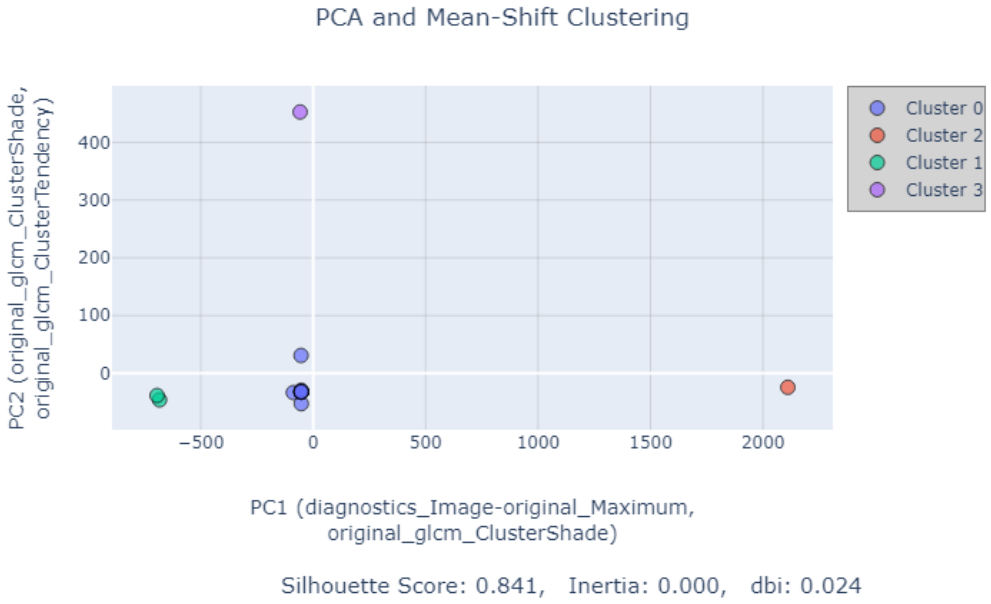
To see more clearly the proportion in the segments separately, comparing them with Liver-ITV, ITV, or PTV, Figure 5.7 (b) shows the example of segment 1, where it is probably over-irradiated for patient No. 2. Regarding the other patients, it is observed that segment 1 is more irradiated in proportion than concerning the entire liver (Liver-ITV). In any case, in both Figure 5.7 (a) and 5.7 (b) we are also unable to see a notable difference between patients with toxicity and those without toxicity (remember that patients 8 to 14 are those with toxicity).

As we are not visually able to detect these differences in this dose proportion, or any of the other variables (i.e., maximum dose, mean dose, or minimum dose), we must perform statistical tests that allow us to find differences between groups (toxicity and non-toxicity) if they exist. Choosing the correct statistical test to compare groups or find differences between them is crucial, and it mainly depends on the nature of our data and the number of groups we are comparing. We have already performed normality tests and concluded that our data assume non-normal distributions, therefore statistical tests such as the t-test or ANOVA/MANOVA cannot help us despite their widespread use and easy interpretability. Because of this, the Multivariate Kruskal-Wallis (MKW), Mann-Whitney U (MW) test, and Kolmogorov-Simirnov (KS) have been conducted.

The results of all the tests are shown in Table 5.2. As can be observed, the MKW test finds differences in **segment 1 - proportion** (stat = 6.87, p-value = 0.008), **segment 5 - minimum dose** (stat = 5.92, p-value = 0.014), **segment 5 - mean dose** (stat = 6.39, p-value = 0.011), **segment 5 - proportion** (stat = 4.64, p-value = 0.031), and **segment 6 - mean dose** (stat = 3.86, p-value = 0.049). MW and KS tests do not find

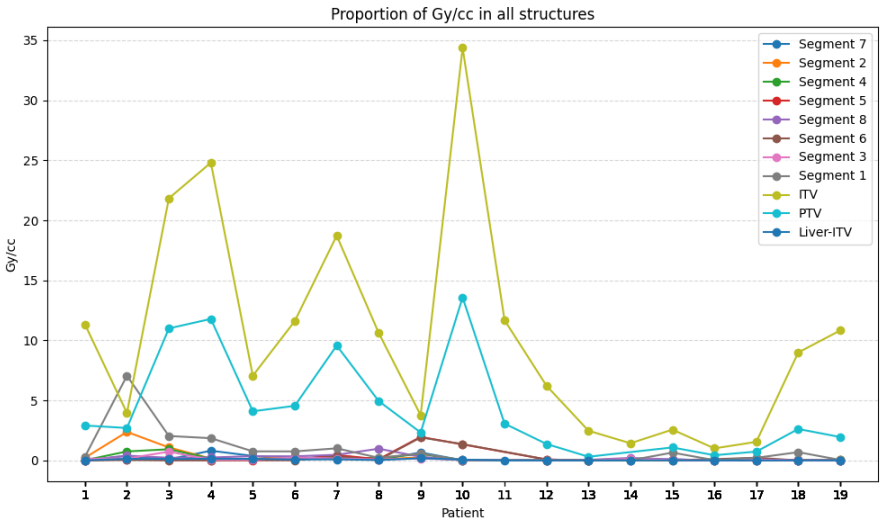


(a) Stratification with MDS and K-Means, all structures

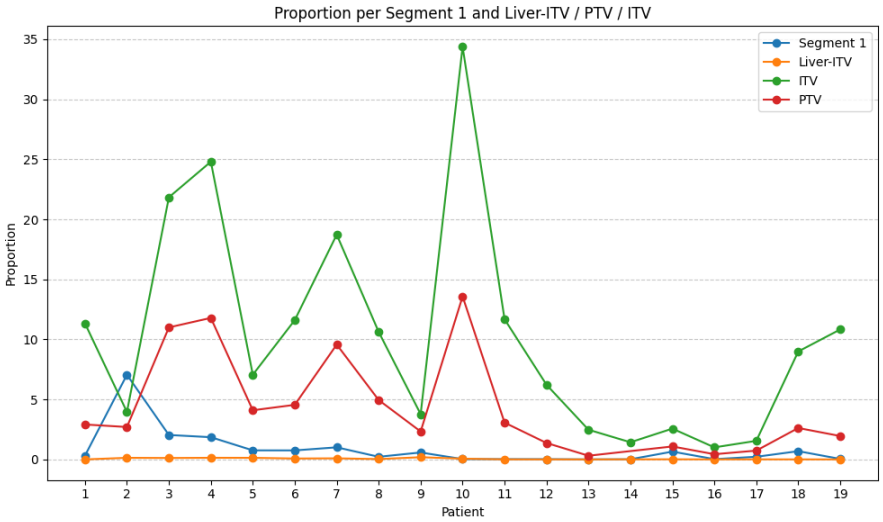


(b) Stratification with PCA and Mean-Shift Clustering, filtering by Segment 3 and cycle 70%

Figure 5.6: Stratification of patients in the Breathing Cycle Couinaud data frame, different configurations. In both cases, the results are good according to the metrics but in (a) two of the clusters contain only one point each, indicating that the patients differ from the cohort.



(a) Proportion: all structures and segments



(b) Proportion: segment 1 and structures

Figure 5.7: Proportion (Gy/cc) along the structures and patients.

	Structure	*Variable*	*stat value*	*p-value*
Multivariate Kruskal-Wallis	Segment 1	Proportion	6.87	0.008
	Segment 5	Min dose,	5.92	0.014
		mean dose,	6.39	0.011
		proportion	4.64	0.031
	Segment 6	Mean dose	3.86	0.049
Mann-Withney and Kolmogorov-Smirnov	Segment 1	Proportion	8.0 (MW), 0.66 (KS)	0.007 (MW), 0.046 (KS)
	Segment 5	Min dose,	62.0 (MW), 0.75 (KS)	0.013 (MW), 0.014 (KS)
		mean dose,	63.0 (MW), 0.66 (KS)	0.009 (MW), 0.046 (KS)
		proportion	59.0 (MW), 0.66 (KS)	0.031 (MW), 0.046 (KS)

Table 5.2: Non-parametric tests, differences found in structures and variables (Toxicity/Non-Toxicity).

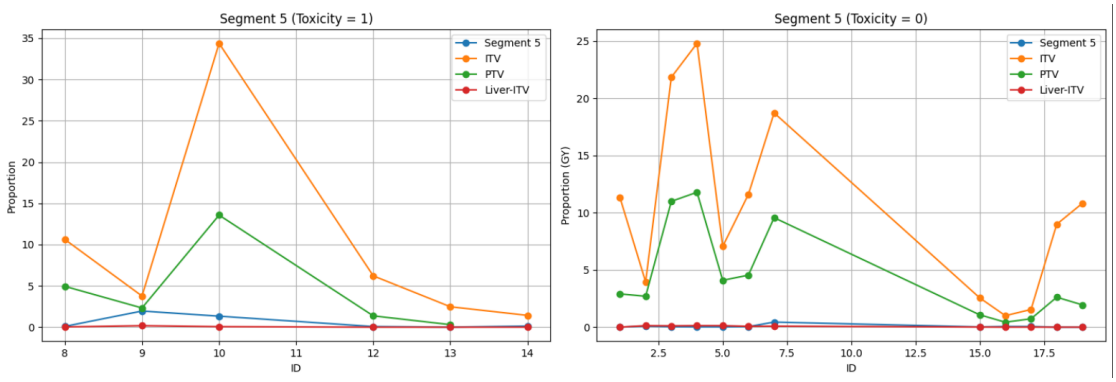


Figure 5.8: Proportion of the Dose across Segment 5, comparing to ITV, PTV, and Liver-ITV. The dose proportion over this segment tends to be higher in patients with toxicity than in those without.

significant differences in segment 6, however, all three tests agree on the differences found for segments 1 and 5, also agreeing on the variables.

In Figure 5.8, we can observe the difference in the dose proportion over segment 5, which tends to be higher in patients with toxicity (Toxicity = 1) than in those without (Toxicity = 0). Furthermore, this can be supported by Figure 5.9, since the average dose and the proportion are dependent measures. Observing this last-mentioned figure, it is also noted that segment 5 tends to be more irradiated in patients who have developed toxicity than in those who have not. The minimum dose (Figures 5.10 and 5.11) also tends to be higher in segment 5, compared to the minimum dose across the liver (Liver-ITV), in patients with toxicity. Finally, it can be observed that for Segment 1, in Figure 5.12, the dose proportion is lower in Segment 1, and for all the patients. Even though this difference was found only by the MKW test, Figure 5.13 shows the average dose in Segment 6, which can be seen to be higher in several of the patients with toxicity, compared to those without toxicity.

In conclusion, differences can be observed between the toxicity and non-toxicity groups,

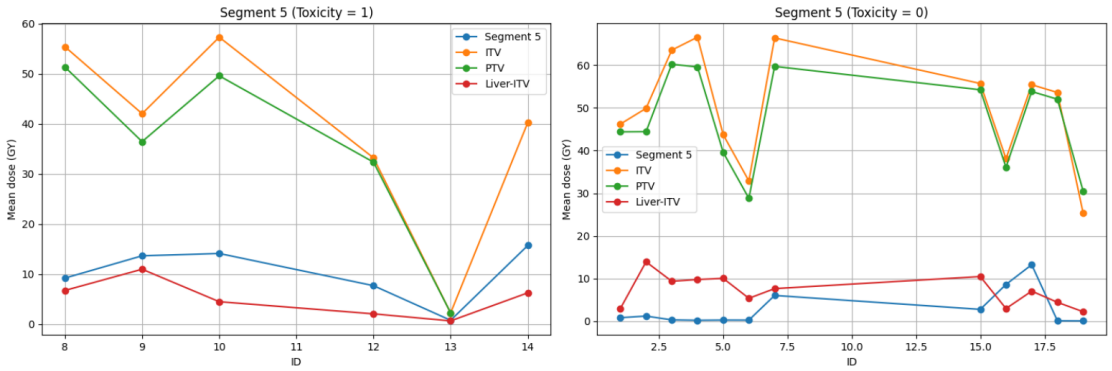


Figure 5.9: Mean of the Dose across Segment 5, comparing to ITV, PTV, and Liver-ITV. The average dose in Segment 5 is higher in patients with toxicity.

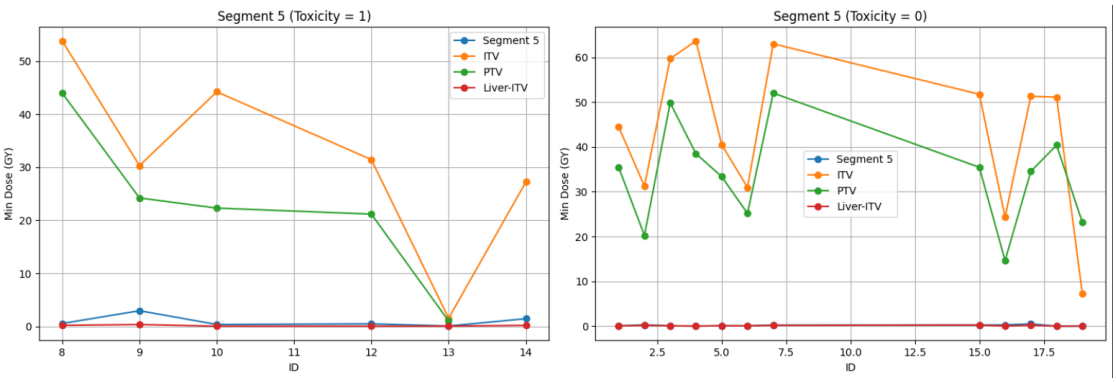


Figure 5.10: Minimum of the Dose across Segment 5, comparing to ITV, PTV, and Liver-ITV. The minimum dose received by Segment 5 is apparently higher in patients with toxicity, above all in patient nº 9.

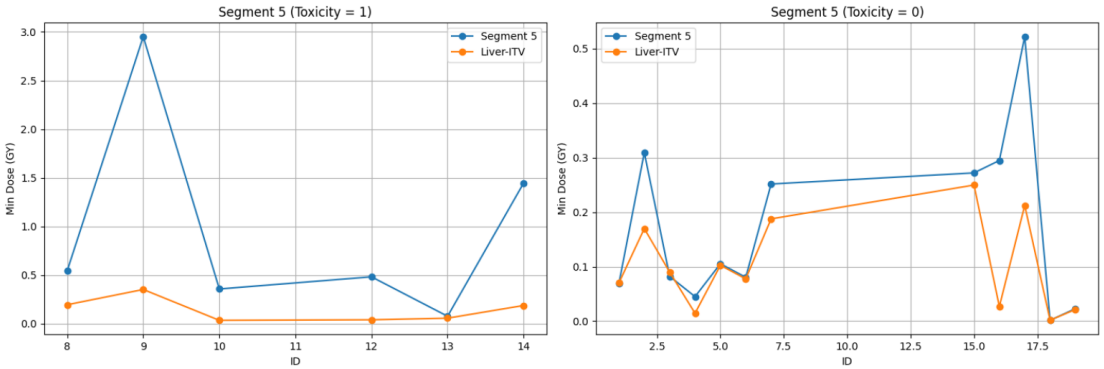


Figure 5.11: Minimum of the Dose across Segment 5, comparing to Liver-ITV. After zooming on Segment 5 and Liver-ITV, we see the difference in the minimum dose received by Segment 5 is higher in patients with toxicity.

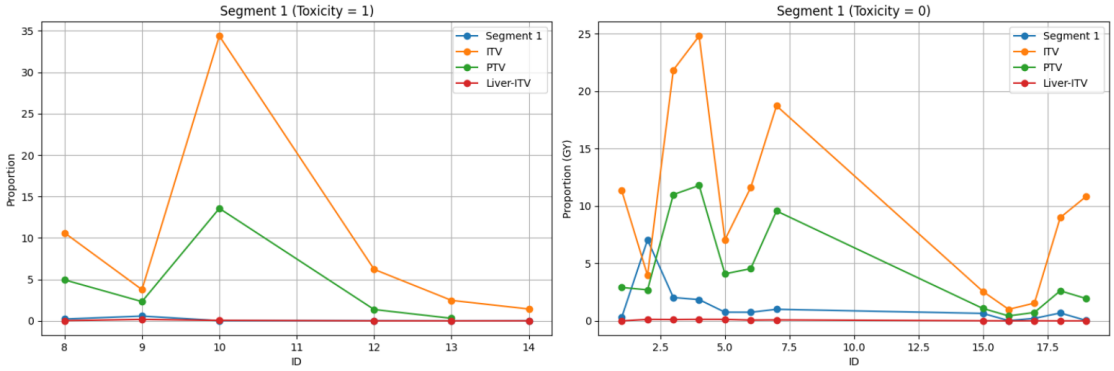


Figure 5.12: Proportion of the Dose across Segment 1, comparing to ITV, PTV, and Liver-ITV. The dose proportion is lower in patients with toxicity.

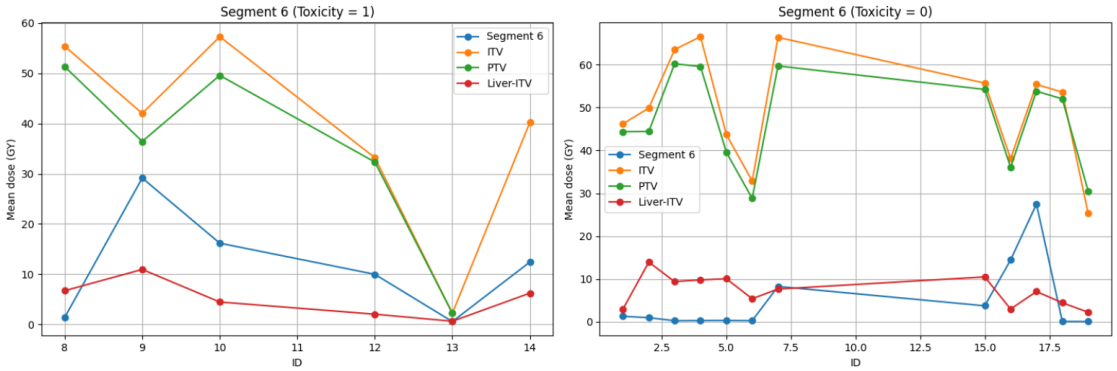


Figure 5.13: Mean of the Dose across Segment 6, comparing to ITV, PTV, and Liver-ITV. The average dose across Segment 6 is higher in patients with toxicity than in patients without toxicity.

based on the doses of radiotherapy received. The relation between both variables is reasonable since a dose higher than suitable in an area or segment of the liver other than where the tumor is located can lead to a side effect of toxicity.

By repeating this procedure for groups of patients with Hepatocellular Carcinoma and Metastatic conditions, we can observe differences between the two groups regarding the dose of radiation received by each, considering that different types of tumors can be treated differently. As shown in Table 5.3, with KW we discover differences in segments 1, 4, 5, 7, and 8, as well as in the Liver-ITV for measurements in the average and minimum dose. Additionally, in the tests performed with MW and KS, they each independently find some further differences in other measures, but always concur on the structure that provides this difference. The most significant differences are found with the KW test in **segment 1 - minimum dose** (stat = 7.89, p-value = 0.005), **Liver-ITV - minimum dose** (stat = 9.25, p-value = 0.002), and with MW/KS tests **Liver-ITV - minimum dose** (stat = 6.0/0.83, p-value = 0.001/0.001).

	Structure	*Variables*	*stat value*	*p-value*
Multivariate Kruskal-Wallis	Segment 1	Min dose	7.89	0.005
	Segment 4	Min dose	7.38	0.006
	Segment 5	Min dose	4.24	0.039
	Segment 7	Min dose	6.87	0.008
	Segment 8	Min dose, mean dose	7.37 5.92	0.006 0.014
	Liver-ITV	Min dose, mean dose	9.25 6.86	0.002 0.008
Mann-Withney and Kolmogorov-Smirnov	Segment 1	Min dose	6.0 (MW), 0.75 (KS)	0.003 (MW), 0.014 (KS)
	Segment 4	Min dose, proportion (KS)	7.0 (MW), 0.75 (KS) 0.66 (KS)	0.004 (MW), 0.014 (KS) 0.046 (KS)
	Segment 5	Min dose (MW)	14.0 (MW)	0.04 (MW)
	Segment 7	Min dose, max dose (KS)	8.0 (MW), 0.66 (KS) 0.66 (KS)	0.006 (MW), 0.046 (KS) 0.046 (KS)
	Segment 8	Min dose, mean dose	7.0 (MW), 0.66 (KS) 10.0 (MW), 0.75 (KS)	0.004 (MW), 0.047 (KS) 0.013 (MW), 0.014 (KS)
	Liver-ITV	Min dose, mean dose (MW)	6.0 (MW), 0.83 (KS) 11.0 (MW)	0.001 (MW), 0.001 (KS) 0.007 (MW)

Table 5.3: Non-parametric tests, differences found in structures and variables (Hepatocarcinoma/Metastasis).

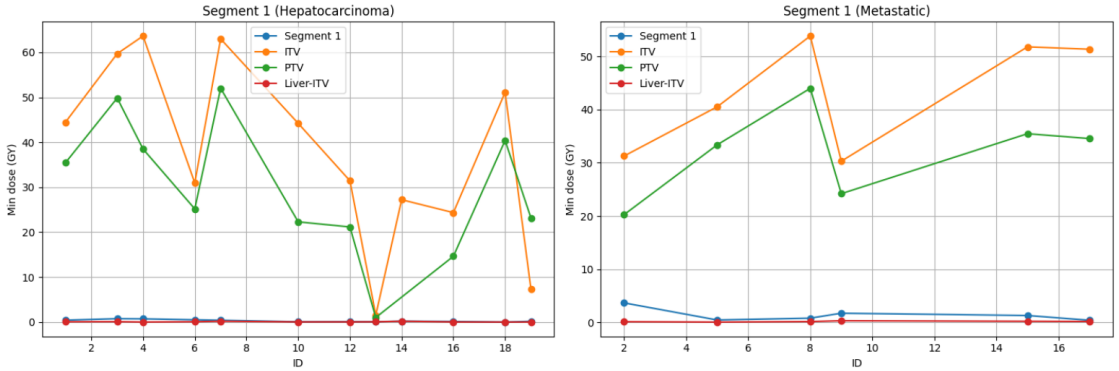


Figure 5.14: Min Dose across Segment 1, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 1 for patients with HCC is practically 0 Gy, while patients with MET have radiation levels above 0 Gy.

From Figures 5.14 to 5.18, we observe the difference between patients with Hepatocellular Carcinoma and Metastatic conditions, through the results obtained in Table 5.2. In Figure 5.14, it is observed that the minimum dose across segment 1 for patients with hepatocellular carcinoma is practically 0 Gy, while patients with metastases (specifically patients 2, 9, and 15) have radiation levels above 0. This can also be seen in the other referenced figures, where for all cases, patients with metastases have a higher minimum radiation dose in these segments. Lastly, in any of the figures, the minimum dose of the Liver-ITV is represented in the same way in all, however, there is no visually significant difference.

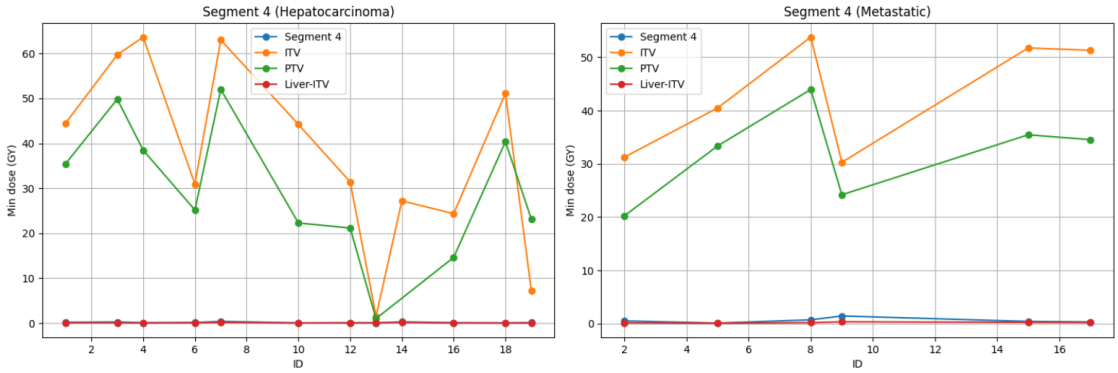


Figure 5.15: Min Dose across Segment 4, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 4 for patients with HCC is lower than in patients with MET.

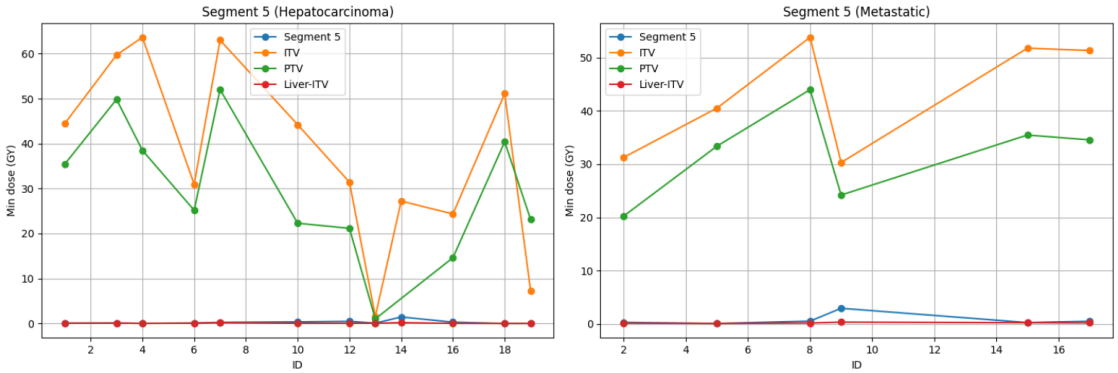


Figure 5.16: Min Dose across Segment 5, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 5 for patients with HCC is lower than in patients with MET.

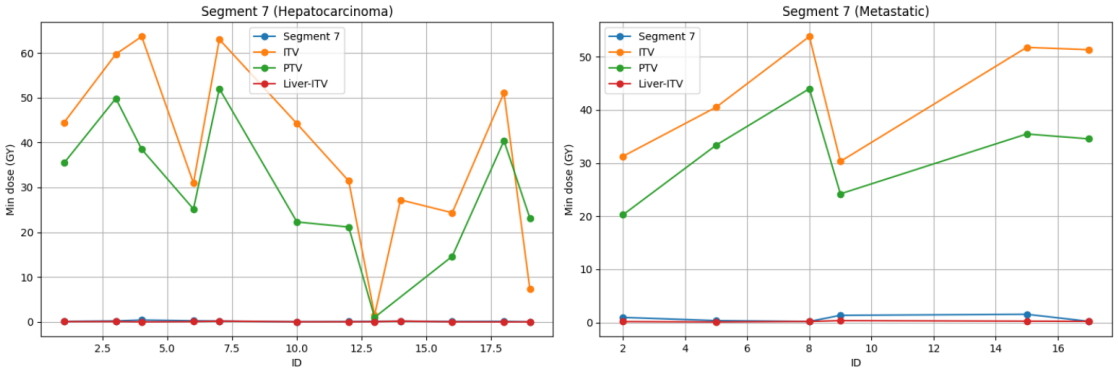


Figure 5.17: Min Dose across Segment 7, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 7 for patients with HCC is lower than in patients with MET.

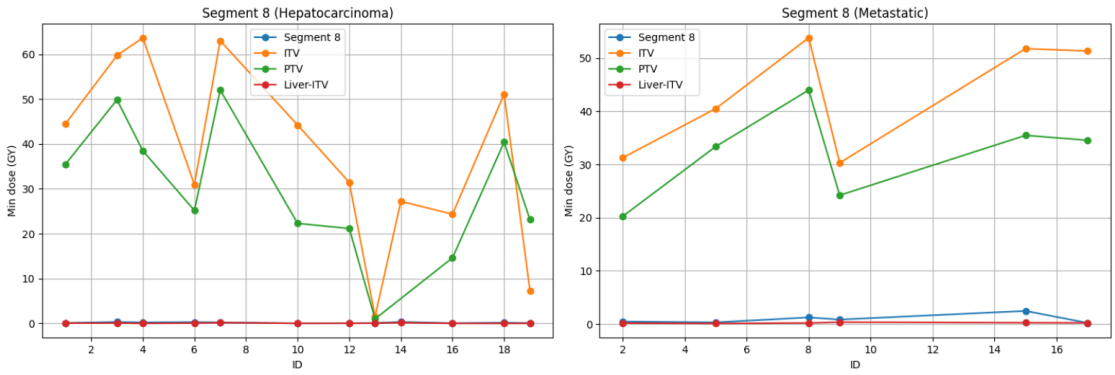


Figure 5.18: Min Dose across Segment 8, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 8 for patients with HCC is lower than in patients with MET.

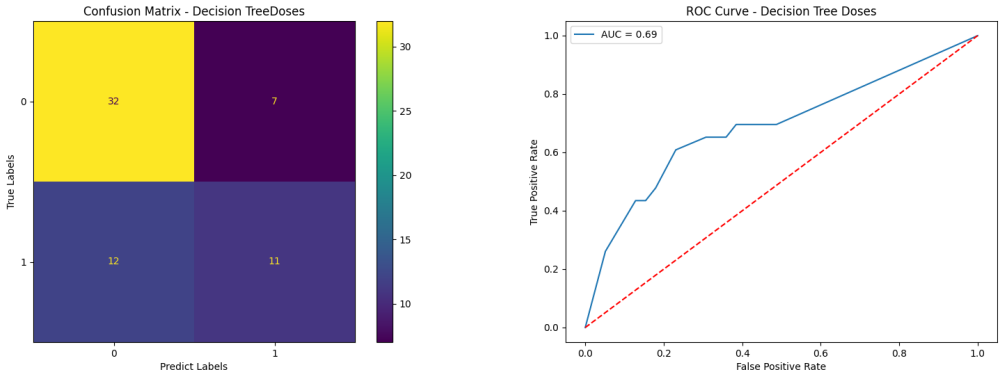
5.1.4 (T5) Predictive Analysis

In this section, we will try to perform a predictive analysis of the patients in our dataset, based on whether they are likely to develop toxicity as a side effect or not, and likewise, the same analysis will be repeated to classify between patients with Hepatocellular Carcinoma and Metastatic conditions. Given that we know there are differences between both groups, and based on both characteristics, we will perform a classification task with Machine Learning algorithms to classify our patients based on radiomic features, and our data with Dosage.

The Machine Learning algorithms used are Decision Tree, Naive Bayes, Random Forest, Support Vector Machine, and Logistic Regression. For this, we have divided each of our dataframes into a training and test ratio of 70-30% respectively, ensuring good prediction accuracy [SHD+22]. To improve the effectiveness of each algorithm, we will use grid search to test the best combinations of hyperparameters, ultimately obtaining the best possible model with these combinations, reflecting its efficiency with measures such as accuracy, ROC curve, and a contingency matrix (sensitivity and specificity). Since we are able to know the weights with which the decision tree and random forest work, we will save the three variables with the most weight used by our models to understand which have been considered as predictors.

Classification of Toxicity according to the Doses

First, classification is based on the dataframe used in the previous section, which contains the minimum, maximum, and average dose, the volume of the structure, and its proportion. We must remove the patient ID as it is not of interest for this task, and also perform One Hot Encoding, as we must assign a value to each row of the dataframe in terms of the structure we are talking about at all times. Without applying OneHotEncoding, we are considering all rows as the same variable (or unique patients), and therefore the model will have very good results given the large number of rows. The training time was



(a) Confusion Matrix

(b) ROC Curve

Figure 5.19: Confusion Matrix and ROC Curve of the Best Model (D.T and the volume removed). The model predicts accurately toxicity and non-toxicity patients according to the dose data frame, but the rates of false positives and negatives result in an AUC of 0.7.

approximately 183 minutes, achieving a maximum accuracy of 67.7% with the Decision Tree and an AUC of 0.62 for the same model.

As is observed in Figure 5.19, the confusion matrices and the values of sensitivity and specificity reveal that the model can accurately predict the non-toxicity of patients, but fail to predict which patients do have toxicity, considering many of them to be non-toxic when they are (False Negatives). The variables that the DT model has considered with the most weight, in this case, are "Max Dose (GY)" and "Volume (cc)", and the Random Forest, which has very similar accuracy to DT, has considered "Max Dose (Gy)" as the first and "Mean Dose (Gy)" as the second variable.

Given that the proportion of the dose, the volume, and the average dose is variables among which there is a linear combination (remember that the proportion is the average dose divided by the volume of the structure), the volume column is removed from the dataframe, and the models are trained again. The training time on this occasion is 55 minutes, considerably lower than the previous case, and the accuracies of almost all models increase.

A comparison between this two trainings is shown in Table 5.4, with the dataframe we are using for prediction, the average accuracy (of the 5 models), and the metrics of the best model (B.M).

Now, the DT and RF have accuracies of 0.69 and 0.67 respectively, and AUCs of 0.69 and 0.65. The DT has decreased its specificity a bit but balanced it with sensitivity. Obtaining the variables with the most weight, it is observed that for both models they are the "Max Dose (Gy)" and "Mean Dose (Gy)", which contrasts with the previous

Dataframe	Average Accuracy	Best Model	Accuracy (B.M)	Sensibility (B.M)	Specificity (B.M)	AUC (B.M)
Doses	61.3%	D.T	67'7 %	0.22	0.95	0.62
Doses (No Volume)	62.2%	D.T	69%	0.48	0.82	0.69

Table 5.4: Results for complete doses data frame and with volume column removed. Decision Trees (D.T) achieve the best results in both cases, increasing accuracy by removing the volume column and also reducing training time.

section in which the importance of both variables for finding differences between groups was explained.

As a final step, we want to study if there is relevance in using some structures over others as predictor variables, instead of using them all at once, which may be creating noise in the model. Therefore, the volume column is removed, and the Dose dataframe will be separated into 7 groups:

- ITV
- PTV
- Liver-ITV
- Tumor (ITV and PTV)
- Liver (liver and Liver-ITV)
- Tumor and Liver
- Couinaud Segments

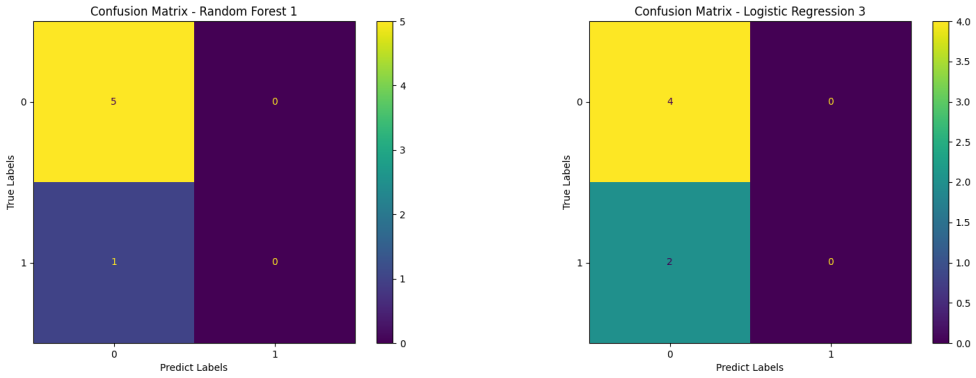
With the independent structures (ITV/PTV/Liver-ITV), good results are obtained with a maximum accuracy of 83.3% for the case of ITV and PTV, with Liver-ITV being the worst case of study with maximum accuracies of 66% and results around 20% for RF and DT. For these three cases, similar to considering all structures, the sensitivity and specificity of the models are very unbalanced. Table 5.5 summarizes the results of this prediction.

Figure 5.20 shows, by having filtered the dataframe by these structures, we are left with few rows (one ITV per patient, one Liver-ITV per patient...etc.), so the amount of data the model has to make the prediction is insufficient. All models are making predictions with 13 samples for training, and 6 for test. In conclusion, it can be obtained that having the same number of rows for ITV and Liver-ITV (i.e., 19 patients), the accuracy is much higher with ITV as shown in Table 5.5.

The results in Table 5.6 show this last conclusion is also reflected when filtering the Dataframe by Tumor/Liver. In the case of the liver, the maximum accuracy obtained is 62.5% for DT, SVM, and LR models, while in the tumor, accuracies of 82%, 73%, and 73% are obtained respectively for the same models. The average accuracy is higher when predicting toxicity with tumor-related structures, although the AUC values and

Dataframe	Average Accuracy	Best Model	Accuracy (B.M)	Sensibility (B.M)	Specificity (B.M)	AUC (B.M)
ITV	83.3%	R.F	83.3 %	0	1	0.7
PTV	69.9%	R.F	83.3%	0	1	0.7
Liver-ITV	46.6%	L.R	66.6%	0	1	0.5

Table 5.5: Results for doses dataframe with the three first filters. Promising results are obtained with the random forest (R.F) and logistic regression (L.R) models, however, the number of data points is very low after filtering by structure, equal to the number of patients.



(a) Confussion Matrix ITV

(b) Confussion Matrix Liver-ITV

Figure 5.20: Confusion Matrices for ITV and Liver-ITV filters. Good results are obtained with the DT for prediction using only the ITV, and mediocre results for the Liver-ITV; however, since we only have 19 patients, most predictions are accurate for non-toxic patients.

Dataframe	Average Accuracy	Best Model	Accuracy (B.M)	Sensibility (B.M)	Specificity (B.M)	AUC (B.M)
Liver	57.5%	D.T	62.5 %	1	0.4	0.5
Tumor	70.9%	D.T	81.8%	0	1	0.7
Liver & Tumor	62%	D.T	89.5%	0.75	0.93	0.5
Cou. Segments	56.3%	S.V.M	63.6%	1	0.3	0.5

Table 5.6: Results for doses dataframe with four last filters. The models yielding the best results are the decision tree (D.T) and support vector machine (SVM), achieving the highest accuracy when using structures related to the tumor and liver (excluding the Couinaud segments).

contingency matrix continue to be imprecise with the prediction of patients with toxicity. Combining these two previous ones, the average accuracy is not higher than the tumor case, however, with DT the highest accuracy so far is obtained, 89.4% with sensitivity and specificity of 0.75 and 0.93 respectively. Finally, with the dataframe filtered only using the Couinaud segments, low accuracies are obtained, with the maximum being 64

Using the dose dataframes, we have found that employing the volume column complicates the task of predicting toxicity, increasing the training time and decreasing the models' accuracy. Moreover, using the dataframe with all structures (four main and eight Couinaud segments) lowers accuracy, ultimately showing that filtering only by the main structures related to liver and tumor yields an accuracy of almost 90%.

Classification of Toxicity according to the Radiomic Features

Since our task is to predict the toxicity and take advantage of the fact that we have radiomic data of the patients, the same study will be performed but with the dataframes of the average and breathing cycle, evaluating the differences between them and trying to make predictions between both groups.

The procedure is analogous to the previous one, performing One Hot Encoding to differentiate between structures (and breathing cycles in the case of the corresponding dataframes), dividing the dataset into a 70-30% split, and obtaining accuracy values, contingency matrices, ROC curves, and the greatest weights in the prediction of the Decision Tree and Random Forest. Additionally, we aim to further contrast the differences between using the MinMaxScaler and the RobustScaler and to see whether it is advisable to use all Radiomic features or to perform PCA for prediction.

Dataframe	Average Accuracy	Best Model	Accuracy (B.M)	Sensibility (B.M)	Specificity (B.M)	AUC (B.M)
Average	60.9%	D.T	68.4 %	0.8	0.55	0.68
Average Cou.	75.42%	L.R	79.5%	0.5	0.93	0.88
B.C	90.73%	R.F	97.7%	0.95	1	0.999
B.C Cou.	89.12%	R.F	99%	0.98	0.996	0.999
Average (PCA)	54.55%	D.T	73.6%	0.6	0.89	0.68
Average Cou. (PCA)	63.14%	L.R	68.18%	0	1	0.566
B.C (PCA)	76.64%	R.F	88.88%	0.82	0.9	0.91
B.C Cou (PCA)	76.67%	R.F	88.7%	0.68	0.99	0.95

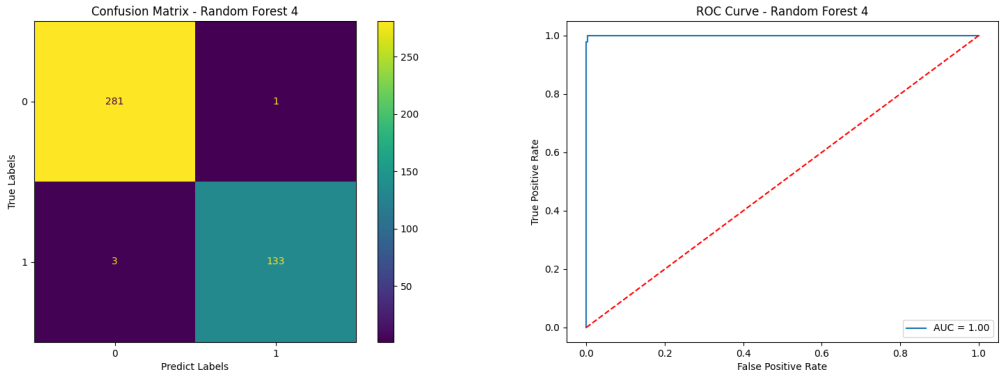
Table 5.7: Results for the four main dataframes, using **MinMaxScaler** and comparison between all radiomic features and dimensionality reduction method. The best-performing models are the decision tree (D.T), logistic regression (L.R), and random forest (R.F), with the highest accuracy reaching 99% and achieved for the B.C Couinaud dataframe without using PCA.

- **MinMaxScaler:** using all features and PCA.

Table 5.7 shows a comparison of the prediction metrics when using all the radiomic features, and after PCA is performed. With the *AVG dataframe*, a maximum accuracy of 68.4% is achieved with the Decision Tree and an average accuracy of 61% across all models, which is higher (though not by much) than the first case study conducted with the Dose dataframe. We encounter the same problems with the AUC and contingency matrices, being unable to correctly detect the True Positives.

Continuing with the *AVG dataframe with Couinaud*, the average accuracy increases to 75% and almost 80% with Logistic Regression. With this model, an AUC of 0.88 is obtained, which means a sensitivity and specificity of 0.5 and 0.9, respectively. It is important to highlight that although there are no substantial differences in the prediction of both models, performing Couinaud segmentation is relevant for prediction since all models increase their accuracy.

Next, the *B.C dataframe* is studied, with which an average accuracy of 90.8% and a maximum of 97.7% with the Random Forest are obtained. Indeed, the sensitivity and specificity are 0.95 and 1, respectively (AUC = 0.99), correctly identifying almost all positives and negatives, with very few false negatives and no false positives. Compared with the previous dataframe, all models achieve better accuracy except for Naive Bayes.



(a) Confussion Matrix B.C Couinaud

(b) ROC Curve B.C Couinaud

Figure 5.21: Confussion Matrix and ROC Curve for B.C Couinaud, 99% Accuracy obtained with Random Forest. MinMaxScaler and all radiomic features.

Lastly, the *BC dataframe with Couinaud* does not exceed the previous one in average accuracy, since Naive Bayes and LR are inferior, although we obtain better results with the other models. The DT, SVM, and RF give us accuracies of 98% for the first two, and 99% for the Random Forest, in all three cases the AUC is practically 1.

The training time for these models with MinMaxScaler has been approximately 55 minutes. By performing PCA with 6 components (as detailed as necessary in previous sections, for the MinMaxScaler case), it is significantly reduced to 19 minutes and 49 seconds. However, accuracies higher than 90% are not achieved, and the average accuracy is lower in all cases, with the best model being the RF for the BC dataframe case with 89% accuracy and AUC = 0.95.

Figure 5.21 shows the confusion matrix and ROC curve for the most accurate model, which is the Random Forest of the Breathing Cycle Couinaud, for the case of MinMaxScaler and all the radiomic features involved. It demonstrates how the identification of true positives and negatives is practically impeccable, especially in the case of patients with toxicity, which until now we had problems detecting correctly. This serves as a confirmation that Couinaud segmentation could be key to predicting this characteristic.

For the four dataframes, the most important radiomic features for prediction have been extracted, resulting in the *"Image Mean"* being the main one for the DT and RF, followed by *"glrlmLongRunLowGrayLevelEmphasis"*, and *"Image Maximum"* (DT) and *"shapeElongation"*, *"shapeMajorAxisLength"* (RF). When observing the variables resulting from the PCA loadings across all dataframes, that is, the variables with the most weight in each component when the dimensionality is reduced, we find *"ImageMean"*, *"gldmGrayLevelNonUniformity"*, and *"glcmSumAverage"* to be significant.

These highlighted features reflect various structural and intensity aspects of the tumor

and surrounding tissue, which are essential for understanding how radiotherapy impacts them. Toxicity can be significantly influenced by tissue heterogeneity, density, shape, and size of the tumor, each of which can alter the distribution and absorption of the radioactive dose [LLD+17b].

Therefore, it can be concluded that performing PCA does not constitute a significant improvement, since despite reducing training times, it only marginally improves accuracy for the AVG Dataframe, and for the rest of the dataframes, it results in a 10% decrease in accuracy for the best models. This observation suggests that while PCA can simplify the model training process by reducing dimensionality and potentially avoiding overfitting, the trade-off in terms of prediction accuracy might not justify its application in all cases.

- **RobustScaler:** using all features and PCA.

Performing the same procedure but using RobustScaler, we encounter a problem: satisfactory classification is achieved for the AVG and AVG with Couinaud dataframes (whose training time is approximately 6 minutes), but for the next two dataframes, convergence to a solution is not achieved, even when using PCA. To address this issue, a significant number of parameters were removed from GridSearch in an attempt to achieve classification for these latter dataframes.

After 28 minutes of training time, the results shown in Table 5.8 are obtained, where it is observed that the average Accuracy has increased for the first two dataframes compared to the MinMaxScaler, and the best models exceed these in accuracy. On the other hand, for BC and BC Cou, the average accuracy has decreased, but the best model of BC has more accuracy compared to MinMaxScaler. Finally, it is observed that performing PCA does not offer a great advantage, and contrary to the MinMaxScaler, the training time has significantly increased (up to 131 minutes). Lower average precisions are obtained, and the precisions of the best models are in all cases considerably lower than when performing PCA with MinMaxScaler.

The most important radiomic features turn out to be *"Image Mean"* for DT and RF, as well as *"Image Maximum"*, *"gldmHighGrayLevelEmphasis"* for DT, and *"glszmHighGrayLevelZoneEmphasis"*, and *"shapeElongation"* for RF. Just like when using the MinMaxScaler, *"Image Mean"*, as well as *"Shape Elongation"*, which is repeated, are of great importance.

These characteristics closely relate to the structural and intensity aspects of the tumor and the surrounding tissue, since the heterogeneity, density, shape, and size of the tumor can significantly influence toxicity [LLD+17b].

Dataframe	Average Accuracy	Best Model	Accuracy (B.M)	Sensibility (B.M)	Specificity (B.M)	AUC (B.M)
Average	63.08%	L.R	78.9%	0.6	1	0.89
Average Cou.	77.21%	L.R	84.1%	0.79	0.87	0.9
B.C	87.38%	D.T	98.3%	0.96	1	0.98
B.C Cou.	84.82%	D.T	98.56%	0.977	0.99	0.98
Average (PCA)	51.4%	D.T	63.15%	0.4	0.88	0.71
Average Cou. (PCA)	59.08%	S.V.M	68.18%	0	1	0.47
B.C (PCA)	60.36%	D.T	68.88%	0.43	0.88	0.66
B.C Cou (PCA)	62.7%	D.T	70.33%	0.5	0.8	0.65

Table 5.8: Results for the four main dataframes, using **RobustScaler** and comparison between all radiomic features and dimensionality reduction method. The best-performing models are the decision tree (D.T), logistic regression (L.R), and support vector machine (S.V.M), with the highest accuracy reaching 98.56% and achieved again for the B.C Couinaud dataframe without using PCA.

As **general conclusions** from the toxicity prediction in our cohort, and comparing between the two scalers or the use of a determined number of variables, we obtain:

- Using the dataframe with the doses and all structures is not effective for predicting toxicity, since the maximum accuracy obtained is 69%.
- Using the dataframe with the doses and filtering by structures or groups of structures results in an accuracy increase of up to 89.5%, making use of the structures Liver, Liver-ITV, ITV, and PTV.
- Using the dataframe with the radiomic features and all structures is effective for predicting toxicity, in addition, to the importance of Couinaud segmentation and preserving all stages of the respiratory cycle, obtaining a model with 99% accuracy.
- Using RobustScaler makes the training process difficult when we use a lot of data, or when we have many different classes at the same time, resulting in convergence problems.
- Using PCA decreases training times for the MinMaxScaler, but increases them for the RobustScaler, and the accuracies are lower compared with using all the radiomic features.

Dataframe	Average Accuracy	Best Model	Accuracy (B.M)	Sensibility (B.M)	Specificity (B.M)	AUC (B.M)
Average	68.21%	D.T	89.47%	0.87	0.91	0.89
Average Cou.	85.01%	L.R	93.18%	0.89	0.94	0.92
B.C	89.98%	R.F	100%	1	1	1
B.C Cou.	89.78%	D.T	100%	1	1	1

Table 5.9: Clasification of Disease: results for the four main dataframes, using **MinMaxScaler** and all radiomic features. The best results are achieved with the decision tree (D.T), logistic regression (L.R), and random forest (R.F) models, achieving 100% accuracy with the data frames from the breathing cycle.

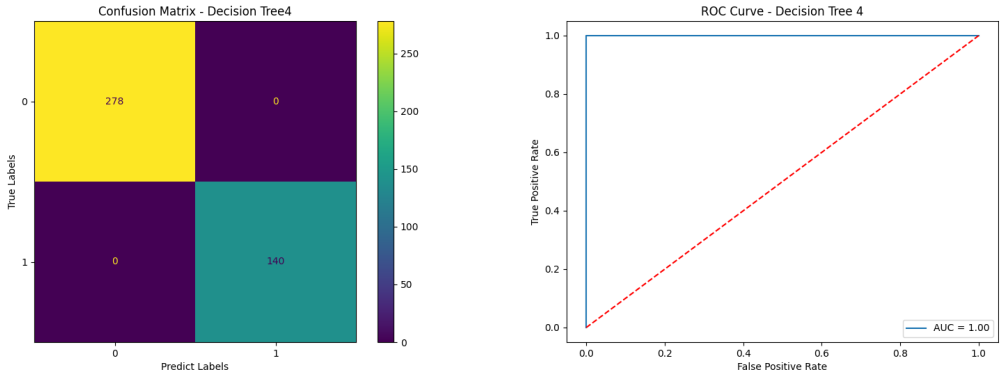
- In almost all cases, the most efficient models for this classification task are the Decision Tree and the Random Forest, considering as predictive variables Image Mean, Shape Elongation, and Image Maximum.

Classification of Disease according to the Radiomic Features

Finally, a short classification task is carried out between patients with Hepatocellular Carcinoma or Metastatic, using MinMaxScaler, all the radiomic features and for the four dataframes analogously to the previous ones. The training time is 50 minutes, but very good results are obtained for all four cases.

We observe in Table 5.9 that very good results are again obtained thanks to the B.C and the B.C Cou, where despite providing additional variability, 100% accuracy is achieved with the R.F and D.T. Figure 5.22 shows the confusion matrix and the ROC curve of this last example mentioned. The most significant predictive features are *"Image Mean"*, *"Image Maximum"*, *"firstorder10Percentile"* for the D.T, and *"Image Mean"*, *"firstorderMedian"*, and *"Image Maximum"* for the R.F model.

As we can see, these characteristics refer to the value of the pixel intensity, reflecting various aspects of the tumor's composition and structure that are critical in distinguishing between hepatocellular carcinoma and metastasis. For instance, the presence of denser regions or areas with different metabolic activity can be captured through these first-order statistics, since hepatocarcinomas and metastases may originate from different cell types and exhibit varying growth patterns and vascularization [GKH15].



(a) Confusion Matrix B.C Couinaud

(b) ROC Curve B.C Couinaud

Figure 5.22: Confusion Matrix and ROC Curve for B.C Couinaud, 100% Accuracy with decision tree. Accuracy is perfect despite being the case with the highest variability or cycle-structure combinations, correctly identifying all patients with Hepatocellular Carcinoma as well as Metastatic based on their radiomic features.

5.1.5 (R1) Which regions of the liver are under- or over-radiated?

Having performed the study of dose variables among the groups, and knowing their values in the Couinaud segments, we can identify which areas of the liver might be over-radiated or under-radiated, also considering the tumor's position. This location is established as previously explained in Table 4.1, there are some cases of patients where the tumor invades one segment more than another, but these differences will not be taken into account because they cannot be quantified exactly or approximately.

In patients with HCC, segment 1 is the least common for tumor location, with segment 8 being the most common as referenced in Renzulli et al. [RBP+22] in Section 2. In our case, it is observed in Figure 5.23 that at a general level of the dataset, segment 1 appears in two of our patients, while segment 8 appears in four of them.

Specifying between groups, in our patients with hepatocellular carcinoma, none have their segment 1 affected by the tumor, while two of the metastatic ones do. On the other hand, segment 8 is affected in one patient with HCC and three patients with metastases. Taking into account whether the patient has developed toxicity or not, it is again observed that segment 8 is repeated more times than 1, but at a general level, **the most repeated segments are 5, 6, and 7, with 7 being the most repeated in 7 patients**.

In Figure 5.24, an example of a patient from each group (toxicity and non-toxicity) can be seen, and the distribution of the maximum dose in each of the segments. According to our dose dataframe, *patient 8* has the tumor affecting segments 1, 7, and 8, while *patient 16* has the tumor in segment 6.

In the case of patient 8 (who developed toxicity), the maximum dose affects segments

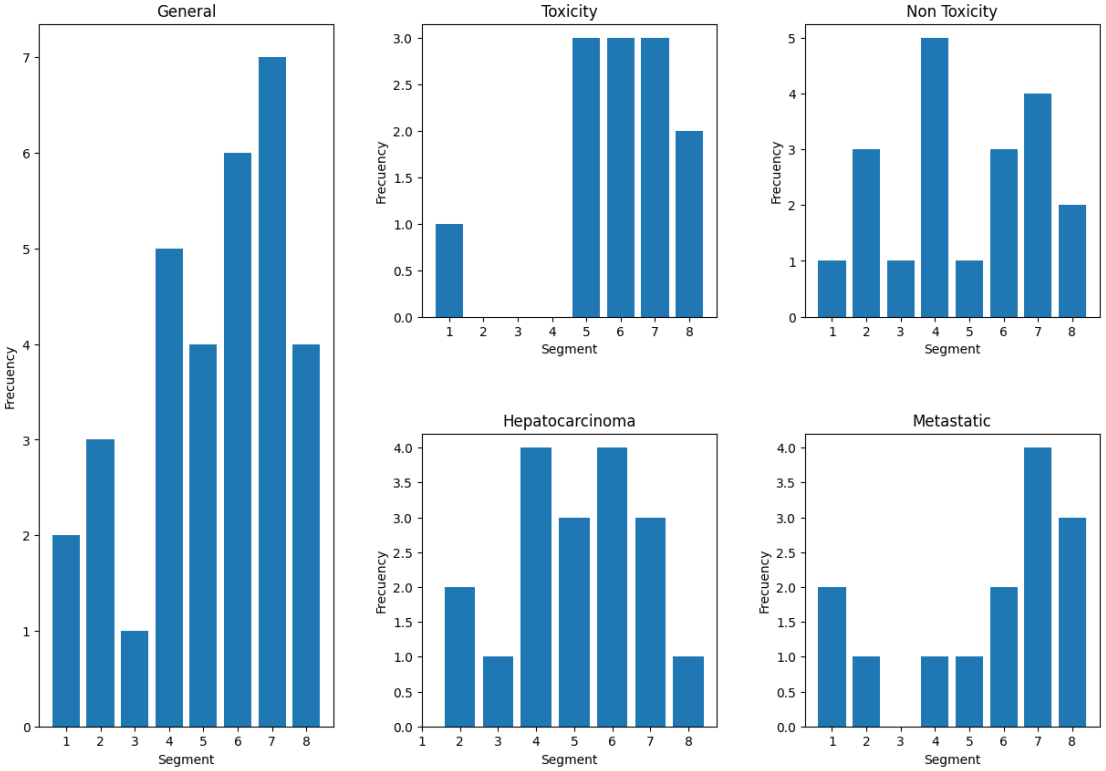


Figure 5.23: Tumor location in Couinaud Segments across all patients and classified by toxicity/disease.

1 and 8 with values of 52 and 57 Gy respectively, however, **segment 5 is treated with a maximum dose of 57 Gy just like segment 8, being a completely over-irradiated area**. Additionally, segment 7, which is also affected by the tumor, has the lowest value of maximum dose, approximately 7 Gy.

In the case of patient 16, segments 6 and 5 are approximately equally treated with the same maximum dose of 40 Gy, and the rest of the segments have very low values. Considering that the tumor is only in segment 6, **segment 5 is over-irradiated unnecessarily**, and despite this, the patient did not develop toxicity.

Another example is shown in Figure 5.25, where the dose ratio (Gy/cc) for patients 9 and 2 is observed. **Patient 9** developed toxicity, with their tumor affecting segments 6 and 7, and **patient 2**, who did not develop toxicity, in segments 1, 2, 4, and 5.

Patient 9 received a dose of nearly 2 Gy/cc in segments 5 and 6, even though their tumor did not affect segment 5, **thus it is over-irradiated**. On the other hand, Patient 2 did receive higher dose ratios in the corresponding segments, likely why they did not develop toxicity.

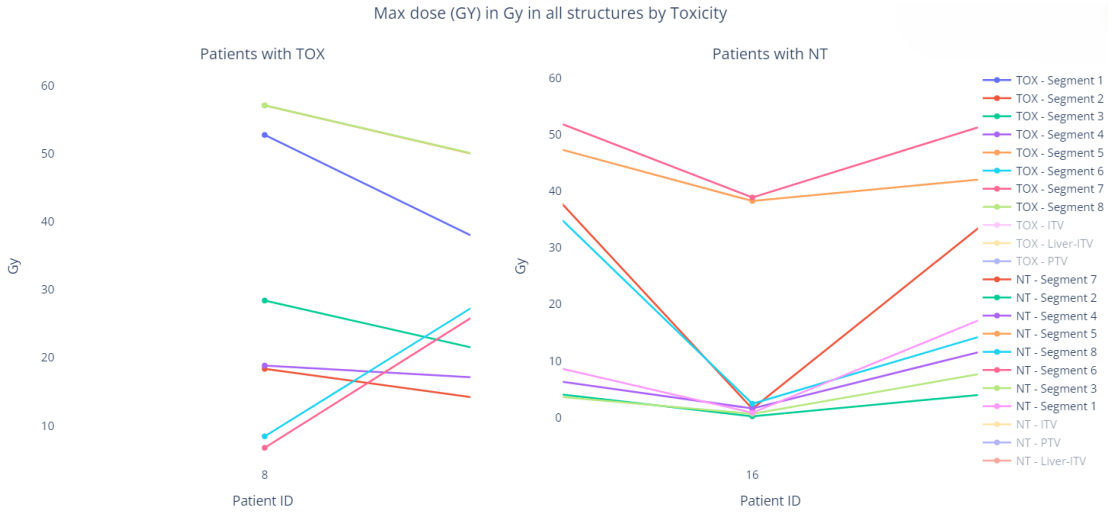


Figure 5.24: Comparative analysis of maximum dose (Gy) in patients exhibiting toxicity versus non-toxicity, in LiverVis Dashboard.

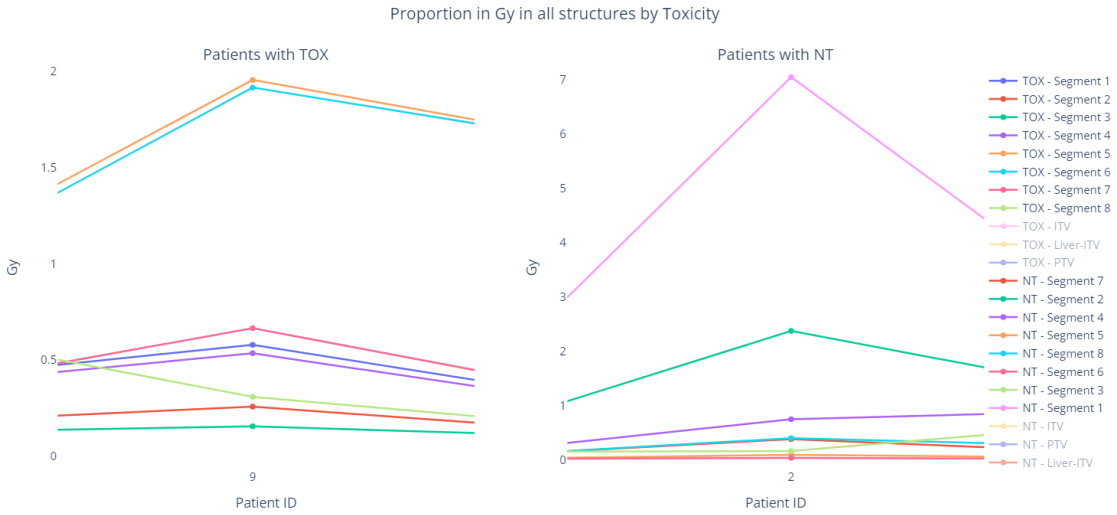


Figure 5.25: Comparative analysis of dose proportion (Gy) in patients exhibiting toxicity versus non-toxicity, in LiverVis Dashboard.

5.1.6 (R2) Which are there disparities observed in the population?

As observed in T4, there are notable differences in both groups regarding toxicity and type of cancer (HCC or MET). The analysis of dose measures across structures demonstrates differences among patients in Segment 5, where the dose proportion is significantly higher in patients who have developed toxicity, which again occurs on the mean dose in the same segment (Figures 5.2 and 5.3). These differences can also be observed in segments 1 and 6, for proportion and mean dose respectively, where the same phenomenon is noted, that is, radiation is higher in patients with toxicity than those without toxicity.

Applying this study analogously to find differences between patients with HCC and MET, differences are found in almost all segments, as well as in the Liver-ITV for the average and minimum doses, with these two variables tending to be higher in the case of patients with metastasis.

Radiomic features turn out to be very useful for the prediction of Toxicity, especially in the case where we consider all respiratory cycles and also Couinaud, where despite adding a lot of variability, accuracies of almost 100% are obtained. During these analyses, it is found that the radiomic features with the most weight in making the prediction are *"Image Mean"*, *"glrlmLongRunLowGrayLevelEmphasis"*, *"shapeElongation"*, *"shapeMajorAxisLength"*, and *"Image Maximum"*.

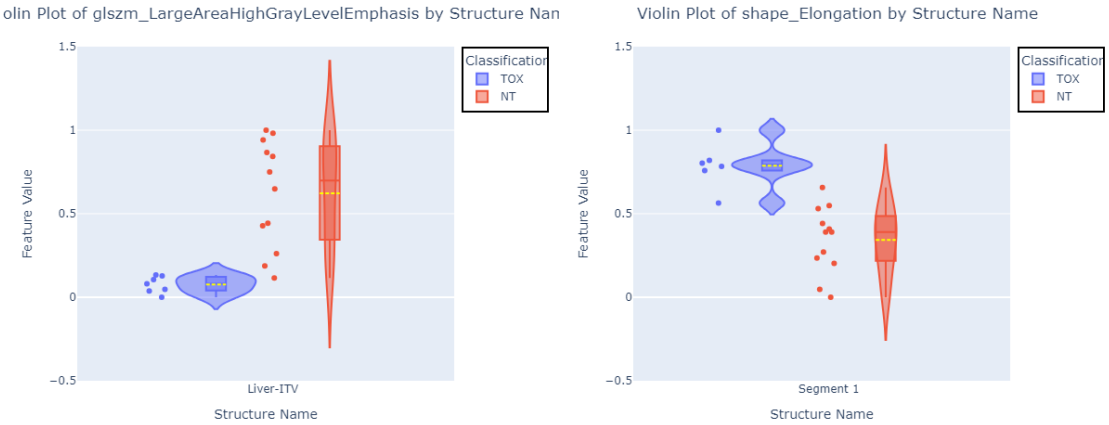
If we try to find differences between both groups in terms of toxicity/disease and based on radiomic features, we can once again perform Multivariate Kruskal-Wallis, Mann-Whitney, and Kolmogorov-Smirnov tests. Additionally, as we are going to work with the dataframe of radiomic features (112 features) instead of 5 variables as with the dose dataframe, to avoid the discovery of false positives, we will perform Bonferroni correction for multiple comparisons. This way, we ensure that the differences we find between structures or structure-cycle combinations are truly significant.

Differences in Toxicity

With the aim of identifying differences between patients with Toxicity and No Toxicity in terms of radiomic features, we will apply the three tests previously mentioned to each of the dataframes, taking into account that due to the large number of features, we will apply the Bonferroni correction.

Applying the Multivariate Kruskal-Wallis test without the Bonferroni correction, we found multiple differences in various structures and for all dataframes. However, when applying the correction, these differences were not sufficient, ultimately indicating that there are no differences in any structure/radiomic feature. On the other hand, the Mann-Whitney and Kolmogorov-Smirnov tests do detect significant differences in the Liver-ITV (average dataframe) and Segment 1 (average Couinaud) after applying the Bonferroni correction.

The significant feature found by the tests in the Liver-ITV is *"glszmLargeAreaHighGrayLevelEmphasis"*, while for Segment 1 it is *"ShapeElongation"*. Using the dashboard



(a) Large Area High Gray Level Emphasis for Liver-ITV

(b) Shape Elongation for Segment 1

Figure 5.26: Comparison between Toxicity and Non-Toxicity, for the features and structures with significant differences after tests are performed.

we have designed, it is possible to observe how these magnitudes vary in the case of toxicity and non-toxicity, as seen in Figure 5.26. In this figure, there is a significant difference in the distribution of both groups, with the group of patients with toxicity having very low values in the case of *"LargeAreaHighGrayLevelEmphasis"* and high in the case of *"ShapeElongation"*. Additionally, the box plot shows how the means are significantly different from each other.

On the other hand, **Large Area High Gray Level Emphasis (LAHGLE)** measures the predominance of zones with high gray levels and large areas in the image, that is, it indicates the presence of large and dense structures in the image [pyr]. High values of LAHGLE, as is the case with patients with Non-Toxicity, indicate a predominance in large areas with high gray levels (solid and dense tissues), while patients with toxicity have low values indicating the absence of large and dense areas (homogeneous tissues or with necrotic characteristics) [TFN+09].

Shape Elongation is the ratio between the principal components or axes of an ROI; therefore, an Elongation value close to 1 indicates that the object is almost circular, while elongations close to 0 indicate that the object is more elongated [pyr]. This suggests that patients with toxicity generally have more circular Segment 1s compared to patients who have not developed toxicity, who have more elongated Segment 1s.

It is necessary to mention that in the models previously described in the predictive analysis (particularly, RF), some of the most recurrent features with more weight in the predictions were both Shape Elongation and LAHGLE, supporting the idea that these variables are important for the development of toxicity in patients.

Differences in Disease

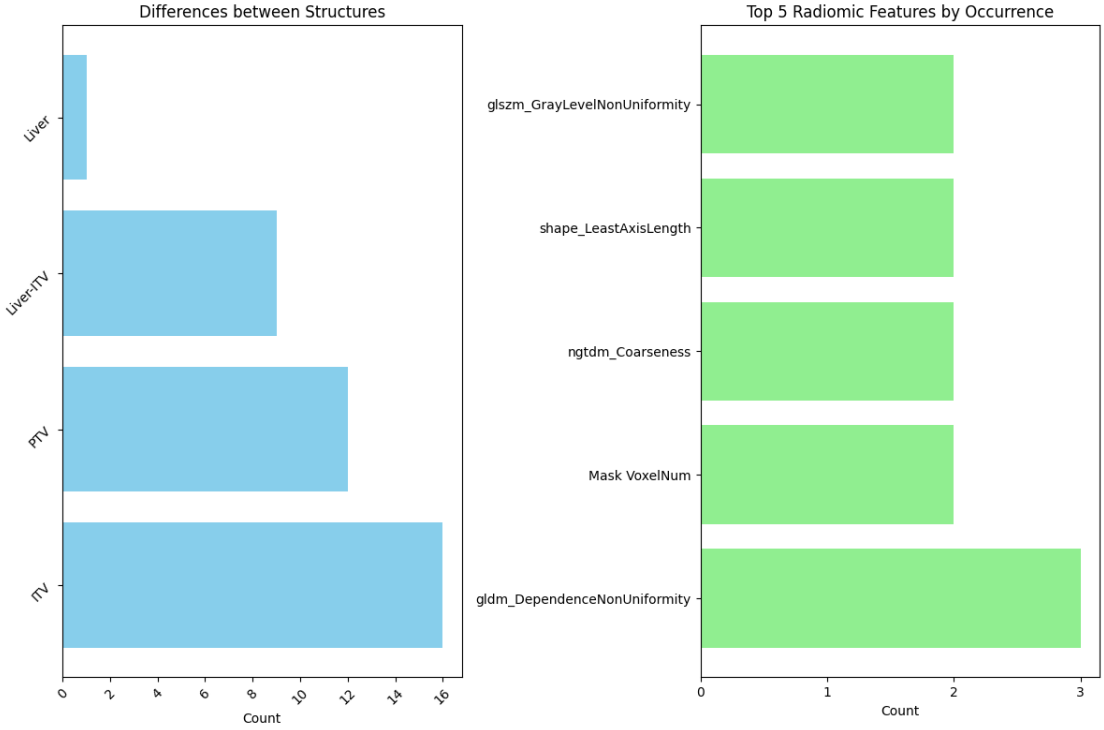
Having applied the three tests to our dataframes, in an attempt to find differences between the two groups and having performed the Bonferroni correction, no significant differences were found for any of them. However, by removing the Bonferroni correction and selecting those differences found by all three tests simultaneously across all dataframes, we obtain the results shown in the following figures.

Figures 5.27 (a) and 5.27 (b) show the counts of differences found both for the structures and the radiomic features associated with them, for the Average (a) and Average Couinaud (b) dataframes. The tests find differences in 16 radiomic features of the ITV and 12 differences in the PTV. Among the features found as significant, *"gldmDependenceNonUniformity"* is common to three structures. On the other hand, the most recurrent segment is Segment 7, and the most repeated features are *"gldmSmallDependenceEmphasis"*.

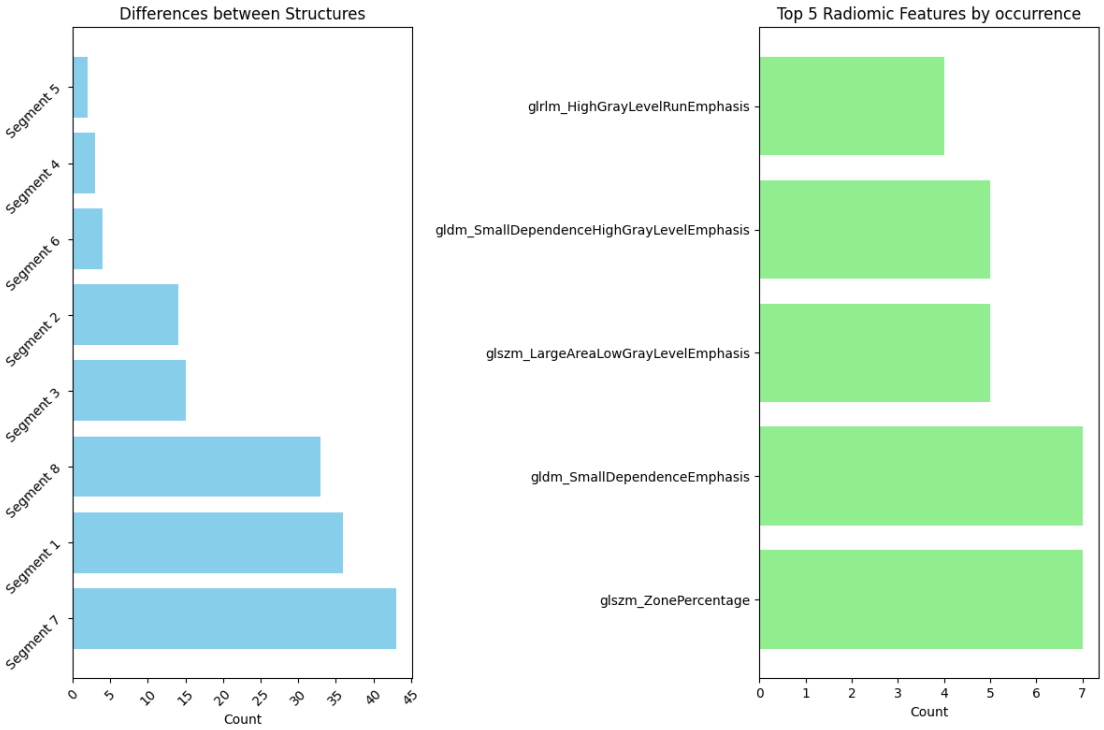
Analogously, Figures 5.28 (a) and 5.28 (b) display the counts for the Breathing Cycle and Breathing Cycle Couinaud dataframes, respectively. In the first case, the structure where the most differences are found again is the ITV, and the most significant features are *"ShapeSurfaceVolumeRatio"* and *"gldmDependenceNonUniformity"*. Finally, for the Couinaud case, Segment 8 is where the most differences are found, in features such as *"glcmLdmn"* and *"gldmSmallDependenceEmphasis"*.

As observed in these found differences, the variable *"gldmDependenceNonUniformity"* is one of the most recurrent for the average and breathing cycle data frames, where our structures are the main focus. Similarly, *"gldmSmallDependenceEmphasis"* is one of the most repeated for the dataframes with Couinaud segmentation. Figure 5.29 shows how the value of *"gldmSmallDependenceEmphasis"* varies, generally being higher for patients with hepatocellular carcinoma compared to those with metastases. On the other hand, Figure 5.30 displays the variation of Dependence Non-Uniformity, generally higher in patients with Hepatocellular Carcinoma, and in the PTV compared to the ITV.

Dependence Non-Uniformity assesses the uniformity of texture dependencies across an image. A lower score suggests a higher uniformity, meaning the dependencies throughout the image are more consistent. **Small Dependence Emphasis** evaluates the occurrence of short dependencies, with a higher value reflecting a prevalence of smaller dependencies, which indicates less uniform textures [pyr]. In both features, the values for Metastatic are lower than those for Hepatocellular carcinoma, indicating therefore that the texture of this type of cancer is more uniform.

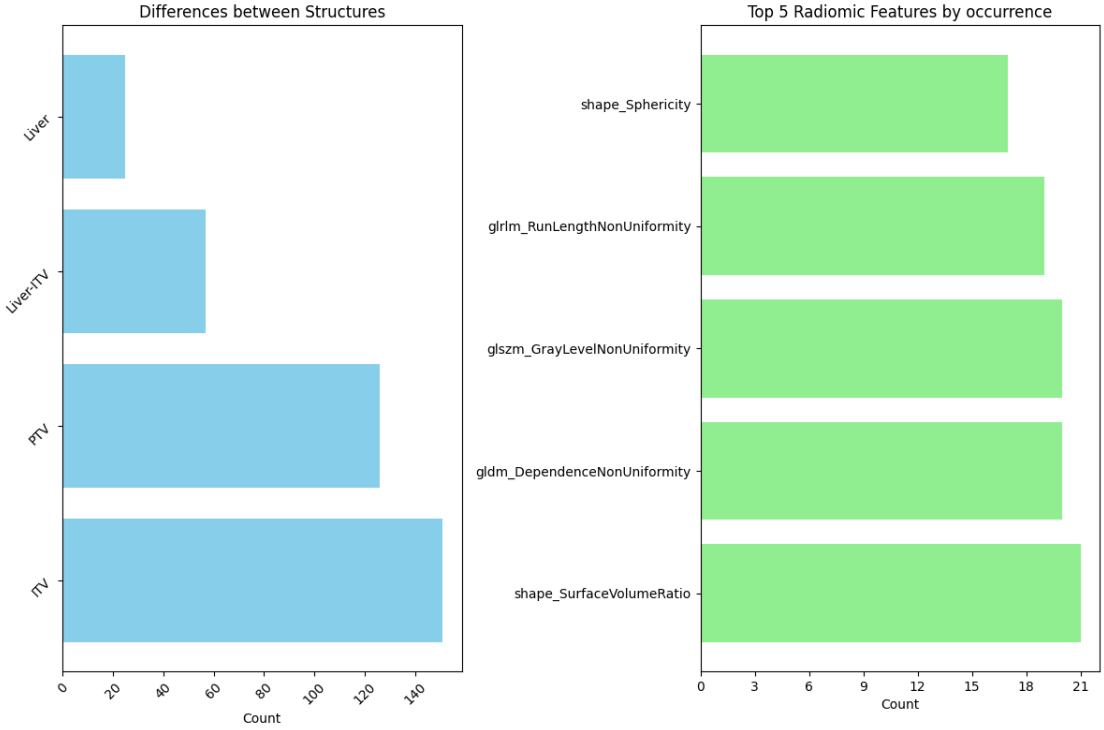


(a) Differences obtained in Average dataframe

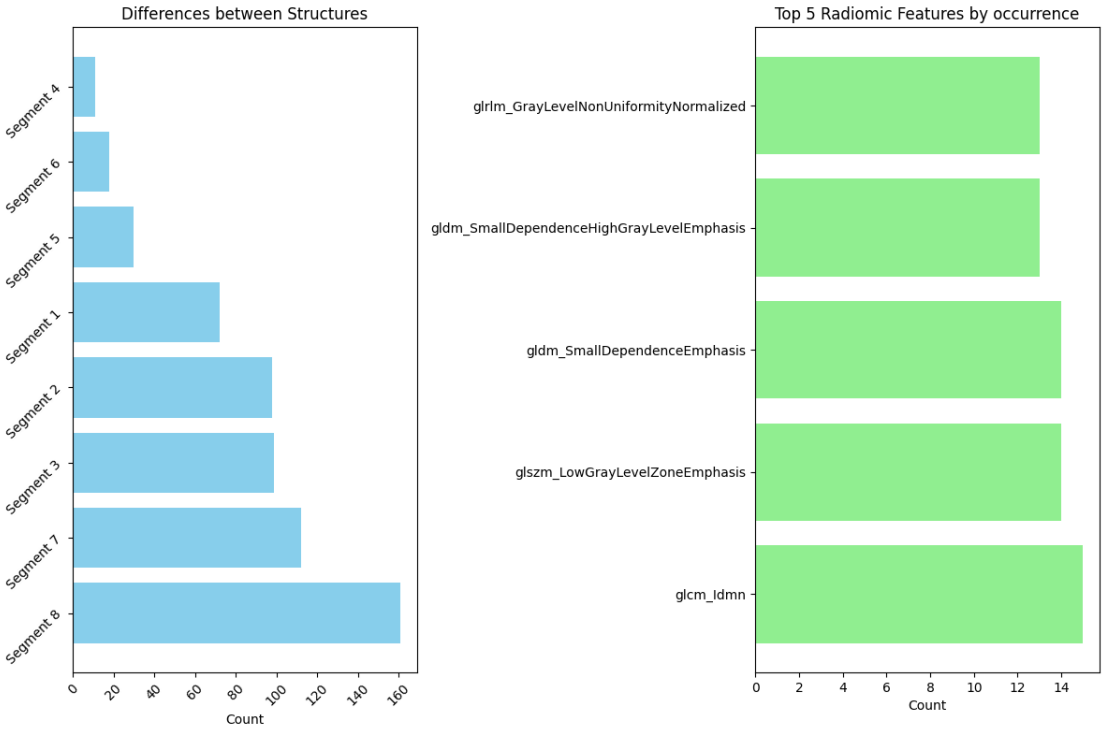


(b) Differences obtained in Average Couinaud data frame

Figure 5.27: Comparison between differences in Toxicity obtained with tests, average and average Couinaud data frames.

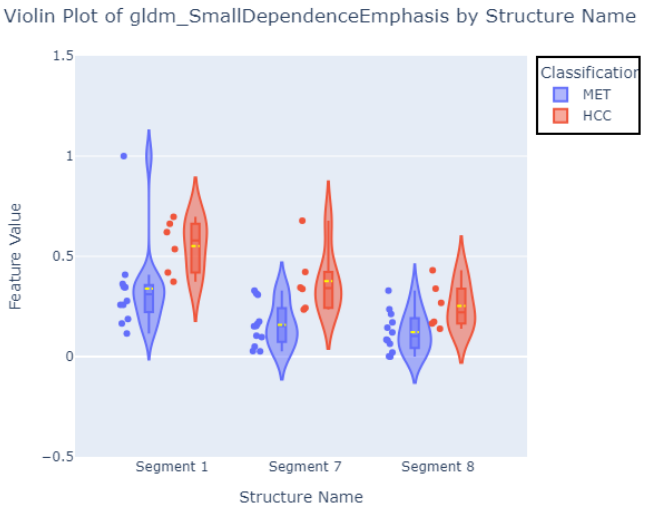


(a) Differences obtained in Breathing Cycle data frame

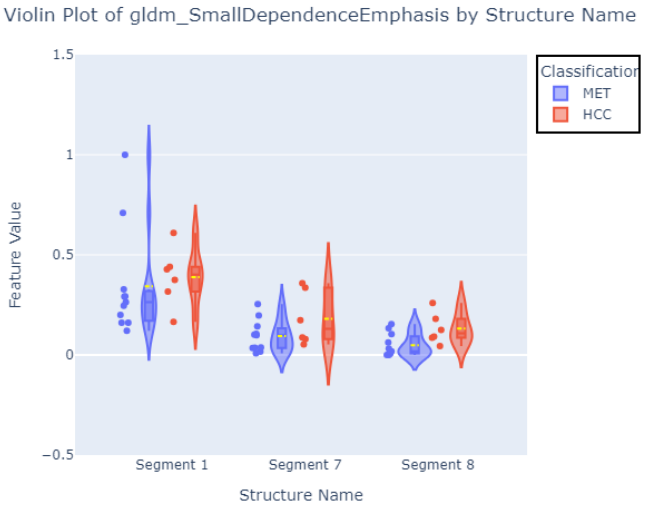


(b) Differences obtained in breathing cycle Couinaud data frame

Figure 5.28: Comparison between differences in toxicity obtained with tests, breathing cycle, and breathing cycle Couinaud data frames.

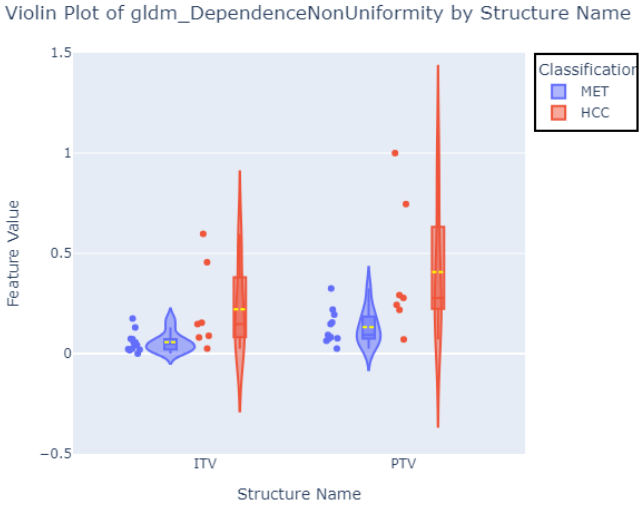


(a) Dependence Non Uniformity for Segments 1,7 and 8, average Couinaud data frame

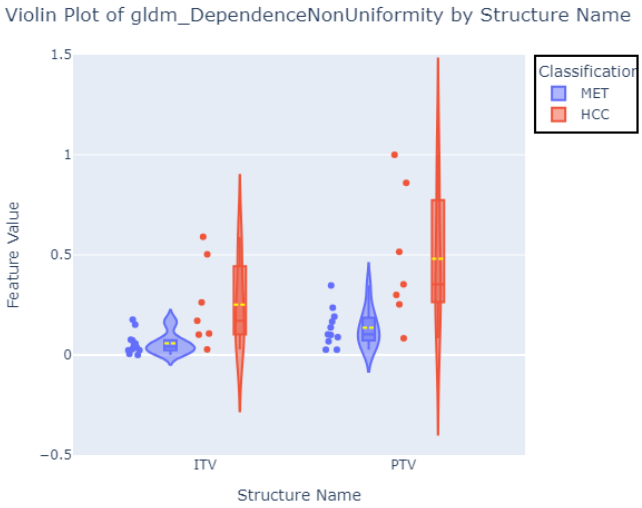


(b) Dependence Non Uniformity for Segments 1,7 and 8, breathing cycle Couinaud data frame

Figure 5.29: Comparison between Hepatocarcinoma and Metastasis, for the features and structures (Couinaud) with significant differences after tests are performed.



(a) Dependence Non Uniformity for ITV and PTV, average data frame



(b) Dependence Non Uniformity for ITV and PTV, breathing cycle data frame

Figure 5.30: Comparison between Hepatocarcinoma and Metastasis, for the features and structures with significant differences after tests are performed.

As **general conclusions** from both research questions, we obtain:

- The use of Couinaud segmentation has proven beneficial not only for adding another modality to our data but also for analyzing how radiation affects specific areas of the liver.
- Couinaud segmentation, coupled with the localization of the tumor position, helps us understand which areas of the liver may be over-irradiated or under-irradiated, establishing differences and patterns between patients with and without toxicity.
- By utilizing dose data on the structures and through statistical testing, we have observed differences in some structures of both populations, particularly in the Couinaud segments.
- Radiomic data aid not only in predicting toxicity or stratifying patients but also in identifying which image characteristics can be crucial when discerning differences between populations, whether in terms of toxicity or tumor type.

CHAPTER 6

Conclusion and Future Work

During the development of this thesis, we investigated and analyzed the radiomic features obtained from CT scans of patients who have undergone radiotherapy for liver cancer. Specifically, we aimed to develop a visual analytics tool for our collaborating specialists at the University of San Diego, enabling them to draw conclusions from their radiomic and clinical datasets. Throughout the development of this tool, numerous stages were involved including data extraction, Couinaud segmentation, and preprocessing of the dataframes such as outlier detection or dimensionality reduction. With the data prepared for use, we proceeded to patient stratification, predictive analysis using Machine Learning models to predict toxicity, and analysis of different measures of radiation therapy doses on the structures. This allowed us to address the two main questions of the thesis: whether there are differences in the populations of our dataset (Toxicity/No Toxicity or Hepatocarcinoma/Metastasis) and whether there are areas of the liver that are over-irradiated or underirradiated considering the tumor position.

6.1 Limitations and Future Work

During the development of this Thesis, we encountered some challenges, many of which have been solved, but would require further investigations in the future.

For the Couinaud segmentation, we attempted to design a Deep Learning model for the automatic segmentation of our livers once extracted. We tried to replicate several architectures mentioned in the papers cited in the methodology, but the high computational cost and the lack of reference codes made the task difficult, yielding very unsatisfactory results from the network output. Instead, manual segmentation was chosen, which, despite being a very simple task given the capabilities of 3D Slicer and the references we have, can be imprecise. Indeed, whether with an algorithm that segments our structures automatically or semi-automatically or with manual segmentation, the results should be

corrected or supervised by an expert, as in both ways, we could be introducing a bias in the radiomic features obtained or in any of the tasks carried out.

Before designing the current tool, during the decision of which work environment was suitable, an attempt was made to create a more complete VA tool with the React JS framework, using Django or Python Flask as a backend. However, lack of knowledge in using this framework and Javascript in general made it a very complex task. For future implementations, we suggest using D3.js, Python Flask, or Django, as they are powerful and flexible tools for data visualization in an interactive, informative, and adaptive manner according to the specific needs of the project.

In articles about the design and development of VA tools, the use of linked visualizations is very common, as it allows changing or interacting once with a component, and having these changes effected in the rest. This could not be achieved due to the limitations of Panel and Python, since the library treats each visualization element individually, with its own buttons for interaction. Using the tools discussed in the previous paragraph, future implementations can easily create interactive visualizations among panels, so that changes in one chart affect the others.

To study which areas are over-irradiated or under-irradiated, we compared the tumor areas with the measurements taken in the radiation therapy doses, always visually. However, scaling this to more patients would be problematic, and we should consider the option of establishing a criterion for over-irradiated and under-irradiated areas automatically. Additionally, extending comparisons to more than two subjects or groups would be ideal in future implementations.

Finally, a future thorough evaluation of the tool would be required. With the feedback of an expert in the field, we could assess the quality of the tool from a more practical standpoint in the field of radiomics and medicine, thereby understanding if our work can truly be useful for researchers and clinicians.

In future work, we want to incorporate more patients into our dataset, so that the conclusions obtained throughout the thesis can be strengthened or contrasted. Additionally, a more robust Couinaud segmentation, i.e., one supervised by an expert, would also lend credibility to the solutions that have been proposed. Finally, the design of the tool will be improved to allow linked visualizations between the different panels provided, as well as the implementation of more types of representations. Thus, an expert in the field could test it and provide feedback on its usefulness.

List of Figures

2.1 Structural organisation of the liver [bio] and the hepatocyte [HJR+14] . . 8
2.2 Comparison between the liver's anatomy (a) [Liv] and Couinaud segmentation (b) [ima] 10
2.3 Dose distribution of SBRT for Hepatocellular Carcinoma [San14]. 12
2.4 Volume concepts used in RT planning [SRM+19]. 15
2.5 Dose-volume histograms comparisons for Liver Metastases [WCH18] . . . 16
2.6 Breathing Cycle and Average, representative diagram. 18
2.7 Workflow of radiomic data, a process that starts with acquiring images and extends to analyzing the extracted features [SGB+21]. 19
2.8 Categories of radiomic features: shape, first order, and texture features [ZZC+23]. 20

3.1 Radiomics Workflow proposed by Huyhn et al. [HCN+16]. 27
3.2 (a) Segmentation with Region Growing method [LPS+20] and Segmentation with Graph Cut method (b) [gra] 27
3.3 TripVista: visual analytics tool for traffic trajectory data [GWY+11]. . . . 31
3.4 iVisClassifier: Visual Analytics tool based on supervised dimension-reduction [CLKP10]. 32
3.5 iVAR: Visual Analytics tool for radiomic features [YJY+17]. 32
3.6 Visual Analytics tool for radiogenomics and Clincal Information [ESNH+23]. 33

4.1 Processes and subprocesses of the implementation pipeline. 36
4.2 Structures of Interest: Liver-ITV (gray), ITV (green), and PTV (yellow) visualization in Slicer 3D. 37
4.3 Comparison of our manual Couinaud segmentation using 3D Slicer (b) with the annotated reference from the GitHub repository (a) [git]. 38
4.4 Location of the tumor in the liver, between segments 2 and 3. Visualization in Slicer 3D. 39
4.5 Pipeline of Dataframes building, a combination of radiomic features and clinical data. 40
4.6 PCA (2 components) with RobustScaler for outliers detection, Breathing Cycle dataframe. 41
4.7 Distribution of 10 random radiomic features, comparison between dataframes. 43

4.8 Comparison of the explained variance between MinMaxScaler and RobustScaler. 46
4.9 Comparison between Hierarchical Clustering for Breathing Cycle, and Average with Couinaud Segments. 48
4.10 Elbow Method for the optimal number of Clusters (K). 49
4.11 Dimensionality Reduction and Clustering for Average and Breathing Cycle Couinaud datasets. 51
4.12 Pipeline of combined dimensionality reduction techniques. 52
4.13 Comparison between two examples of combined dimensionality reduction, for average data frame (a) and breathing cycle Couinaud data frame (b). . . 52
4.14 Predictive analysis workflow: training datasets, ML models and evaluation. 54
4.15 Visualization of DICOM RT Structures in 3D Slicer, example with one patient. 56
4.16 Software used for implementation, 3D Slicer and Python. 57
4.17 Pipeline of how to structure Panel's coding. 59
4.18 General overview of the LiverVis Tool: Main screen (A), exploratory analysis (B), machine learning analysis (C), comparative analysis (D). 60
4.19 Desing of Logo and Favicon for the LiverVis Tool, based on Hepatocytes shape. 61
4.20 Comparison between Sunburst plot of the whole dataset (a), and classified by toxicity (b). 62
4.21 Number of patients according to clinical data (gender, disease, and toxicity) and number of available structures in the dataset. 63
4.22 Parallel charts: categorical coordinates (a) and parallel coordinates (b). . 64
4.23 Violin plot for *shape elongation* feature, patients classified by toxicity and specific structures. 65
4.24 Circos Plot, values of 5 radiomic features for six patients. 66
4.25 Visualization of the doses across CT images, interactive selection of patients and slices. 66
4.26 Visualization of the DVHs of patient's structures and measures of the Liver-ITV doses. 67
4.27 Visualization of the variable 'Proportion', patients classified by toxicity (left) and non-toxicity (right). 67
4.28 Heatmap of correlation between radiomic Features, breathing cycle data frame, and no correlation filtering. 68
4.29 Cohort stratification for average (a) and breathing cycle data frame (b), comparative analysis in LiverVis Dashboard 70

5.1 Comparison between dimensionality reduction methods, using MinmaxScaler and RobustScaler in Average dataframe. Being the data frame with the fewest data points, the PCA and UMAP algorithms are capable of clearly separating into two distinct groups when using MinMaxScaler, whereas RobustScaler is better at detecting outliers. 73

5.2 Comparison between dimensionality reduction methods, using MinmaxScaler and RobustScaler in Breathing Cycle Couinaud dataframe. Being the data frame with the most data points, the algorithms are not capable of separating when using MinMaxScaler, whereas RobustScaler can detect outliers. . . . 74
5.3 Stratification of patients in the average data frame, different configurations. In (a) UMAP reduces the dimensionality in two differentiated groups and metrics indicate clusters are dense and well-separated from each other. In (b) PCA separates into groups, but clusters have internal dispersion. 76
5.4 Stratification of patients in the average Couinaud data frame, different configurations. Both methods provide inefficient values of metrics, indicating that the separation of clusters is not optimal. 78
5.5 Stratification of patients in the Breathing Cycle data frame, different configurations. UMAP and Mean-Shift (a) separate our data into four consistent clusters, while PCA and K-Means (b) separate into three clusters but are not compact. 79
5.6 Stratification of patients in the Breathing Cycle Couinaud data frame, different configurations. In both cases, the results are good according to the metrics but in (a) two of the clusters contain only one point each, indicating that the patients differ from the cohort. 81
5.7 Proportion (Gy/cc) along the structures and patients. 82
5.8 Proportion of the Dose across Segment 5, comparing to ITV, PTV, and Liver-ITV. The dose proportion over this segment tends to be higher in patients with toxicity than in those without. 83
5.9 Mean of the Dose across Segment 5, comparing to ITV, PTV, and Liver-ITV. The average dose in Segment 5 is higher in patients with toxicity. 84
5.10 Minimum of the Dose across Segment 5, comparing to ITV, PTV, and Liver-ITV. The minimum dose received by Segment 5 is apparently higher in patients with toxicity, above all in patient nº 9. 84
5.11 Minimum of the Dose across Segment 5, comparing to Liver-ITV. After zooming on Segment 5 and Liver-ITV, we see the difference in the minimum dose received by Segment 5 is higher in patients with toxicity. 84
5.12 Proportion of the Dose across Segment 1, comparing to ITV, PTV, and Liver-ITV. The dose proportion is lower in patients with toxicity. 85
5.13 Mean of the Dose across Segment 6, comparing to ITV, PTV, and Liver-ITV. The average dose across Segment 6 is higher in patients with toxicity than in patients without toxicity. 85
5.14 Min Dose across Segment 1, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 1 for patients with HCC is practically 0 Gy, while patients with MET have radiation levels above 0 Gy. 86
5.15 Min Dose across Segment 4, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 4 for patients with HCC is lower than in patients with MET. 87

5.16 Min Dose across Segment 5, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 5 for patients with HCC is lower than in patients with MET. 87
5.17 Min Dose across Segment 7, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 7 for patients with HCC is lower than in patients with MET. 87
5.18 Min Dose across Segment 8, compared to ITV, PTV, and Liver-ITV. The minimum dose across segment 8 for patients with HCC is lower than in patients with MET. 88
5.19 Confusion Matrix and ROC Curve of the Best Model (D.T and the volume removed). The model predicts accurately toxicity and non-toxicity patients according to the dose data frame, but the rates of false positives and negatives result in an AUC of 0.7. 89
5.20 Confusion Matrices for ITV and Liver-ITV filters. Good results are obtained with the DT for prediction using only the ITV, and mediocre results for the Liver-ITV; however, since we only have 19 patients, most predictions are accurate for non-toxic patients. 91
5.21 Confussion Matrix and ROC Curve for B.C Couinaud, 99% Accuracy obtained with Random Forest. MinMaxScaler and all radiomic features. 94
5.22 Confusion Matrix and ROC Curve for B.C Couinaud, 100% Accuracy with decision tree. Accuracy is perfect despite being the case with the highest variability or cycle-structure combinations, correctly identifying all patients with Hepatocellular Carcinoma as well as Metastatic based on their radiomic features. 98
5.23 Tumor location in Couinaud Segments across all patients and classified by toxicity/disease. 99
5.24 Comparative analysis of maximum dose (Gy) in patients exhibiting toxicity versus non-toxicity, in LiverVis Dashboard. 100
5.25 Comparative analysis of dose proportion (Gy) in patients exhibiting toxicity versus non-toxicity, in LiverVis Dashboard. 100
5.26 Comparison between Toxicity and Non-Toxicity, for the features and structures with significant differences after tests are performed. 102
5.27 Comparison between differences in Toxicity obtained with tests, average and average Couinaud data frames. 104
5.28 Comparison between differences in toxicity obtained with tests, breathing cycle, and breathing cycle Couinaud data frames. 105
5.29 Comparison between Hepatocarcinoma and Metastasis, for the features and structures (Couinaud) with significant differences after tests are performed. 106
5.30 Comparison between Hepatocarcinoma and Metastasis, for the features and structures with significant differences after tests are performed. 107

List of Tables

2.1 Comparison of Radiotherapy Techniques for Liver Cancer. 13

4.1 Doses data frame: structures, measures, clinical data, and tumor location. 40

5.1 Detection of outliers with local, global, and combination of methods. . . . 72
5.2 Non-parametric tests, differences found in structures and variables (Toxicity/Non-Toxicity). 83
5.3 Non-parametric tests, differences found in structures and variables (Hepatocarcinoma/Metastasis). 86
5.4 Results for complete doses data frame and with volume column removed. Decision Trees (D.T) achieve the best results in both cases, increasing accuracy by removing the volume column and also reducing training time. 90
5.5 Results for doses dataframe with the three first filters. Promising results are obtained with the random forest (R.F) and logistic regression (L.R) models, however, the number of data points is very low after filtering by structure, equal to the number of patients. 91
5.6 Results for doses dataframe with four last filters. The models yielding the best results are the decision tree (D.T) and support vector machine (SVM), achieving the highest accuracy when using structures related to the tumor and liver (excluding the Couinaud segments). 92
5.7 Results for the four main dataframes, using **MinMaxScaler** and comparison between all radiomic features and dimensionality reduction method. The best-performing models are the decision tree (D.T), logistic regression (L.R), and random forest (R.F), with the highest accuracy reaching 99% and achieved for the B.C Couinaud dataframe without using PCA. 93
5.8 Results for the four main dataframes, using **RobustScaler** and comparison between all radiomic features and dimensionality reduction method. The best-performing models are the decision tree (D.T), logistic regression (L.R), and support vector machine (S.V.M), with the highest accuracy reaching 98.56% and achieved again for the B.C Couinaud dataframe without using PCA. 96

5.9 Clasification of Disease: results for the four main dataframes, using **MinMaxScaler** and all radiomic features. The best results are achieved with the decision tree (D.T), logistic regression (L.R), and random forest (R.F) models, achieving 100% accuracy with the data frames from the breathing cycle. . 97

Acronyms

AI Artificial Intelligence. 2

CT Computed Tomography. 2

HCC Hepatocellular carcinoma. 8

ITV Internal Target Volume. 2

ML Machine Learning. 2

OARs Organs at Risk. 1

PCA Principal Component Analysis. 5

PTV Planning Target Volume. 2

RT Radiotherapy. 11

SBRT Stereotactic Body Radiation Therapy. 1

VA Visual Analytics. 2

Bibliography

[AHT20] Shaeela Ayesha, Muhammad Kashif Hanif, and Ramzan Talib. Overview and comparative study of dimensionality reduction techniques for high dimensional data. *Information Fusion*, 59:44–58, 2020.

[AMB10] Sherif R Z Abdel-Misih and Mark Bloomston. Liver anatomy. *Surg. Clin. North Am.*, 90(4):643–653, August 2010.

[Apt15] Udayan Apte. Chapter 1 - liver regeneration: An introduction. In Udayan Apte, editor, *Liver Regeneration*, pages 2–11. Academic Press, Boston, 2015.

[ARJA21] Zobair Arya, Gerard Ridgway, Arun Jandor, and Paul Aljabar. *Deep Learning-Based Landmark Localisation in the Liver for Couinaud Segmentation*, pages 227–237. 07 2021.

[ARZN+19] Anan Abu Rmilah, Wei Zhou, Erek Nelson, Li Lin, Bruce Amiot, and Scott L. Nyberg. Understanding the marvels behind liver regeneration. *WIREs Developmental Biology*, 8(3):e340, 2019.

[Asl20] Muhammad Aslam. Introducing kolmogorov–smirnov tests under uncertainty: An application to radioactive data. *ACS Omega*, 5(1):914–917, 2020. PMID: 31956845.

[BAC+20] Julien Bellec, Florian Arab, Joël Castelli, C. Lafond, and E. Chajon. Itv versus mid-ventilation for treatment planning in lung sbrt: A comparison of target coverage and ptv adequacy by using in-treatment 4d cone beam ct. *Radiation Oncology*, 15, 03 2020.

[Baz23] Peter J. Bazira. Anatomy of the liver. *Surgery (Oxford)*, 41(6):313–318, 2023.

[BB19] Aditya Hari Bawono and Fitra Abdurrachman Bachtiar. Outlier detection with supervised learning method. In *2019 International Conference on Sustainable Information Engineering and Technology (SIET)*, pages 306–309, 2019.

[BBW77] A Blouin, R P Bolender, and E R Weibel. Distribution of organelles and membranes between hepatocytes and nonhepatocytes in the rat liver parenchyma. A stereological study. *Journal of Cell Biology*, 72(2):441–455, 02 1977.

[BC09] Zakaria Babutsidze and Robin Cowan. Inertia, interaction and clustering in demand. *United Nations University, Maastricht Economic and social Research and training centre on Innovation and Technology, UNU-MERIT Working Paper Series*, 01 2009.

[BCD+10] Søren M Bentzen, Louis S Constine, Joseph O Deasy, Avi Eisbruch, Andrew Jackson, Lawrence B Marks, Randall K Ten Haken, and Ellen D Yorke. Quantitative analyses of normal tissue effects in the clinic (QUANTEC): An introduction to the scientific issues. *Int. J. Radiat. Oncol. Biol. Phys.*, 76(3):S3–S9, March 2010.

[BG05] Ingwer Borg and Patrick Groenen. *Modern Multidimensional Scaling: Theory and Applications (Springer Series in Statistics)*. 08 2005.

[bio] Liver Structure | BioNinja — ib.bioninja.com.au. `https://ib.bioninja.com.au/options/option-d-human-physiology/d3-functions-of-the-liver/liver-structure.html`. [Accessed 21-01-2024].

[Bis14] Henri Bismuth. A new look on liver anatomy: Needs and means to go beyond the couinaud scheme. *Journal of Hepatology*, 60(3):480–481, 2014.

[BMC+21] Alexandru Blidisel, Iasmina Marcovici, Dorina Coricovac, Florin Hut, Cristina Dehelean, and Octavian Cretu. Experimental models of hepatocellular carcinoma—a preclinical perspective. *Cancers*, 13, 08 2021.

[BMR+10] I. Barshack, E. Meiri, S. Rosenwald, D. Lebanony, M. Bronfeld, S. Aviel-Ronen, K. Rosenblatt, S. Polak-Charcon, I. Leizerman, M. Ezagouri, M. Zepeniuk, Norberto Shabes, L. Cohen, S. Tabak, D. Cohen, Z. Bentwich, and N. Rosenfeld. Differential diagnosis of hepatocellular carcinoma from metastatic tumors in the liver using microrna expression. *The international journal of biochemistry cell biology*, 42 8:1355–62, 2010.

[BR12] Reto Bürgin and Gilbert Ritschard. Categorical parallel coordinate plot. 01 2012.

[BRDC04] F. Bosch, J. Ribes, Mireia Díaz, and R. Clèries. Primary liver cancer: worldwide incidence and trends. *Gastroenterology*, 127 5 Suppl 1:S5–S16, 2004.

[Bre01] L Breiman. Random forests. *Machine Learning*, 45:5–32, 10 2001.

[BVZ01] Yuri Boykov, Olga Veksler, and Ramin Zabih. Fast approximate energy minimization via graph cuts. ieee pami, 23(11), 1222-1239. *Pattern Analysis and Machine Intelligence, IEEE Transactions on*, 23:1222–1239, 12 2001.

[BZA20] Azzedine Boukerche, Lining Zheng, and Omar Alfandi. Outlier detection: Methods, models, and classification. *ACM Comput. Surv.*, 53(3), jun 2020.

[CAC10] Santhana Chaimontree, Katie Atkinson, and Frans Coenen. Best clustering configuration metrics: Towards multiagent based clustering. pages 48–59, 11 2010.

[cana] Estadísticas importantes sobre el cáncer de hígado — cancer.org. `https://www.cancer.org/es/cancer/tipos/cancer-de-higado/acerca/que-es-estadisticas-clave.html`. [Accessed 21-01-2024].

[canb] Liver cancer | Causes, Symptoms & Treatments — cancer.org.au. `https://www.cancer.org.au/cancer-information/types-of-cancer/liver-cancer`. [Accessed 21-01-2024].

[canc] Radiation Therapy for Cancer — cancer.gov. `https://www.cancer.gov/about-cancer/treatment/types/radiation-therapy#LDL`. [Accessed 21-01-2024].

[cand] What Is Cancer? — cancer.gov. `https://www.cancer.gov/about-cancer/understanding/what-is-cancer`. [Accessed 21-01-2024].

[CCX+20] Ya Cui, Zhe Cui, Jianfeng Xu, Dapeng Hao, Jiejun Shi, Dan Wang, Hui Xiao, Xiaohong Duan, Runsheng Chen, and Wei Li. Ng-circos: next-generation circos for data visualization and interpretation. *NAR Genomics and Bioinformatics*, 2, 09 2020.

[CG20] M. Chetan and F. Gleeson. Radiomics in predicting treatment response in non-small-cell lung cancer: current status, challenges and future perspectives. *European Radiology*, 31:1049 – 1058, 2020.

[CHM19] Luca Cozzi, Ben J M Heijmen, and Ludvig P Muren. Advanced treatment planning strategies to enhance quality and efficiency of radiotherapy. *Phys. Imaging Radiat. Oncol.*, 11:69–70, July 2019.

[CJB+21] Matthijs Cysouw, Bernard Jansen, Tim Brug, Daniela Oprea-Lager, Elisabeth Pfaehler, Bart de Vries, Reindert Moorselaar, Otto Hoekstra, Andre Vis, and Ronald Boellaard. Machine learning-based analysis of [18f]dcfpyl pet radiomics for risk stratification in primary prostate cancer. *European Journal of Nuclear Medicine and Molecular Imaging*, 48, 02 2021.

[CLKP10] Jaegul Choo, Hanseung Lee, Jaeyeon Kihm, and Haesun Park. ivisclassifier: An interactive visual analytics system for classification based on supervised dimension reduction. pages 27 – 34, 11 2010.

[CM02] Dorin Comaniciu and Peter Meer. Meer, p.: Mean shift: A robust approach toward feature space analysis. ieee transactions on pattern analysis and machine intelligence 24(5), 603-619. *Pattern Analysis and Machine Intelligence, IEEE Transactions on*, 24:603 – 619, 06 2002.

[col] Liver Functions, Location, Anatomy and Disease | Columbia Surgery — columbiasurgery.org. [Accessed 21-01-2024].

[CT07] B. Chang and R. Timmerman. Stereotactic body radiation therapy: A comprehensive review. *American Journal of Clinical Oncology*, 30:637–644, 2007.

[Cui19] Wenqiang Cui. Visual analytics: A comprehensive overview. *IEEE Access*, 7:1–1, 06 2019.

[CWH+02] Jason Chia-Hsien Cheng, Jian-Kuen Wu, Chao-Ming Huang, Hua-Shan Liu, David Y. Huang, Skye Hongiun Cheng, Stella Y. Tsai, James Jer-Min Jian, Yu-Mong Lin, Tsun-I Cheng, Cheng-Fang Horng, and Andrew T. Huang. Radiation-induced liver disease after three-dimensional conformal radiotherapy for patients with hepatocellular carcinoma: dosimetric analysis and implication. *International Journal of Radiation Oncology*Biology*Physics*, 54(1):156–162, 2002.

[DAVY08] Jamshid Dehmeshki, Hamdan Amin, Manlio Valdivieso, and Xujiong Ye. Segmentation of pulmonary nodules in thoracic ct scans: A region growing approach. *Medical Imaging, IEEE Transactions on*, 27:467 – 480, 05 2008.

[DC14] Nieves Doménech García and Natalia Cal Purriños. Biobancos y su importancia en el ámbito clínico y científico en relación con la investigación biomédica en españa. *Reumatología Clínica*, 10(5):304–308, 2014.

[dCC23] Lucas B.V. de Amorim, George D.C. Cavalcanti, and Rafael M.O. Cruz. The choice of scaling technique matters for classification performance. *Applied Soft Computing*, 133:109924, 2023.

[DEC06] Laura A Dawson, Cynthia Eccles, and Tim Craig. Individualized image guided iso-NTCP based liver cancer SBRT. *Acta Oncol.*, 45(7):856–864, January 2006.

[dic] dicomstandard.org. `https://www.dicomstandard.org/News-dir/ftsup/docs/sups/sup11.pdf`. [Accessed 05-02-2024].

[DPP+19] Stefanos Diamantopoulos, Kalliopi Platoni, Georgios Patatoukas, Pantelis Karaiskos, Vassilis Kouloulias, and Efstathios Efstathopoulos. Treatment plan verification: A review on the comparison of dose distributions. *Physica Medica*, 67:107–115, 2019.

[ED23] Oumer Ebrahim and Getachew Derbew. Application of supervised machine learning algorithms for classification and prediction of type-2 diabetes disease status in afar regional state, northeastern ethiopia 2021. *Scientific Reports*, 13, 05 2023.

[edu] Expediente cl&xED;nico: caracter&xED;sticas, componentes, importancia, ejemplo — definicion.edu.lat. `https://definicion.edu.lat/academia/6360F155D23DFE44CFBB2C95418714D8.html`. [Accessed 18-03-2024].

[ESNH+23] Sarah El-Sherbiny, Jing Ning, Brigitte Hantusch, Lukas Kenner, and Renata Raidou. Visual analytics for the integrated exploration and sensemaking of cancer cohort radiogenomics and clinical information. In *VCBM 2023: Eurographics Workshop on Visual Computing for Biology and Medicine*, page 13. The Eurographics Association, September 2023.

[FBJ11] Mary Feng and Edgar Ben-Josef. Radiation therapy for hepatocellular carcinoma. *Semin. Radiat. Oncol.*, 21(4):271–277, October 2011.

[FC23] Luís Ferreira and Paulo Cortez. Autooc: A python module for automated multi-objective one-class classification. *Software Impacts*, 18:100590, 2023.

[Fek13] Jean-Daniel Fekete. Visual analytics infrastructures: From data management to exploration. *Computer*, 46:22–29, 2013.

[FF21] CARLOS ERNESTO F. and KARLA LISSETTE F. Pruebas para comprobar la normalidad de datos en procesos productivos: Anderson-darling, ryan-joiner, shapiro-wilk y kolmogÓrov-smirnov. *Societas. Revista de Ciencias Sociales y Humanísticas*, 23:83–106, 2021.

[FGM+20] Katarína Furmanová, Nicolas Grossmann, Ludvig P. Muren, Oscar Casares-Magaz, Vitali Moiseenko, John P. Einck, M. Eduard Gröller, and Renata G. Raidou. Vapor: Visual analytics for the exploration of pelvic organ variability in radiotherapy. *Computers Graphics*, 91:25–38, 2020.

[FHI17] Gong Tang Triptish Bhatia Stewart J. Anderson Mary Amanda Dew Robert Krafty Vishwajit Nimgaonkar Smita Deshpande Martica Hall Fanyin He, Sati Mazumdar and Charles F. Reynolds III. Non-parametric manova approaches for non-normal multivariate outcomes with missing values. *Communications in Statistics - Theory and Methods*, 46(14):7188–7200, 2017. PMID: 29416225.

[Fis05] Brian Fisher. *Illuminating the Path: An RD Agenda for Visual Analytics*, pages 69–104. 01 2005.

[GAA+13] Matthias Guckenberger, Nicolaus Andratschke, H Alheit, R Holy, C. Moustakis, Ursula Nestle, and Otto Sauer. Definition of stereotactic body radiotherapy principles and practice for the treatment of stage i non-small cell lung cancer. *Strahlentherapie und Onkologie : Organ der Deutschen Rontgengesellschaft ... [et al]*, 190, 09 2013.

[GCDGO04] C Greco, G Catalano, A Di Grazia, and R Orecchia. Radiotherapy of liver malignancies. from whole liver irradiation to stereotactic hypofractionated radiotherapy. *TUMORI*, 90(1):73–79, JAN-FEB 2004.

[GFC+14] T. Germain, S. Favelier, J.-P. Cercueil, A. Denys, D. Krausé, and B. Guiu. Liver segmentation: Practical tips. *Diagnostic and Interventional Imaging*, 95(11):1003–1016, 2014.

[Gir] Heena Girdher. Hierarchical Clustering — medium.com. `https://medium.com/analytics-vidhya/hierarchical-clustering-d2d92835280c`. [Accessed 13-03-2024].

[git] GitHub - GLCUnet/dataset — github.com. `https://github.com/glcunet/dataset`. [Accessed 06-02-2024].

[GJT08] Bin Gao, Won-Il Jeong, and Zhigang Tian. Liver: An organ with predominant innate immunity. *Hepatology*, 47(2):729–736, 2008.

[GKH15] Robert Gillies, Paul Kinahan, and Hedvig Hricak. Radiomics: Images are more than pictures, they are data. *Radiology*, 278:151169, 11 2015.

[gra] GitHub - AmarJ/GraphCut: Kaptur is an image segmentation tool that uses a graph cut algorithm to segment the foreground from an image. — github.com. `https://github.com/AmarJ/GraphCut`. [Accessed 22-03-2024].

[GWY+11] Hanqi Guo, Zuchao Wang, Bowen Yu, Huijing Zhao, and Xiaoru Yuan. Tripvista: Triple perspective visual trajectory analytics and its application on microscopic traffic data at a road intersection. pages 163–170, 03 2011.

[Ham10] Dong-Han Ham. The state of the art of visual analytics. pages 213–222, 09 2010.

[Har18] Anoop Haridass. Developments in stereotactic body radiotherapy. *Cancers (Basel)*, 10(12):497, December 2018.

[Har21] Jenine Harris. Primer on binary logistic regression. *Family Medicine and Community Health*, 9:e001290, 12 2021.

[HCN+16] Elizabeth Huynh, Thibaud P. Coroller, Vivek Narayan, Vishesh Agrawal, Ying Hou, John Romano, Idalid Franco, Raymond H. Mak, and Hugo J.W.L. Aerts. Ct-based radiomic analysis of stereotactic body radiation therapy patients with lung cancer. *Radiotherapy and Oncology*, 120(2):258–266, 2016.

[HJR+14] Syeda H. Afroze, Kendal Jensen, Kinan Rahal, Fanyin Meng, Gianfranco Alpini, and Shannon S. Glaser. Chapter 26 - liver regeneration: The stem cell approach. In Giuseppe Orlando, Jan Lerut, Shay Soker, and Robert J. Stratta, editors, *Regenerative Medicine Applications in Organ Transplantation*, pages 375–390. Academic Press, Boston, 2014.

[HK98] S. Hojjat and J Kittler. Region growing: A new approach. *IEEE transactions on image processing : a publication of the IEEE Signal Processing Society*, 7:1079–84, 02 1998.

[HMI+22] [1] Harshada, C Mandhare, S Idate, Asso Professor, and Sonali Mali. A comparative study of cluster based outlier detection, distance based outlier detection and density based outlier detection techniques. 08 2022.

[HN98] Jerry L. Hintze and Ray D. Nelson. Violin plots: A box plot-density trace synergism. *The American Statistician*, 52(2):181–184, 1998.

[hol] Getting Started &x2014; Panel v1.3.8 — panel.holoviz.org. `https://panel.holoviz.org/getting_started/index.html`. [Accessed 06-03-2024].

[HTVR17] M. Hatt, F. Tixier, D. Visvikis, and C. Cheze Le Rest. Radiomics in pet/ct: More than meets the eye? *The Journal of Nuclear Medicine*, 58:365 – 366, 2017.

[HZTM+22] P. Hoegen, Kevin Zhang, Eric Tonndorf-Martini, F. Weykamp, S. Regnery, P. Naumann, Kristin Lang, J. Ristau, Stefan Koerber, Constantin Dreher, C. Buchele, Carolin Rippke, C. Katharina Renkamp, K. Paul, Laila König, C. Büsch, Johannes Krisam, Oliver Sedlaczek, H.-P Schlemmer, and J. Hörner-Rieber. Mr-guided adaptive versus itv-based stereotactic body radiotherapy for hepatic metastases (maestro): a randomized controlled phase ii trial. *Radiation Oncology*, 17, 03 2022.

[ICR] `https://journals.sagepub.com/home/cru`. [Accessed 15-02-2024].

[ima] Hepatic segmentation - e-Anatomy - IMAIOS — imaios.com. `https://www.imaios.com/en/e-anatomy/anatomical-structure/hepatic-segmentation-121126600`. [Accessed 21-01-2024].

[ITS+22] Ali Idrus, Naf'An Tarihoran, Ucup Supriatna, Ahmad Tohir, Suwarni Suwarni, and Robbi Rahim. Distance analysis measuring for clustering using k-means and davies bouldin index algorithm. *TEM Journal*, 11:1871–1876, 11 2022.

[Jol02] I.T. Jolliffe. *Principal Component Analysis*. Springer Series in Statistics. Springer, 2002.

[Jos] Rohan Joseph. Grid Search for model tuning — towards-datascience.com. https://towardsdatascience.com/grid-search-for-model-tuning-3319b259367e#:~:text=Grid%20search%20builds%20a%20model,to%20find%20the%20best%20solution. [Accessed 15-03-2024].

[JWF15] Jimmy Johansson Westberg and Camilla Forsell. Evaluation of parallel coordinates: Overview, categorization and guidelines for future research. *IEEE Transactions on Visualization and Computer Graphics*, 22:1–1, 11 2015.

[KAF+08a] Daniel Keim, Gennady Andrienko, Jean-Daniel Fekete, Carsten Görg, Jörn Kohlhammer, and Guy Melançon. Visual analytics: Definition, process, and challenges. 03 2008.

[KAF+08b] Daniel Keim, Gennady Andrienko, Jean-Daniel Fekete, Carsten Görg, Jörn Kohlhammer, and Guy Melançon. Visual analytics: Definition, process, and challenges. 03 2008.

[KGB+12] Virendra Kumar, Yuhua Gu, Satrajit Basu, Anders Berglund, Steven A. Eschrich, Matthew B. Schabath, Kenneth Forster, Hugo J.W.L. Aerts, Andre Dekker, David Fenstermacher, Dmitry B. Goldgof, Lawrence O. Hall, Philippe Lambin, Yoganand Balagurunathan, Robert A. Gatenby, and Robert J. Gillies. Radiomics: the process and the challenges. *Magnetic Resonance Imaging*, 30(9):1234–1248, 2012. Quantitative Imaging in Cancer.

[KNM+23] Atsuto Katano, Tomoyuki Noyama, Kosuke Morishima, Yuki Nozawa, and Hideomi Yamashita. Dosimetric comparison study between free breathing and breath hold techniques in patients treated by liver-directed stereotactic body radiation therapy. *Cureus*, June 2023.

[KOW+21] K. Kraus, M. Oechsner, J. Wilkens, Kerstin Kessel, Stefan Münch, and Univ.-Prof. Dr. Stephanie Combs. Patient individual phase gating for stereotactic radiation therapy of early stage non-small cell lung cancer (nsclc). *Scientific Reports*, 11:5870, 03 2021.

[KSB+09] Martin Krzywinski, Jacqueline Schein, Inanç Birol, Joseph Connors, Randy Gascoyne, Doug Horsman, Steven Jones, and Marco Marra. Circos: an

information aesthetic for comparative genomics. *Genome research*, 19:1639–45, 07 2009.

[KWZH16a] Florence K. Keane, Jennifer Y. Wo, Andrew X. Zhu, and Theodore S. Hong. Liver-Directed Radiotherapy for Hepatocellular Carcinoma. *Liver Cancer*, 5(3):198–209, 05 2016.

[KWZH16b] Florence K. Keane, Jennifer Y. Wo, Andrew X. Zhu, and Theodore S. Hong. Liver-directed radiotherapy for hepatocellular carcinoma. *LIVER CANCER*, 5(3):198–209, 2016.

[KZK+15] Maria-Aggeliki Kalogeridi, Anna Zygogianni, George Kyrgias, John Kouvaris, Sofia Chatziioannou, Nikolaos Kelekis, and Vassilis Kouloulias. Role of radiotherapy in the management of hepatocellular carcinoma: A systematic review. *WORLD JOURNAL OF HEPATOLOGY*, 7(1):101–112, JAN 27 2015.

[LCC15] Chun Yu Liu, Kuen-Feng Chen, and Pei-Jer Chen. Treatment of liver cancer. *Cold Spring Harbor perspectives in medicine*, 5, 07 2015.

[LH06] Kongkuo Lu and William Higgins. Improved 3d live-wire method with application to 3d ct chest image analysis. *Proceedings of SPIE - The International Society for Optical Engineering*, 6144, 03 2006.

[Liv] liver ligaments | Human liver anatomy, Diagnostic medical sonography, Liver anatomy — pinterest.com. `https://www.pinterest.com/pin/52002570678885071/`. [Accessed 18-03-2024].

[LKT23] Guillaume Landry, Christopher Kurz, and Alberto Traverso. The role of artificial intelligence in radiotherapy clinical practice. *BJR|Open*, 5, 10 2023.

[LLD+17a] Philippe Lambin, Ralph Leijenaar, Timo Deist, Jurgen Peerlings, Evelyn de Jong, Janita Van Timmeren, Sebastian Sanduleanu, Ruben Larue, Aniek Even, Arthur Jochems, Yvonka Wijk, Henry Woodruff, Johan Soest, Tim Lustberg, Erik Roelofs, Wouter Elmpt, André Dekker, Felix Mottaghy, Joachim Wildberger, and Seán Walsh. Radiomics: The bridge between medical imaging and personalized medicine. *Nature Reviews Clinical Oncology*, 14, 10 2017.

[LLD+17b] Philippe Lambin, Ralph Leijenaar, Timo Deist, Jurgen Peerlings, Evelyn de Jong, Janita Van Timmeren, Sebastian Sanduleanu, Ruben Larue, Aniek Even, Arthur Jochems, Yvonka Wijk, Henry Woodruff, Johan Soest, Tim Lustberg, Erik Roelofs, Wouter Elmpt, André Dekker, Felix Mottaghy, Joachim Wildberger, and Seán Walsh. Radiomics: The bridge between medical imaging and personalized medicine. *Nature Reviews Clinical Oncology*, 14, 10 2017.

[Loh] H. Lohninger. PCA - Loadings and Scores — statistics4u.com. http://www.statistics4u.com/fundstat_eng/cc_pca_loadscore.html. [Accessed 13-03-2024].

[LPS+20] Vincent Lemarteleur, Matthieu Peycelon, Jean-Louis Sablayrolles, Patrick Plaisance, Alaa El-Ghoneimi, and Pierre-francois Ceccaldi. Realization of open software chain for 3d modeling and printing of organs in simulation centers: Example of renal pelvis reconstruction. *Journal of Surgical Education*, 78, 07 2020.

[LRVL+12] Philippe Lambin, Emmanuel Rios-Velazquez, Ralph Leijenaar, Sara Carvalho, Ruud G.P.M. van Stiphout, Patrick Granton, Catharina M.L. Zegers, Robert Gillies, Ronald Boellard, André Dekker, and Hugo J.W.L. Aerts. Radiomics: Extracting more information from medical images using advanced feature analysis. *European Journal of Cancer*, 48(4):441–446, 2012.

[Lux04] Ulrike Luxburg. A tutorial on spectral clustering. *Statistics and Computing*, 17:395–416, 01 2004.

[LW07] S. Larsson and A. Wolk. Overweight, obesity and risk of liver cancer: a meta-analysis of cohort studies. *British Journal of Cancer*, 97:1005 – 1008, 2007.

[Mas23] Ryota Masuzaki. Liver cancer: Improving standard diagnosis and therapy. *Cancers*, 15(18), 2023.

[Mat23] Yukinori Matsuo. Stereotactic body radiotherapy for hepatocellular carcinoma: A brief overview. *Curr. Oncol.*, 30(2):2493–2500, February 2023.

[Max14] Brent Maxfield. Chapter 8 - plotting. In Brent Maxfield, editor, *Essential PTC Mathcad® Prime® 3.0*, pages 199–261. Academic Press, Boston, 2014.

[May16] Nova Mayasari. Comparison of support vector machine and decision tree in predicting on-time graduation (case study : Universitas pembangunan panca budi). *International Journal of Recent Trends in Engineering Research*, 2:2016, 12 2016.

[MG10] A. Martin and A. Gaya. Stereotactic body radiotherapy: A review. *Clinical Oncology*, 22(3):157–172, 2010.

[MHF+23] Joao Miranda, Natally Horvat, Gilton Marques Fonseca, Jose de Arimateia Batista Araujo-Filho, Maria Clara Fernandes, Charlotte Charbel, Jayasree Chakraborty, Fabricio Ferreira Coelho, Cesar Higa Nomura, and Paulo Herman. Current status and future perspectives of radiomics in hepatocellular carcinoma. *World J. Gastroenterol.*, 29(1):43–60, January 2023.

[MHGS+14] James D Murphy, Jona Hattangadi-Gluth, William Y Song, Emily Vollans, Marie-Laure Camborde, Robert Kosztyla, Shaun K Loewen, Catherine Crumley, and Vitali Moiseenko. Liver toxicity prediction with stereotactic body radiation therapy: The impact of accounting for fraction size. *Pract. Radiat. Oncol.*, 4(6):372–377, November 2014.

[MHM20] Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction, 2020.

[MML+20] Marius E. Mayerhoefer, Andrzej Materka, Georg Langs, Ida Häggström, Piotr Szczypiński, Peter Gibbs, and Gary Cook. Introduction to radiomics. *Journal of Nuclear Medicine*, 61(4):488–495, 2020.

[MMT+14] Pietro Majno, Gilles Mentha, Christian Toso, Philippe Morel, Heinz O. Peitgen, and Jean H.D. Fasel. Anatomy of the liver: An outline with three levels of complexity – a further step towards tailored territorial liver resections. *Journal of Hepatology*, 60(3):654–662, 2014.

[MNR07] P Mayles, A Nahum, and J C Rosenwald, editors. *Handbook of radiotherapy physics*. CRC Press, Boca Raton, FL, June 2007.

[MNRP08] Philip Mayles, Alan Nahum, J. Rosenwald, and Niko Papanikolaou. Handbook of radiotherapy physics: Theory and practice. *Medical Physics*, 35:4281, 08 2008.

[MVM+21] Moyed Miften, Yevgeniy Vinogradskiy, Vitali Moiseenko, Jimm Grimm, Ellen Yorke, Andrew Jackson, Wolfgang A. Tomé, Randall K. Ten Haken, Nitin Ohri, Alejandra Méndez Romero, Karyn A. Goodman, Lawrence B. Marks, Brian Kavanagh, and Laura A. Dawson. Radiation dose-volume effects for liver sbrt. *International Journal of Radiation Oncology*Biology*Physics*, 110(1):196–205, 2021. A Red Journal Special Issue: HyTEC.

[OFC08] Dário Oliveira, Raul Feitosa, and Mauro Correia. Automatic couinaud liver and veins segmentation from ct images. pages 249–252, 01 2008.

[PGND16] I. Popp, A. Grosu, G. Niedermann, and D. Duda. Immune modulation by hypofractionated stereotactic radiation therapy: Therapeutic implications. *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology*, 120 2:185–94, 2016.

[PKD+10] Charlie C. Pan, Brian D. Kavanagh, Laura A. Dawson, X. Allen Li, Shiva K. Das, Moyed Miften, and Randall K. Ten Haken. Radiation-associated liver injury. *International Journal of Radiation Oncology*Biology*Physics*, 76(3, Supplement):S94–S100, 2010. Quantitative Analyses of Normal Tissue Effects in the Clinic.

[PR21] Gonzalez-Del Portillo Elisabet Rodriguez-Gutierrez Ana Matias-Perez Angela Perez-Romasanta, Luis A. Stereotactic radiotherapy for hepatocellular carcinoma, radiosensitization strategies and radiation-immunotherapy combination. *Cancers (Basel)*, 13(2):192, January 2021.

[PSDK15] Paul C. Parsons, K. Sedig, Arman Didandeh, and Arash Khosravi. Interactivity in visual analytics: Use of conceptual frameworks to support human-centered design of a decision-support tool. *2015 48th Hawaii International Conference on System Sciences*, pages 1138–1147, 2015.

[pyr] Welcome to pyradiomics documentation! &x2014; pyradiomics v3.1.0rc2.post5+g6a761c4 documentation — pyradiomics.readthedocs.io. `https://pyradiomics.readthedocs.io/en/latest/#`. [Accessed 06-02-2024].

[RBGR18] Oliver Reiter, Marcel Breeuwer, Eduard Gröller, and Renata Georgia Raidou. Comparative Visual Analysis of Pelvic Organ Segmentations. In Jimmy Johansson, Filip Sadlo, and Tobias Schreck, editors, *EuroVis 2018 - Short Papers*. The Eurographics Association, 2018.

[RBP+22] Matteo Renzulli, Nicolò Brandi, Anna Pecorelli, Luigi Vincenzo Pastore, Alessandro Granito, Giuseppe Martinese, Francesco Tovoli, Mario Simonetti, Elton Dajti, Antonio Colecchia, and Rita Golfieri. Segmental distribution of hepatocellular carcinoma in cirrhotic livers. *Diagnostics (Basel)*, 12(4):834, March 2022.

[RBV17] Renata G. Raidou, Marcel Breeuwer, and Anna Vilanova. Visual Analytics for Digital Radiotherapy: Towards a Comprehensible Pipeline. 2017.

[RCMA+18] Renata Raidou, Oscar Casares-Magaz, Aleksandr Amirkhanov, Vitali Moiseenko, L.P. Muren, John Einck, Anna Vilanova, and Eduard Gröller. Bladder runner : Visual analytics for the exploration of rt-induced bladder toxicity in a cohort study. *Computer Graphics Forum*, 37:205–216, 06 2018.

[RDRS+19] Anna Rossetto, Valli De Re, Agostino Steffan, Gianmaria Miolo, Patrizia Leone, Vito Racanelli, Alessandro Uzzau, Umberto Baccarani, and Matteo Cescon. Carcinogenesis and metastasis in liver: Cell physiological basis. *Cancers*, 11:1731, 11 2019.

[REE+09] G. Robertson, D. Ebert, S. Eick, D. Keim, and K. Joy. Scale and complexity in visual analytics. *Information Visualization*, 8:247 – 253, 2009.

[Ris01] Irina Rish. An empirical study of the naïve bayes classifier. *IJCAI 2001 Work Empir Methods Artif Intell*, 3, 01 2001.

[RLC+10] Ricarda Ruehl, Lutz Luedemann, Anna Czarnecka, Florian Streitparth, Max Seidensticker, Konrad Mohnike, Maciej Pech, Peter Wust, and Jens Ricke. Radiobiological restrictions and tolerance doses of repeated single-fraction hdr-irradiation of intersecting small liver volumes for recurrent hepatic metastases. *RADIATION ONCOLOGY*, 5, MAY 27 2010.

[Rud14] David A. Rudnick. Chapter 25 - liver regeneration: The developmental biology approach. In Giuseppe Orlando, Jan Lerut, Shay Soker, and Robert J. Stratta, editors, *Regenerative Medicine Applications in Organ Transplantation*, pages 353–374. Academic Press, Boston, 2014.

[Rus16] Daniel Russell. Simple is good: Observations of visualization use amongst the big data digerati. pages 7–12, 06 2016.

[RYY+21] Amir Masoud Rahmani, Efat Yousefpoor, Mohammad Sadegh Yousefpoor, Zahid Mehmood, Amir Haider, Mehdi Hosseinzadeh, and Rizwan Ali Naqvi. Machine learning (ml) in medicine: Review, applications, and challenges. *Mathematics*, 9(22), 2021.

[Saj] Basil Saji. Elbow Method for Finding the Optimal Number of Clusters in K-Means — analyticsvidhya.com. `https://www.analyticsvidhya.com/blog/2021/01/in-depth-intuition-of-k-means-clustering-algorithm-in-machine-learnin` [Accessed 12-03-2024].

[San14] Naoko Sanuki. Role of stereotactic body radiation therapy for hepatocellular carcinoma. *World J. Gastroenterol.*, 20(12):3100, 2014.

[SBL12] Cheng Saw, Shanglian Bao, and Sicong Li. A review on the technical and dosimetric aspects of stereotactic body radiation therapy (sbrt). *Journal of Radiation Oncology*, 1, 12 2012.

[scia] Getting Started — scikit-learn.org. `https://scikit-learn.org/stable/getting_started.html`. [Accessed 28-03-2024].

[scib] SciPy - — scipy.org. `https://scipy.org/`. [Accessed 28-03-2024].

[SCR+10] S. Son, B. Choi, M. Ryu, Y. Kang, Jisun Jang, S. Bae, S. Yoon, I. Choi, K. Kang, and H. Jang. Stereotactic body radiotherapy for patients with unresectable primary hepatocellular carcinoma: dose-volumetric parameters predicting the hepatic complication. *International journal of radiation oncology, biology, physics*, 78 4:1073–80, 2010.

[SDK+21] Joshua D. Shur, Simon J. Doran, Santosh Kumar, Derfel ap Dafydd, Kate Downey, James P. B. O'Connor, Nikolaos Papanikolaou, Christina Messiou, Dow-Mu Koh, and Matthew R. Orton. Radiomics in oncology: A practical guide. *RadioGraphics*, 41(6):1717–1732, 2021. PMID: 34597235.

[Sed12] Philip Sedgwick. Pearson's correlation coefficient. *BMJ*, 345:e4483–e4483, 07 2012.

[Sem20] Manojkumar Semwal. Khan's the physics of radiation therapy. *J. Med. Phys.*, 45(2):134, 2020.

[SFS+21] Hyuna Sung, Jacques Ferlay, Rebecca L. Siegel, Mathieu Laversanne, Isabelle Soerjomataram, Ahmedin Jemal, and Freddie Bray. Global cancer statistics 2020: Globocan estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, 71(3):209–249, 2021.

[SGB+21] Camilla Scapicchio, Michela Gabelloni, Andrea Barucci, Dania Cioni, Luca Saba, and Emanuele Neri. A deep look into radiomics. *La Radiologia medica*, 126, 07 2021.

[SHD+22] Po-Chien Shen, Wen-Yen Huang, Yang-Hong Dai, Cheng-Hsiang Lo, Jen-Fu Yang, Yu-Fu Su, Ying-Fu Wang, Chia-Feng Lu, and Chun-Shu Lin. Radiomics-based predictive model of radiation-induced liver disease in hepatocellular carcinoma patients receiving stereo-tactic body radiotherapy. *Biomedicines*, 10(3), 2022.

[SJY23] Ajay P. Singh, Vansh S. Jain, and John-Paul J. Yu. Diffusion radiomics for subtyping and clustering in autism spectrum disorder: A preclinical study. *Magnetic Resonance Imaging*, 96:116–125, 2023.

[SK21] Meshal Shutaywi and Nezamoddin Kachouie. Silhouette analysis for performance evaluation in machine learning with applications to clustering. *Entropy*, 23:759, 06 2021.

[sli] 3D Slicer image computing platform — slicer.org. `https://www.slicer.org/`. [Accessed 06-02-2024].

[Smi20] Abir Smiti. A critical overview of outlier detection methods. *Computer Science Review*, 38:100306, 2020.

[SMM+17] Evis Sala, E. Mema, E. Mema, Yuki Himoto, H. Veeraraghavan, James D. Brenton, A. Snyder, B. Weigelt, and H. A. Vargas. Unravelling tumour heterogeneity using next-generation imaging: radiomics, radiogenomics, and habitat imaging. *Clinical radiology*, 72 1:3–10, 2017.

[SNM+22] Jordan Sack, Jennifer Nitsch, Hans Meine, Ron Kikinis, Michael Halle, and Anna Rutherford. Quantitative analysis of liver disease using mri-based radiomic features of the liver and spleen. *Journal of Imaging*, 8(10), 2022.

[SPB12] K. Sedig, Paul C. Parsons, and Alex Babanski. Towards a characterization of interactivity in visual analytics. *J. Multim. Process. Technol.*, 3:12–28, 2012.

[SRM+19] Matthias Schlachter, Renata Raidou, L.P. Muren, Bernhard Preim, P.M. Putora, and Katja Bühler. State-of-the-art report: Visual computing in radiation therapy planning. *Computer Graphics Forum*, 38:753–779, 06 2019.

[SSC+19] Ryan Schulze, Micah Schott, Carol Casey, Pamela Tuma, and Mark McNiven. The cell biology of the hepatocyte: A membrane trafficking machine. *The Journal of Cell Biology*, 218:jcb.201903090, 06 2019.

[TCNBC+17] Nathalia Tunissiolli, Márcia Castanhole-Nunes, Patrícia Biselli-Chicote, Érika Pavarino, Renato da Silva, Rita de Cassia da Silva, and Eny Goloni-Bertollo. Hepatocellular carcinoma: a comprehensive review of biomarkers, clinical aspects, and therapy. *Asian Pacific journal of cancer prevention : APJCP*, 18:863–872, 04 2017.

[TFN+09] Guillaume Thibault, Bernard FERTIL, Claire Navarro, Sandrine Pereira, Nicolas Lévy, Jean Sequeira, and Jean-Luc Mari. Texture indexes and gray level size zone matrix application to cell nuclei classification. 11 2009.

[TLSX19] Jiang Tian, Li Liu, Zhongchao Shi, and Feiyu Xu. *Automatic Couinaud Segmentation from CT Volumes on Liver Using GLC-UNet*, pages 274–282. 10 2019.

[TRY+17] Tetsuya Tsujikawa, Tasmiah Rahman, Makoto Yamamoto, Shizuka Yamada, Hideaki Tsuyoshi, Yasushi Kiyono, Hirohiko Kimura, Yoshio Yoshida, and Hidehiko Okazawa. 18f-fdg pet radiomics approaches: comparing and clustering features in cervical cancer. *Annals of Nuclear Medicine*, 31, 08 2017.

[TSKK18] Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, and Vipin Kumar. *Introduction to Data Mining (2nd Edition)*. Pearson, 2nd edition, 2018.

[TWX+23] Shengnan Tang, Jin Wu, Shanshan Xu, Qi Li, and Jian He. Clinical-radiomic analysis for non-invasive prediction of liver steatosis on non-contrast ct: A pilot study. *Frontiers in Genetics*, 14, 2023.

[uzh] Mann-Whitney-U-Test — methodenberatung.uzh.ch. `https://www.methodenberatung.uzh.ch/de/datenanalyse_spss/unterschiede/zentral/mann.html`. [Accessed 21-02-2024].

[vdMH08] Laurens van der Maaten and Geoffrey Hinton. Viualizing data using t-sne. *Journal of Machine Learning Research*, 9:2579–2605, 11 2008.

[vGFP+17] Joost van Griethuysen, Andrey Fedorov, Chintan Parmar, Ahmed Hosny, Nicole Aucoin, Vivek Narayan, Regina Beets-Tan, Jean-Christophe Fillion-Robin, Steve Pieper, and Hugo Aerts. Computational radiomics system to decode the radiographic phenotype. *Cancer Research*, 77:e104–e107, 11 2017.

[VOA19] Victoria Vickerstaff, Rumana Omar, and Gareth Ambler. Methods to adjust for multiple comparisons in the analysis and sample size calculation of randomised controlled trials with multiple primary outcomes. *BMC Medical Research Methodology*, 19, 06 2019.

[WCH18] Wei-Jun Wang, Jeng-Fong Chiou, and Yaoru Huang. Treatment of liver metastases using an internal target volume method for stereotactic body radiotherapy. *Journal of Visualized Experiments*, 2018, 05 2018.

[Wij13] J. V. Wijk. Evaluation: A challenge for visual analytics. *Computer*, 46:56–60, 2013.

[WJL+22] Manyang Wang, Renchao Jin, Jiayi Lu, Enmin Song, and Guangzhi Ma. Automatic ct liver couinaud segmentation based on key bifurcation detection with attentive residual hourglass-based cascaded network. *Computers in Biology and Medicine*, 144:105363, 2022.

[WLW+24] Meilong Wu, Liping Liu, Xiaojuan Wang, Ying Xiao, Shizhong Yang, and Jiahong Dong. Radiomic features on contrast-enhanced images of the remnant liver predict the prognosis of hepatocellular carcinoma after partial hepatectomy. *iLIVER*, page 100079, 2024.

[WRB20] Jacob S Witt, Stephen A Rosenberg, and Michael F Bassetti. MRI-guided adaptive radiotherapy for liver tumours: visualising the future. *Lancet Oncol.*, 21(2):e74–e82, February 2020.

[XWY+21] Ruizhi Xiang, Wencan Wang, Lei Yang, Shiyuan Wang, Chaohan Xu, and Xiaowen Chen. A comparison for dimensionality reduction methods of single-cell rna-seq data. *Frontiers in Genetics*, 12, 2021.

[YJY+17] Lina Yu, Hengle Jiang, Hongfeng Yu, Chi Zhang, Josiah McAllister, and Dandan Zheng. ivar: Interactive visual analytics of radiomics features from large-scale medical images. pages 3916–3923, 12 2017.

[YS] B.Pharm. Yolanda Smith. Radiation Therapy Dosage — news-medical.net. `https://www.news-medical.net/health/Radiation-Therapy-Dosage.aspx`. [Accessed 23-01-2024].

[ZKW+23] Jun Zhou, Minglei Kang, Yinan Wang, Kristin A Higgins, Charles B Simone, Pretesh Patel, Mark W McDonald, Liyong Lin, and Duncan Bohannon. Proton liver stereotactic body radiation therapy: Treatment techniques and dosimetry feasibility from a single institution. *Journal of Radiosurgery and SBRT*, 9(1):33–42, 2023.

[ZZC+23] Yuan-Peng Zhang, Xin-Yun Zhang, Yu-Ting Cheng, Bing Li, Xin-Zhi Teng, Jiang Zhang, Saikit Lam, Ta Zhou, Zong-Rui Ma, Jia-Bao Sheng, Victor Tam, Lee Shara, Hong Ge, and Jing Cai. Artificial intelligence-driven

radiomics study in cancer: the role of feature engineering and modeling. *Military Medical Research*, 10:22, 05 2023.